

NuKAT Preliminary Design Review (PDR)

May 4th, 6th, and 8th, 2026

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PDR Panelist Introduction

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- **Dr. Amir Bahadori**
 - Professor and Program Director, Kansas State University Nuclear Engineering
 - Former NASA civil servant in the Space Radiation Analysis Group (SRAG)
- **Aiden Bourman**
 - PhD student, K-State Nuclear Engineering
 - Research focus: Micro-Pocket Fission detectors
- **Chris Brandt**
 - President, K-State's Research Foundation
- **Dave Copeland**
 - Space Communications System Engineer, John Hopkins University Applied Physics Laboratory
 - Co-Founder, Silversat



PDR Panelist Introduction

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- **Glen Fountain**
 - Project Manager, John Hopkins University Applied Physics Laboratory – Retired
 - Former Industry Advisor, Mike Wiegers Department of Electrical and Computer Engineering, Kansas State University
- **Chris Hersman**
 - Mission Systems Engineer, John Hopkins University Applied Physics Laboratory
- **Ashleigh Mauler**
 - M.S. student, K-State Nuclear Engineering
 - Research focus: Health Physics
- **Zach Metzinger**
 - Engineer, METECS, Lunar Gateway Mission
 - Member, AMSAT Board of Directors
 - Volunteer, AMSAT GOLF Program



PDR Panelist Introduction

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- **Dr. Taylor Ochs**
 - Nuclear Engineer, RDT
 - Award-winning research on the Dual-Sided MSND
- **Dr. Matt O'Keefe**
 - Professor, Kansas State University Chemical Engineering
- **Jeffrey Sheldon**
 - Director, K-State's Sunderland Foundation Innovation Lab
 - FCC Trustee, K-State's Amateur Radio Club (W0QQQ)
- **Dr. Tim Steffensmeier**
 - Assistant Vice President and Director of Engagements and Outreach, Kansas State University
- **Kyle Tidball**
 - Engineer Specialist, Electronics Development Group, Textron Aviation
- **Tim Trenkle**
 - Engineer (Retired), NASA Goddard Space Flight Center



Kevin and Stacie Needham

- President – Kiewit Power Engineering
- Senior Vice President – Power Markets & Strategy
- **Thank you for your support and generous donation**

Wildcat Space Program

- **Thank you to those who have supported us through WSP fundraising drives**

Purpose of PDR

Objective – Preliminary Design Review

Provide increased assurance that the proposed design approach, and the manufacturing and test implementation plans, will result in an acceptable product, with minimal project risk.

Outcomes

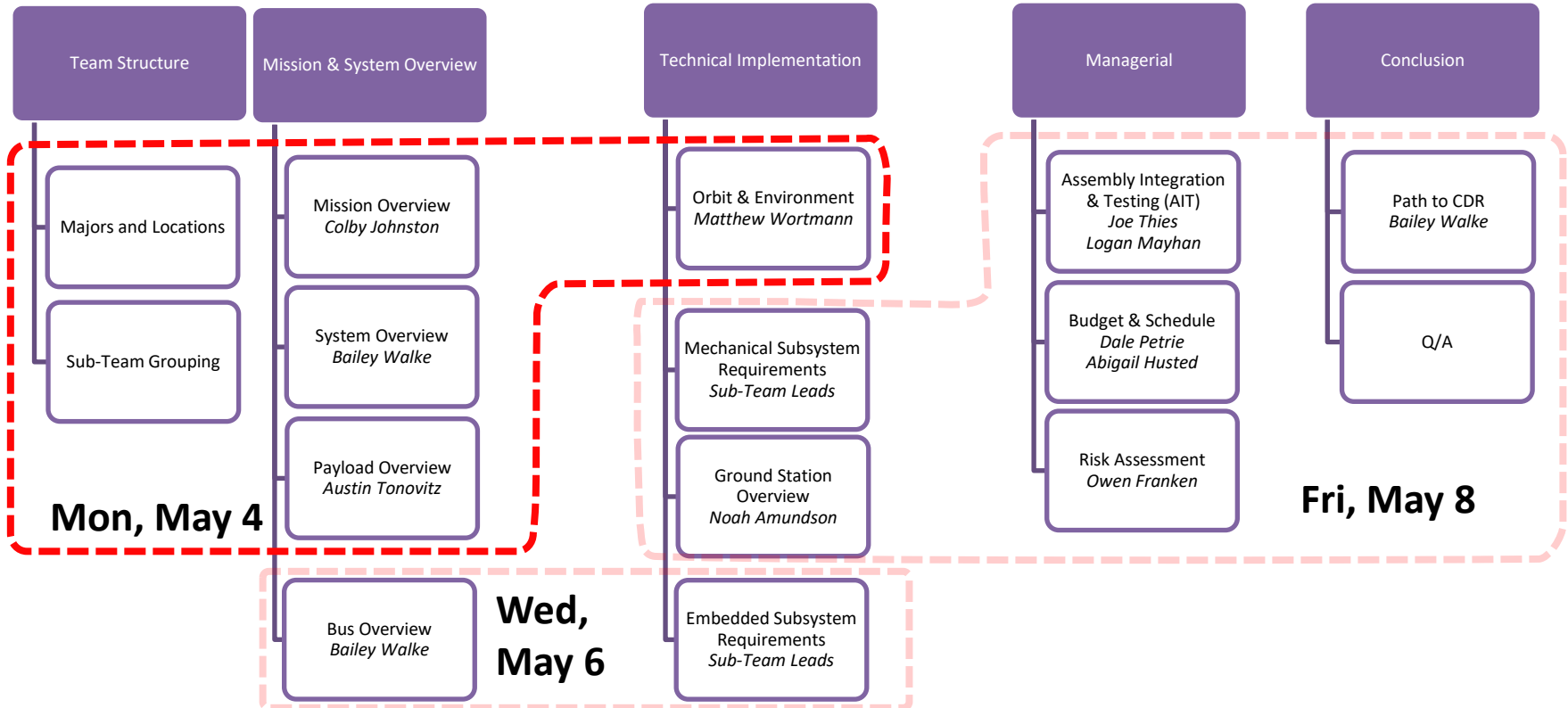
- Confirmation of technical feasibility within budget and schedule
- Confirmation that design meets all requirements and to verify allocations
- Assessment of high risks
- Complete and approved technical baseline

Review Focus Areas

- Mission objectives and rationale
- System/subsystem design and interfaces
- Orbit and environment
- Schedule, budget and risk management
- Readiness for complete design review

Agenda

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Presenters

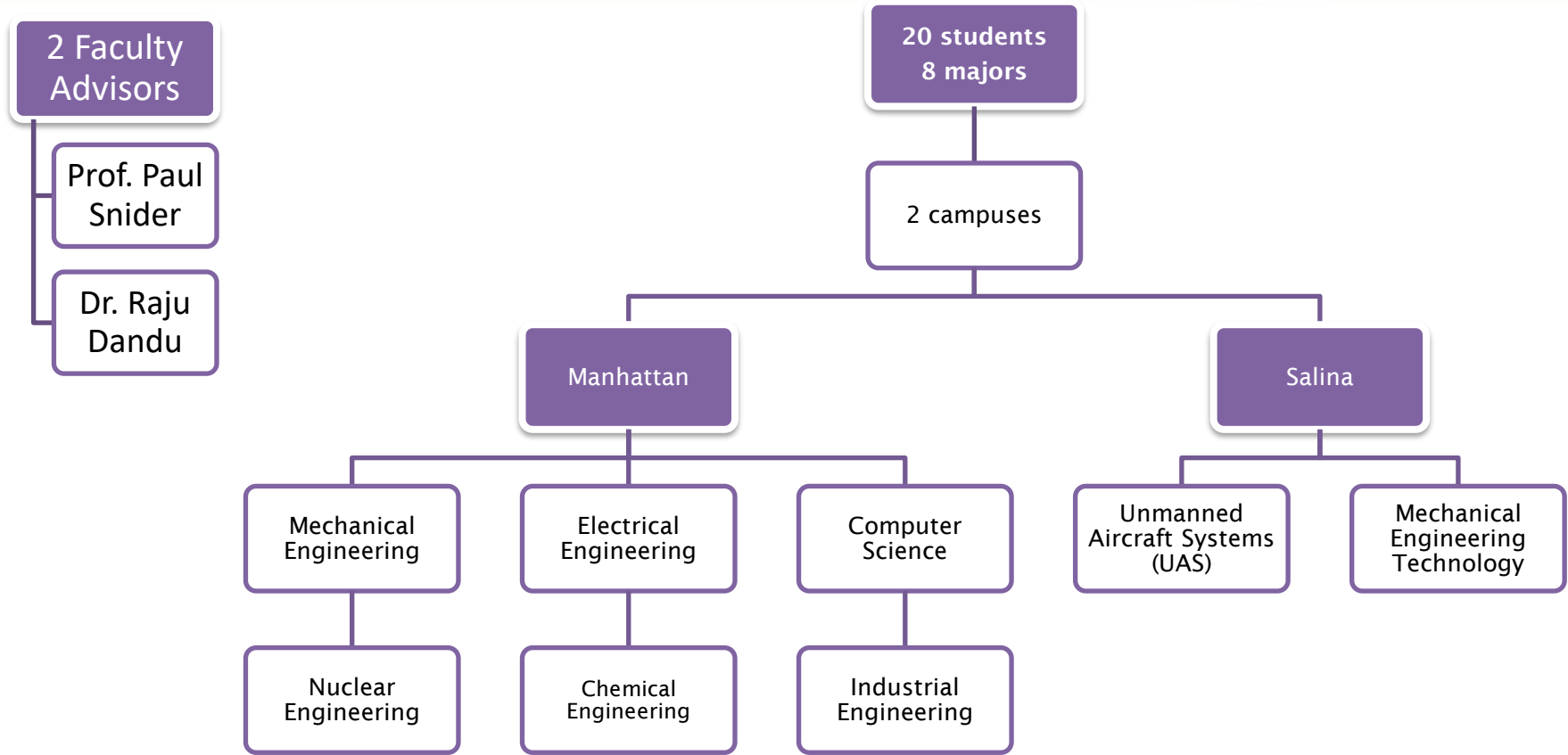
8

Colby Johnston	Principal Investigator
Bailey Walke	Chief Systems Engineer
Matthew Wortmann	Orbit & Environment Lead
Austin Tonovitz	Payload Lead



Team Acknowledgement

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MISSION OVERVIEW

Neutron detector: K-State Analysis Tool

Colby Johnston – NuKAT Principal Investigator
Mechanical and Nuclear Engineering | Spring 2026



Mission Background

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Typical Neutron Detector Drawbacks

- Pressure vessels
- Large volume
- High voltages

Microstructured Semiconductor Neutron Detectors (MSNDs)

- Low power
- Solid state
- Small volume
- High thermal neutron efficiency

MSND Platforms

- **Standard Domino** (four MSNDs) – TRL 9
 - NeRDI-1A and -1B missions
- **DSC Domino** (four MSNDs with integrated driver board interface) – TRL 5

Mission Importance

Space missions need flexibility in component characteristics (reduced SWAP, avoiding pressure vessels, “plug-and-play” modularity, etc.). The current neutron detector gaps are answered by MSNDs and the DSC Domino; developing its flight heritage results in better options for the future of space use and exploration.

Solution and Success Criteria

Mission Statement

Advancing space-based neutron detection using MSNDs in low Earth orbit and validating the DSC Domino as a space-qualified detector platform.

Minimum mission success

DSC Domino flight heritage – TRL 7

Two months of science operations

Reduced data package

Complete mission success

DSC Domino Space Qualification – TRL 8

One year of science operations

Full data package

Mission Success

NuKAT will achieve minimum mission success by building DSC Domino **flight heritage** and producing location-correlated measurements over **two months**.

NuKAT will achieve complete mission success from DSC Domino **space qualification** and satellite-tagged location-correlated measurements over **one year**.

Objectives and Stakeholders

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Mission Goals

Design, build and launch a CubeSat

DSC Domino space qualification*

K-State CubeSat program pathfinder

STEM education, engagement, and outreach

Provide neutron counts and orbital location*

Stakeholders

Core

- K-State
- Radiation Detection Technologies (RDT)

Secondary

- NASA's Space Radiation Analysis Group (SRAG)

Mission Requirements

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NUKAT REQUIREMENTS VERIFICATION MATRIX

Req. ID	Title	Requirement	Risk Score	Status
NuKAT LEVEL 1 REQUIREMENTS: Quantitative Mission-Level				
NU-1.01	Neutron Detector Flight Heritage	The neutron detector shall gather thermal and epithermal counts for comparison to accepted orbital readings and detector longevity qualification.	6	Partial Compliance
NU-1.02	Results Communication	The NuKAT mission shall write and distribute a publically-available paper detailing mission results following end of satellite operations.	4	Partial Compliance
NU-1.03	Operational Integration and Success	NuKAT shall have a fully operational mission (science) lifetime of at least 2 months with payload supported and integrated using a satellite bus system and ground station.	12	Not Compliant
NU-1.04	Operational Optimization	NuKAT shall ensure volume, power budget, data accuracy, data processing, link budget, orbit, and financial budget are optimized to ensure at least minimum mission success.	15	Partial Compliance
NU-1.06	Launch Survivability	All components and systems of NuKAT shall be capable of surviving the conditions present in travel from K-State to the launch provider, while waiting integration, during integration with the launch vehicle, and during launch up until release into its final orbit.	10	Partial Compliance



Mission Requirements

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NUKAT REQUIREMENTS VERIFICATION MATRIX

Req. ID	Title	Requirement	Risk Score	Status
NuKAT LEVEL 1 REQUIREMENTS: Quantitative Mission-Level				
NU-1.07	Space Environment Survivability	All components and systems of NuKAT shall be capable of surviving the space environment at the selected orbit and altitude.	10	Compliant
NU-1.08	Satellite Compliance	NuKAT shall be compliant with applicable guidelines from necessary governing bodies and launch providers.	5	Compliant
NU-1.09	Satellite Guidelines	NuKAT should be designed to meet the requirements specified in Caly Poly CubeSat Design Specifications (CP-CDS-R14.1) and University NanoSat Program Satellite Development Guidelines (EXTDOC-315).	4	Partial Compliance
NU-1.10	Contamination and Hazardous Materials	NuKAT systems shall NOT use cadmium parts, cadmium-plated parts, zinc plating, mercury, and/or compounds containing mercury, pure tin or tin electroplate, and silicone rubber or room-temperature-vulcanizing (RTV) silicones with probability or transfer to Co-payload(s) or Launch Vehicle hardware.	4	Compliant
NU-1.11	K-State Radio Station Use	NuKAT shall utilize the K-State radio station as a primary ground station for data downlink/uplink	6	Partial Compliance
NU-1.12	Deorbit Time	NuKAT shall be able to deorbit 5 years after end of science mission.	3	Compliant



Mission Updates/Justifications

Orbital Drivers



Requirement Compliance

- Pending budgets closing – there is a path forward
- Pending activities beyond PDR scope/timeline

Subsystems Compatibility

- Some ongoing trade studies

There is a Path Toward CDR

Mission Meets Main Objectives

In light of this, the mission should proceed toward CDR with further effort to close subsystem designs and improve detector measurement confidence.

SYSTEM OVERVIEW

Bailey Walke – NuKat Chief Systems Engineer

Mechanical Engineering Masters | Spring 2026



System Requirements Cont. 20

Req. ID	Title	Requirement	Rationale	Risk Score	Status
2.1 Satellite Requirements					
SYS-2.01a	Orbital Parameters	NuKAT shall successfully launch into a low earth orbit at an altitude of 500 km and with an LTAN between 8:00pm and 10:00pm.	This orbit will allow us to gather data for all stakeholders in the region of interest. Furthermore, this orbit allows us sufficient ground station access	4	Compliant
SYS-2.01b	Deorbit Time	NuKAT shall be able to deorbit 5 years after end of science mission.	Compliance with industry and NASA guidelines.	3	Compliant
SYS-2.01c	Altitude Range	NuKAT shall not have an initial altitude lower than 475 km.	Any orbit lower than 475 would make our mission life shorter than 2 years	3	Compliant
SYS-2.01d	Orbital Radiation	NuKAT shall survive a radiation environment of 7.25krad.	Will be interacting with high radiation environments due to mission objectives.	10	Compliant
SYS-2.02	Altitude Trace	NuKAT altitude shall be known with ± 1 km	Needed for stamped data	4	Partial Compliance
SYS-2.03	Location Trace	NuKAT latitude and longitude shall be known within 5 degrees of certainty	Needed for stamped data	8	Partial Compliance
SYS-2.04	Time Trace	Detector data downlinked by NuKAT shall provide time and date data.	Needed for stamped data	4	Partial Compliance
SYS-2.05	Satellite Orientation Range	NuKAT orientation shall be kept within +/- 20 degrees of margin off Nadir.	Detectors need to be oriented in a certain way to collect accurate counts	12	Partial Compliance
SYS-2.06	Operational Modes	NuKAT shall utilize operational modes to optimize power usage and data storage.	The satellite will experience high activity periods while downlinking data, then lower activity levels and idle periods.	8	Partial Compliance
SYS-2.09	System Monitoring	NuKAT shall monitor all subsystems and payloads to ensure they are nominally operational at all modes of operation.	Housekeeping data - information used to support the mission itself (SME).	10	Compliant
SYS-2.10	Satellite Size and Weight	NuKAT shall be 2U in size and not exceed 4kg.	Ensure that satellite is cost effective but still able to successfully complete mission objectives.	4	Compliant



System Requirements Cont. 21

Req. ID	Title	Requirement	Rationale	Risk Score	Status
2.1 Satellite Requirements					
SYS-2.11	Satellite Center of Gravity Range	NuKAT shall have a center of gravity within range specified by CP-CDS-R14.1 Table 2.	CDS-2.2.11	2	Compliant
SYS-2.14	System Cost	NuKAT shall cost less than \$200,000 USD.	Cost effective satellite	9	Compliant
SYS-2.15	Mission Lifetime	NuKAT shall be fully operational and supporting payload functions for at least 1 year.	Determined by mission success criteria	12	Partial Compliance
SYS-2.16	Satellite Survivability	NuKAT shall be designed to withstand the launch and on-orbit environments of the launch vehicle without failure that results in damage of the launch vehicle and its contents or failure that causes injury to the ground handling crew.	UNP11-1	15	Partial Compliance
SYS-2.17	Data Interpretation	NuKAT's payload readings and locational/time trace readings shall be correctly correlated and packaged for downlinking.	Data needs to be time traced and location tagged to be considered useful to stakeholders.	12	Compliant
SYS-2.18	Data Timeliness	Interpreted data shall be published within 4 months of science mission completion	The overall system is able to downlink and interpret data that is useful to the stakeholders and distribute it within a certain time.	6	N/A
SYS-2.22	Nominal Operation Satellite Orientation	Following initialization and apengage deployment, NuKAT shall be oriented with the -y face in the nadir direction and +y in the zenith direction.	Solar panel and antenna configuartion and lack of active attitude control	15	Partial Compliance
SYS-2.23	Operational Delays	NuKAT should be designed for variable delays after separation from the launch vehicle of the following: start of Radio Frequency (RF) transmission, active deployable actuation, attitude maneuvers, and orbital maneuvers.	UNP11-82	3	Compliant
SYS-2.24	Launch Vehicle Compliance	NuKAT System shall be compliant with selected launch provider dispenser system and regulations	Launch vehicle and dispenser complaine to be awarded launch opportunity	5	Compliant



Passive Attitude Control System

- Power budget – decrease power usage
- No need for yaw control about nadir axis

Solar Panel Configuration

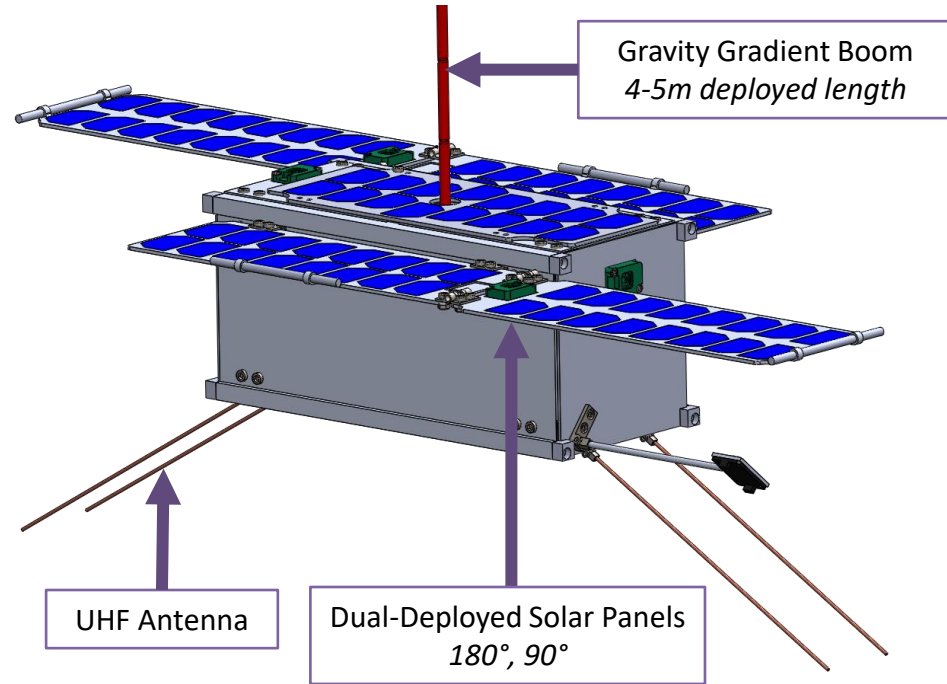
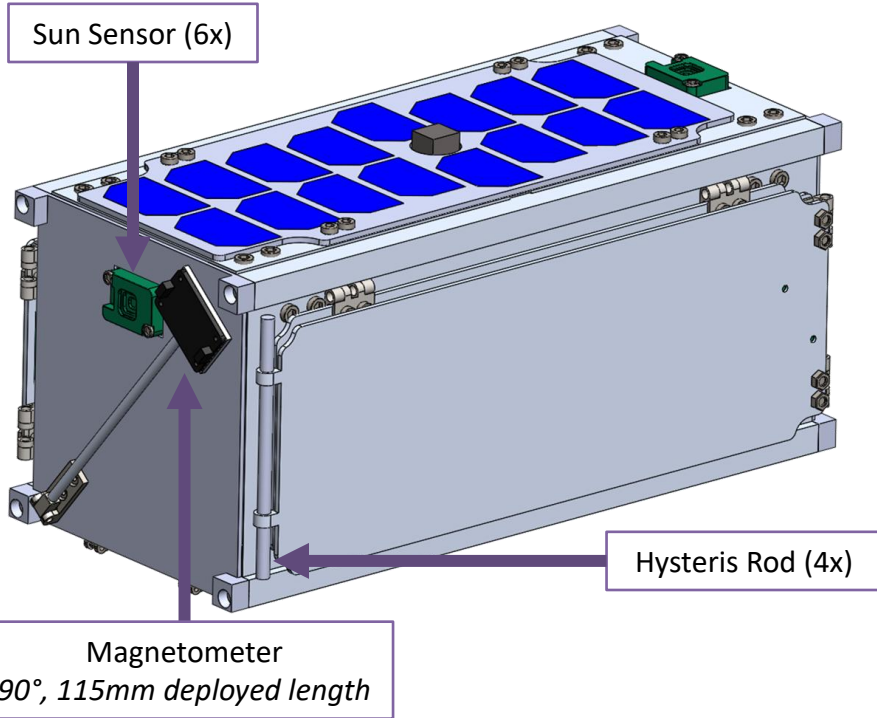
- Power budget – increase solar panel area
- “Hot dog” flight orientation

Internal Layout

- Payload spacing requirements
- Reduced number of subsystem components

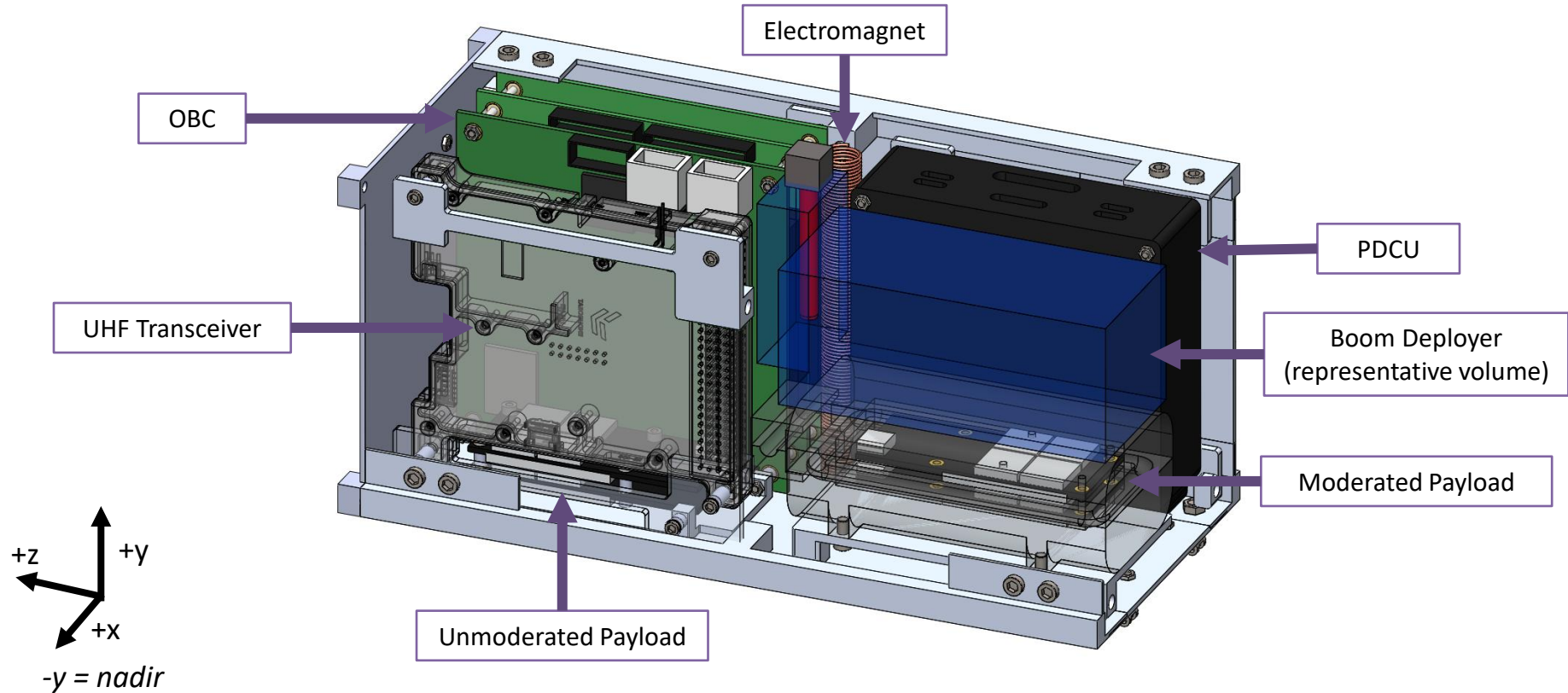
External Hardware Layout

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Internal Hardware Layout

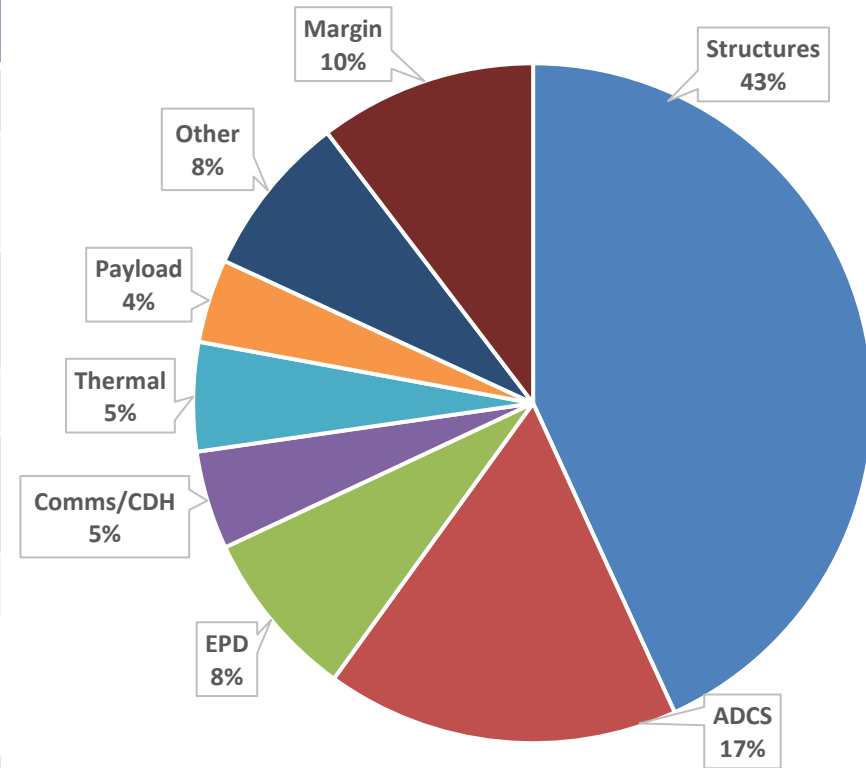
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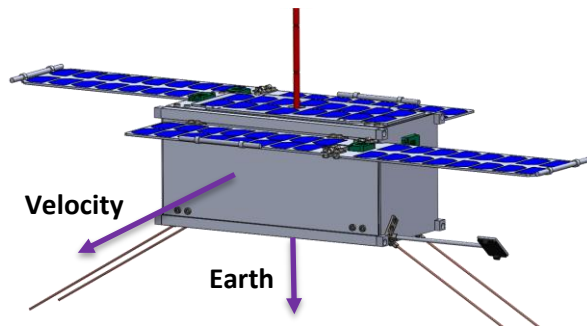
Mass Budget

25

Subsystem	Mass (g)	Percentage
Structures & Mechanisms	1667.12	43.16%
Attitude Determination & Control	649	16.80%
Electrical Power & Distribution	311.52	8.07%
Thermal	200	5.18%
Communications, Commands and Data Handling	181.04	4.69%
Payload	153.64	3.98%
Other (wiring, coatings)	300	7.77%
Margin	400	10.36%
Total	3862.32	



SATELLITE



Fast/thermal neutron counts

1 count every 60 seconds (**0.017Hz**); 300km ground distance between readings; **400kB** of science data each day

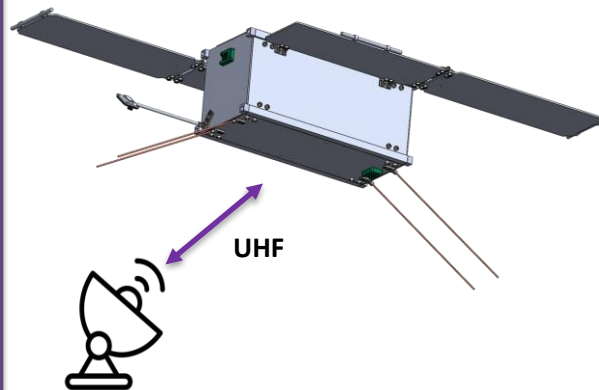
Science data tagging

Satellite orientation at count; data and time at count; latitude and longitude and count

Satellite health data

Critical component temperatures; battery level; operation mode log; data storage

SATELLITE/GROUND STATION



Data downlink

Downlinking **36 kB/min** (4.8 kbps) via UHF amateur radio; average contact time of around **12 minutes/day**; ground station contact every **7-12.5 hours**

Periodic command uplink

Manual mode command or specific data requests via UHF amateur radio

GROUND STATION



Science data post processing

Processing, validation and production of desired science data format

Science data distribution

Aggregated raw data and mission report distributed to stakeholders and published

Satellite health monitoring

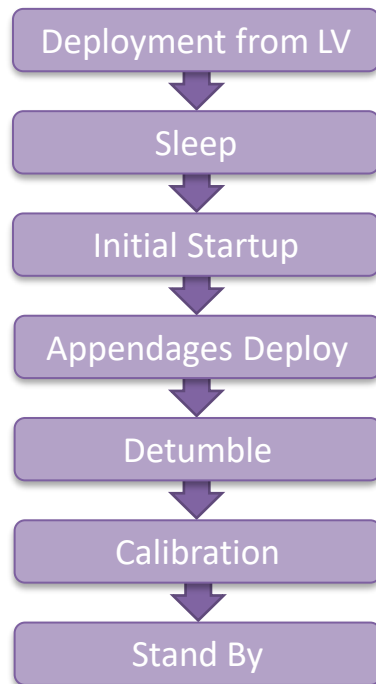
Reviewing health beacon packages to respond with uplink commands or software updates if necessary

Spacecraft Modes

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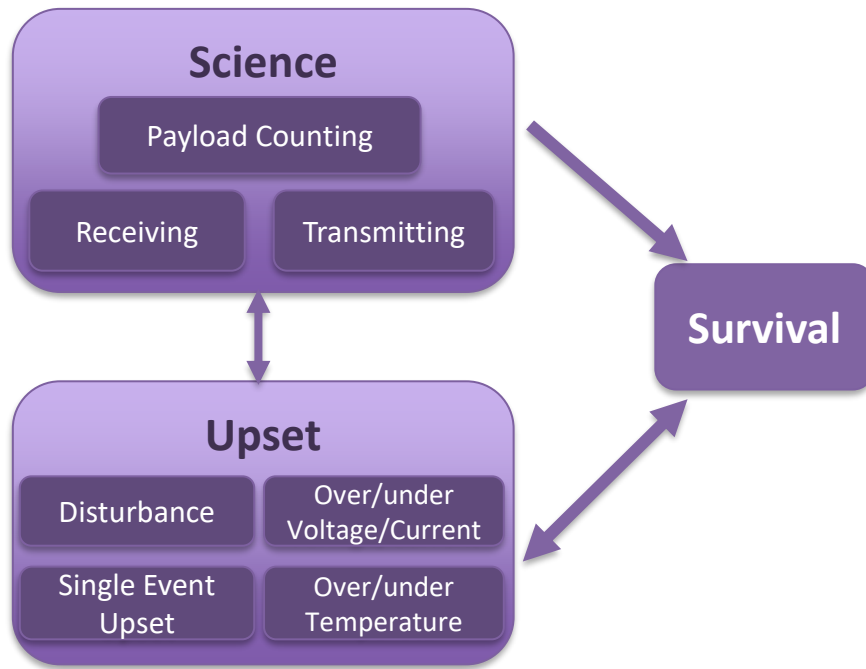
LAUNCH & INITIALIZATION

1 Month



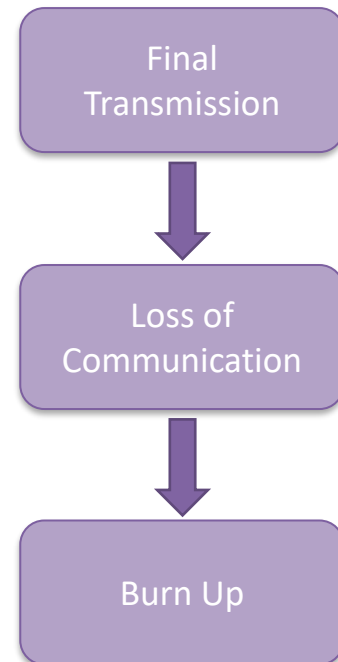
MISSION OPERATIONS

>12 Months



DECOMMISSIONING

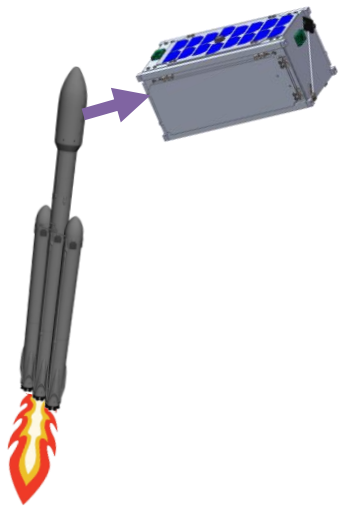
< 5 years



Initialization Phases

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LV Deployment

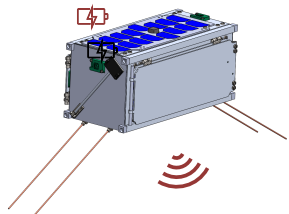


- Rail ejection from launch vehicle at specified altitude
- Maximum tumble rate of $60^\circ/\text{s}$

Appendage Deployment

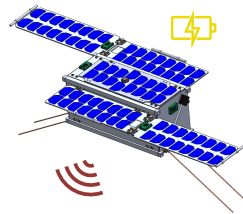
- Triggered automatically
- Sequential deployment (antenna, 180° panel, 90° panel)
- Tumbling assumed
- Thermal control within *survivable* temperature
- Full health package recorded and stored

UHF Antenna



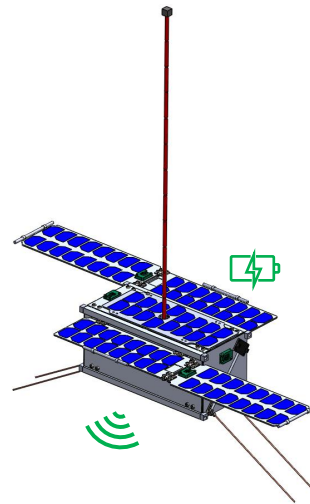
- Initiate connection with ground
- Limited contact time due to tumbling
- Battery level downlinked

Solar Panels



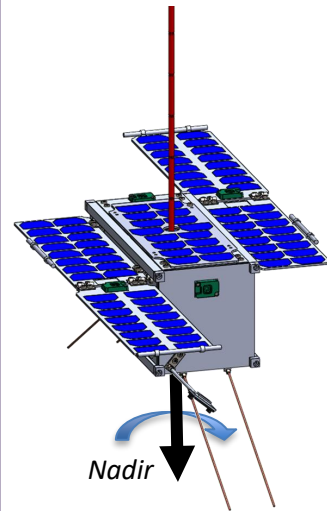
- Increasing sun exposed area for charging
- Increasing MOI decreases angular velocity

Boom Deployment



- Triggered automatically
- Deploy over poles with boom zenith facing
- Lock rotation about pitch and roll axes (x and z)

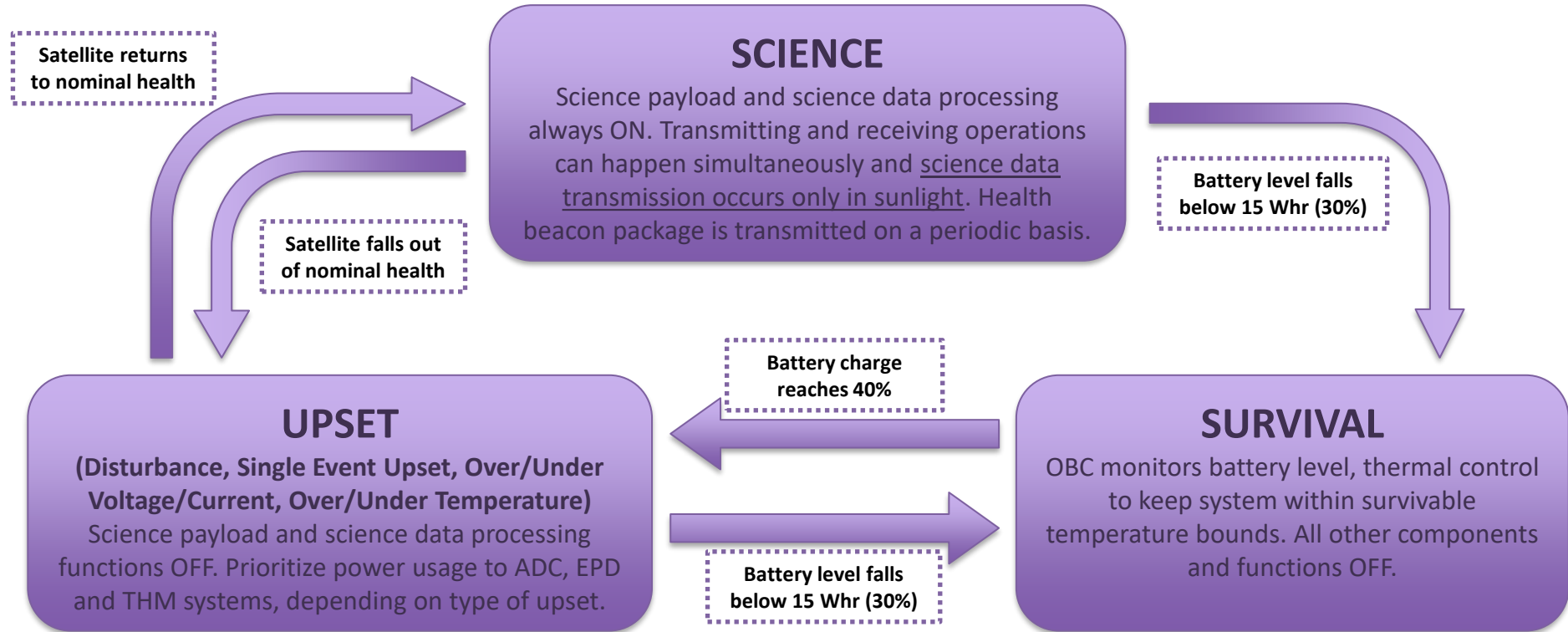
Stand By



- Triggered manually
- Periodic transmission of full health beacon package
- Spinning rate about nadir axis $<10^\circ/\text{s}$

Concept of Operations

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Requirement Compliance

- Most requirements that are able to be addressed at this time are compliant
- Environmental survivability requirements need further analysis and testing to determine compliance

System Compatibility

- Further design for deployment systems and layout
- FlatSat configuration to test subsystem compatibility
- Better understanding of ADC boom deployment mechanisms and timing

The System has a Path Toward CDR

The System is Capable of Supporting NuKAT Mission

In light of this, the system should proceed toward CDR with further effort to develop the layout and component placement to ensure mission requirements are met.

ORBIT & ENVIRONMENT

Matthew Wortmann – Orbit & Environment Lead
Chemical Engineering | Spring 2028



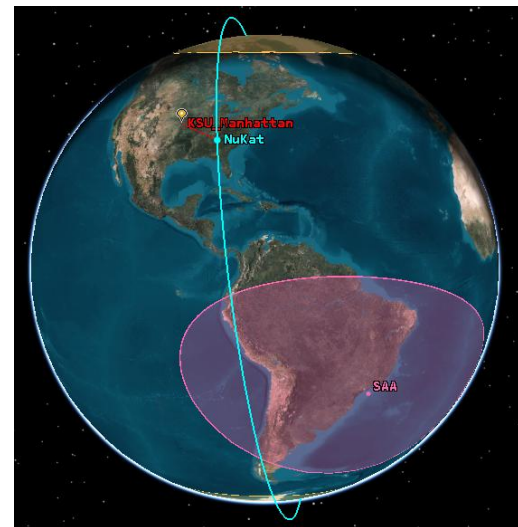
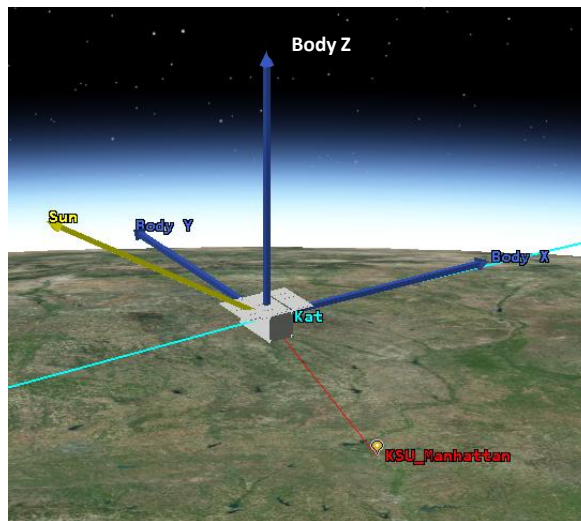
Points of Interest

Ground Stations	Science
KSU Manhattan	South Atlantic Anomaly (SAA)
KSU Salina	Earth Poles



Orbit

Type	Polar/Sun Synchronous
Altitude	500-600km
Inclination	$\sim 97^\circ \pm 5^\circ$
LTAN/LTDN	9pm / 9am



Left Figure: NuKAT body vectors with KSU access
Right Figure: Global view of orbit and points of interest

Matthew Wortmann

Orbit

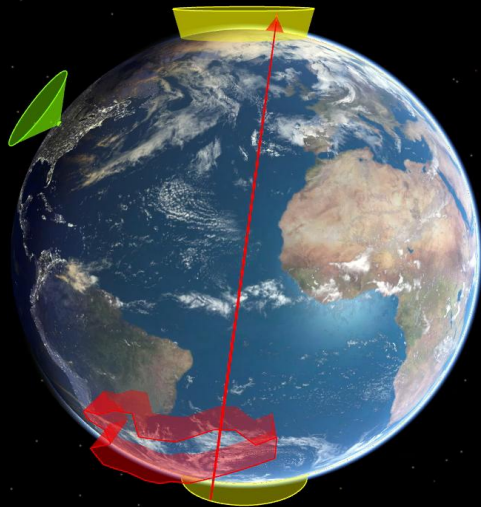
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Oct 01 2027 10:34:56.000000000 UTC

Target: Earth

Source: Earth(143° RA, 15° Dec, 30000 km Radius)

FOV: 45°



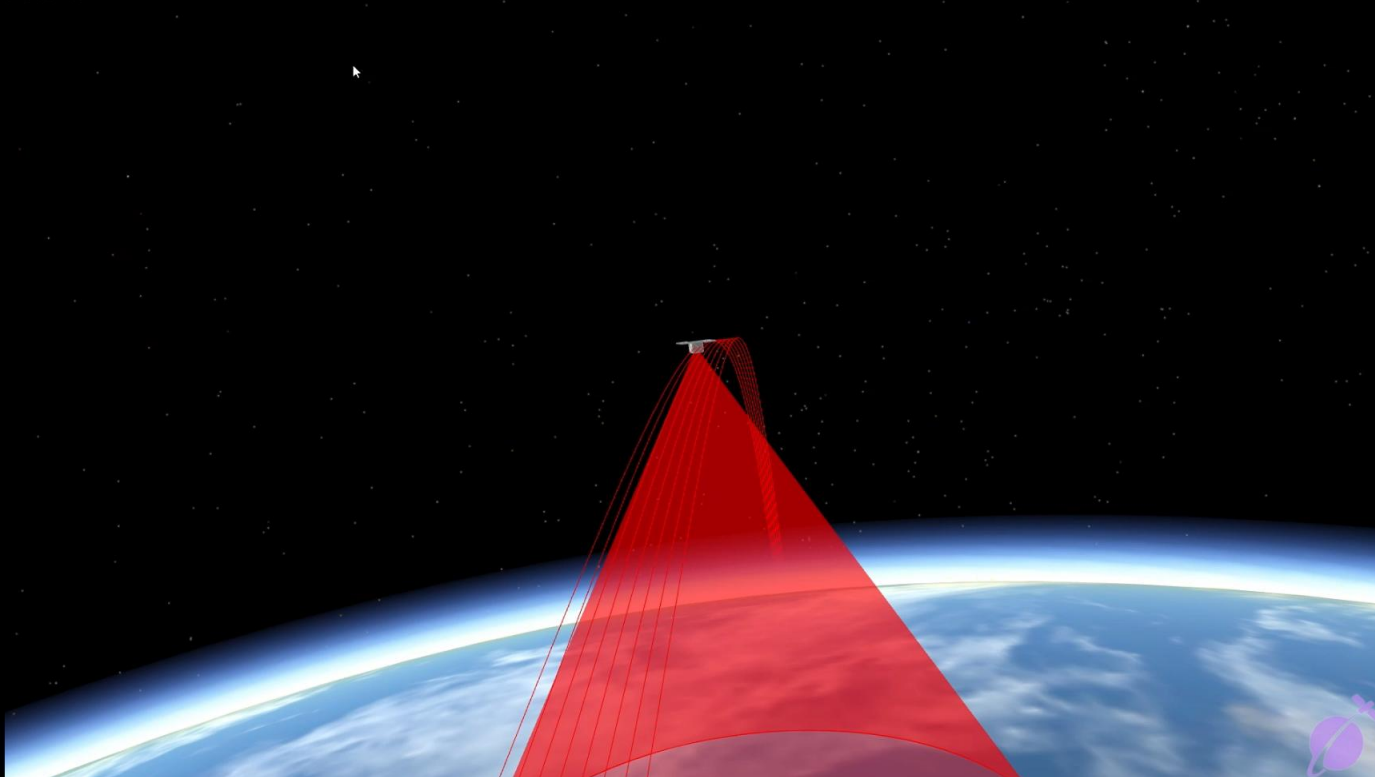
Pluto



Orbit

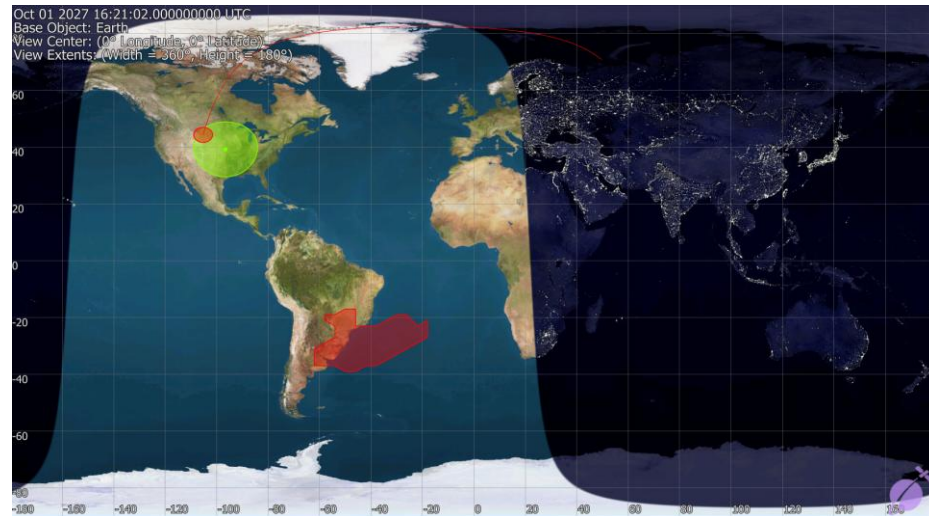
34

Oct 01 2027 12:23:26.000000000 UTC
Target: NuKAT
Source: NuKAT(320° RA, 42° Dec, 1028 km Radius)
FOV: 45°



Orbit

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Access for NuKAT to:	KSU Manhattan	SAA	North Pole	South Pole
Avg. Contact Time	5.8min	11.3min	11.7min	11.7min
Avg. Revisit Time	7.5-13hrs	1.4-6.25hrs	1.4hrs	1.4hrs



KANSAS STATE
UNIVERSITY

Carl R. Ice
College of Engineering

Left Figure: NuKAT ascending to KSU in light
Right Figure: NuKAT descending to KSU in dark

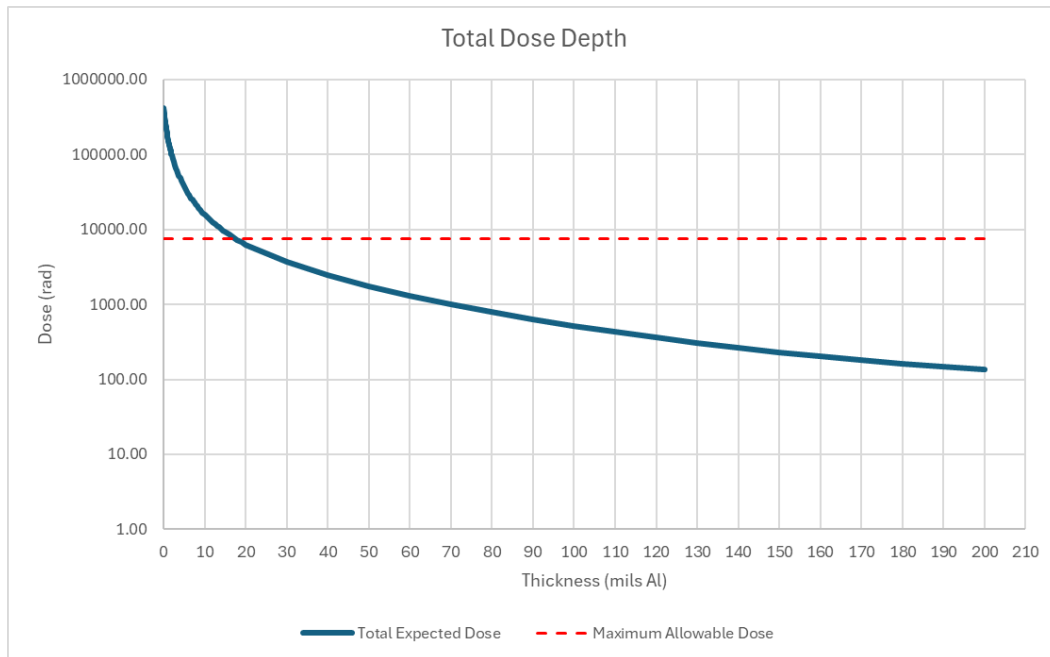
Matthew Wortmann

Radiation Risks

Mechanism	Effects	Source
Total Ionizing Dose	Degradation of electronics	- Trapped electrons & protons - Solar protons
Displacement Damage Dose	- Degradation of optical components and electronics - Degradation of solar panels	- Trapped electrons & protons - Solar protons - Neutrons
Single Event Effects	- Data corruption & noise on images - System shutdowns - Electronic component damage	- Trapped electrons & protons - Solar protons - Neutrons - Galactic cosmic rays
Surface Erosion	Degradation of components	- Particle radiation - Ultraviolet - Atomic oxygen

Space Environment

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Shielding Thickness	Dose (500 km)
0.01 mils AL "no shield"	~417.25 krad
10.5 mils AL (0.2667 mm)	~14.85 krad
18.0 mils AL (0.4572 mm)	~7.27 krad
60 mils AL (1.524 mm)	~1.30 krad

NASA recommends for reliability to use a Factor of Safety of 2 for determining shielding

Current Max Allowable Dose is 15 krad
With a FOS of 2, and that becomes 7.5 krad.

Space Environment

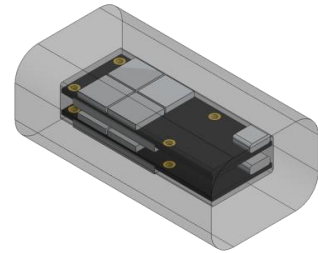
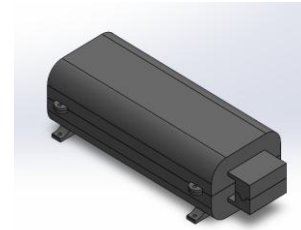
38

Environmental Risks

Mechanism	Effects	Source
Surface Charging	• Biasing of instrument readings • Power drains	• Plasma
Deep Dielectric Charging	• Biasing of instrument readings • Electrical discharges causing physical damage	• Trapped electrons
Structure Impacts	• Structural Damage	• Orbital Debris
Drag	• Torques • Orbital Decay*	• Thermosphere
Outgassing	• Condensing of volatile materials on vital bus components	• Satellite Components



- Material outgassing is measured in three criteria:
 - TML --- Total Mass Lost
 - "The mass loss of the sample is determined from the weights before and after the 398 K (125° C) exposure." [1]
 - CVCM --- Collected Volatile Condensable Material
 - "The difference between the weight of a clean collector and of the collector having condensed materials will provide the mass of condensables." [1]
 - WVR --- Water Vapor Recovered
 - "This value is obtained as a percentage of the starting mass is that amount of water readsorbed/reabsorbed in 24 hours while the sample is exposed to 25° C, and 50-percent relative humidity." [1]
- NASA standards require:
 - $TML \leq 1.0\%$ [2]
 - $CVCM \leq 0.10\%$ [2]
- Currently, the only material that lies outside of these standards is the neutron moderator – High Density Polyethylene or HDPE.
 - $TML = 0.58\%$
 - $CVCM = 0.19\% > 0.10\%$
- To bring the CVCM criteria into spec, we will need to use a vacuum-bake out process.



Several possible launch opportunities

- Earliest launch: October 2027

Analysis based on Falcon 9 requirements – *Details in backup slides*

- Fundamental Vibration Frequencies
- Acceleration Range
- Shock Loading
- Acoustic Environment
- Thermal Considerations
- Ascent Venting

Integration Parameters

- Dispenser requires all access ports on a single 2×1U side
- Environmental compatibility met in collaboration with launch provider
- Rail deployment

Launch System

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- Cost: \$75,000-90,000
- Tumble rate: $<60^\circ/\text{s}$
- Volume: 2U, optional tuna can
- AIT compliance: TVAC testing, shock & vibe testing
- Mass: 3.67kg, max 4.33+
- Compliance: N/A will resolve after provider is selected (depends on mission support, cost, schedule)
 - Currently Arrow Sci Tech or Launch 42

Req. ID	Title	Requirement	Rationale	Risk Score	Status
3.13 Launch System Requirements					
LNC-3.01	Deployment Tumble	The launch dispenser shall deploy NuKAT with a tumbling rate of less than 60 deg/s.	LNC must not create excessive tumbling, due to STR deployment and ADC detumble.	4	Not Compliant
LNC-3.02	Volume Compliance	LNC shall be capable of launching a 2U satellite into the desired orbit.	Constraint based on system parameters.	5	Compliant
LNC-3.03	Mass Compliance	LNC shall be capable of launching at least 2.66 kg into the desired orbit.	Constraint based on NASA CubeSat definition of 2kg/U and best practice of 1.33kg/U.	5	Compliant
LNC-3.04	Launch Provider Compliance	LNC shall provide STR with dispenser specifications and testing methods, to ensure compability with the provider's deployment system.	Launch provider affects STR deployment design.	5	Not Compliant
LNC-3.05	Cost	LNC shall cost less than \$x.	Budget allocated for launch.	3	N/A
4.13 Launch System Requirements					
LNC-4.01	Integration Compliance	All satellite components must comply with SpaceX Transporter and provider launch system requirements.	Dispenser and integration (LNC-3.01, 3.02, 3.03, 3.05) managed by launch provider. The only NuKAT requirement is that we meet company regulations.	5	Partial Compliance



PAYLOAD OVERVIEW

Austin Tonovitz – Payload System Lead

Mechanical and Nuclear Engineering | Spring 2027



Payload Requirements

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Level 2 Requirements:

Req. ID	Title	Requirement	Rationale	Risk Score	Status
PL-2.06	DSC-Domino keep-out zone	The two unmoderated platforms shall have their nadir- and zenith-faces unobstructed by the HDPE moderator within a cone of 70-degrees relative to zenith/nadir	To ensure that the moderator is not impacting the unmoderated detectors, they should be located in the 1U allocated for the bus systems where there is no HDPE directly above or below the detector. Ideally, they're on the opposite side of the craft from the moderator	16	Partial Compliance

Level 3 Requirements:

Req. ID	Title	Requirement	Rationale	Risk Score	Status
PL-3.01	Neutron Detector Payload Accuracy	The neutron detector payload shall detect neutrons within 10% of the accepted orbital neutron profile (after proper efficiency corrections from moderator choice/thickness and detector efficiency).	To determine the accuracy of the detector, it must be compared to the accepted measurements	12	Not Compliant
PL-3.03	Charged Particle Rejection (proton, beta, GCRs, gamma, etc.)	The non-backfilled MSNDs shall enable non-neutron particle rejection within "x"% accuracy	To ensure only neutrons are being counted, a rejection technique must be employed (coincidence detection, temporal discrimination, etc.)	8	Not Compliant
PL-3.07	Neutron Detector Thermal Limits	The neutron detector payload shall be kept between -20 °C and +55 °C during operation and -40°C to +100°C during survival	The optimal operating temperature is 30 °C, but must be in these bounds for operation.	12	Partial Compliance



Payload Requirements

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Level 3 Requirements continued:

Req. ID	Title	Requirement	Rationale	Risk Score	Status
PL-3.09	Neutron Detector Vibration Limits	The neutron detector shall be integrated such that it does not exceed [70%] its [TBD vibration amplitude and frequency] limit	The detector can only handle this maximum vibration load, so mounting, integration, and other factors must ensure an acceptable factor of safety is maintained	16	Compliant
PL-3.17	DSC-Domino data interface	The dominos shall use a usb-c cable to receive commands and relay data	The platform comes with a built-in USB-C port for all communication needs	9	Partial Compliance
PL-3.18	Detector power distribution {if needed for separate power and data interfaces}	There shall be a 5 Volt rail capable of supplying 0.012 amps to each detector (62 mW to each detector, or 0.248 W for the system)	In the case of compartmentalized power and data interfaces with the detector, a 5 V rail capable of supplying enough power for all four detectors will be needed for the subsystem	16	Compliant
PL-3.19	DSC-Domino thermal interface	The dominos shall have thermal couples to the heater(s) and radiator(s) to ensure they are kept in thermal limits (-20 °C and 55 °C) during operation	The detectors must be kept within thermal limits when operating to keep thermal noise to a minimum (from high temperature) and ensure hardware functions (from low temperature)	12	Not Compliant

Level 4 Requirements:

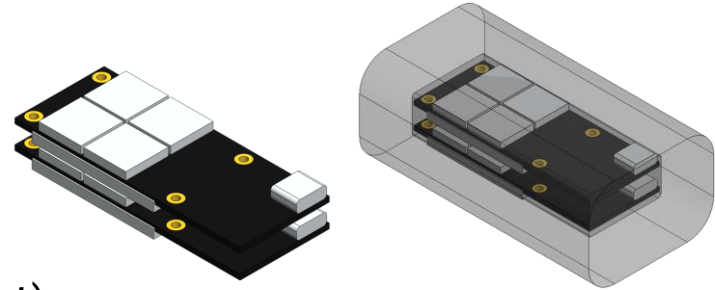
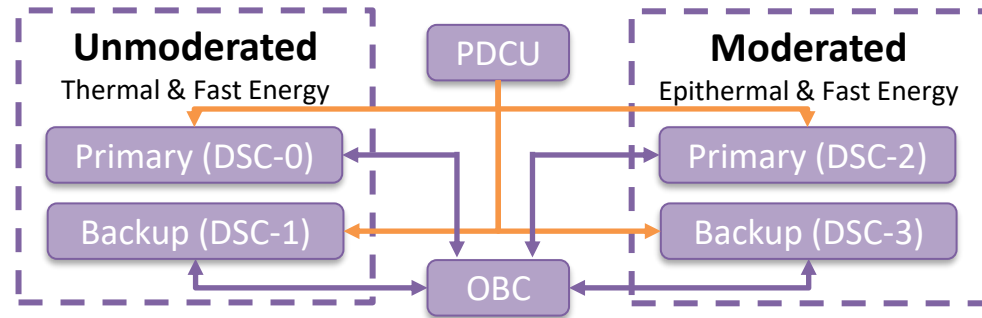
Req. ID	Title	Requirement	Rationale	Risk Score	Status
PL-4.03	Neutron Moderator Size	The neutron moderator shall encase 2 of the detectors with a thickness of 10 mm on the "top" side of the board and 10 mm on the bottom side of the board.	Two of the detectors shall be encased in the moderator to shift the energy spectrum as needed	3	Partial Compliance



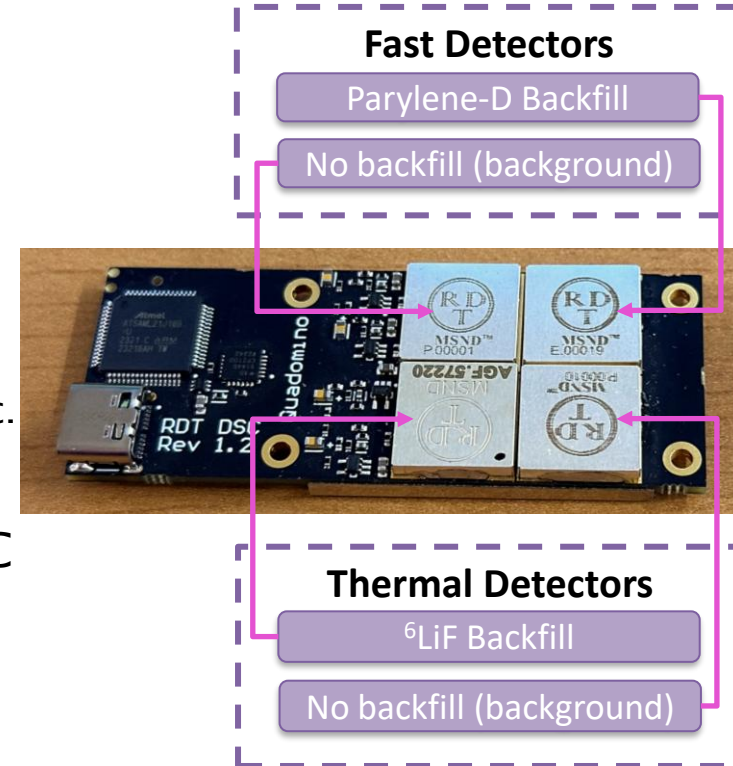
Payload Details

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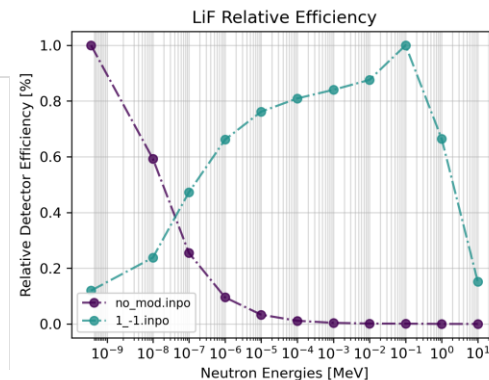
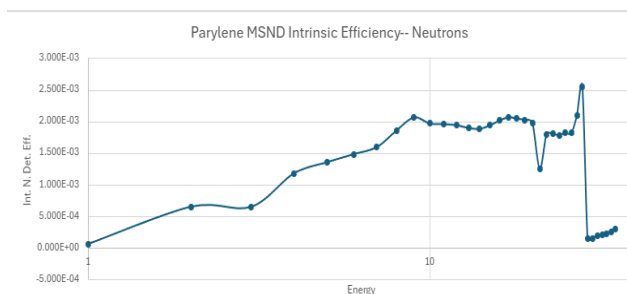
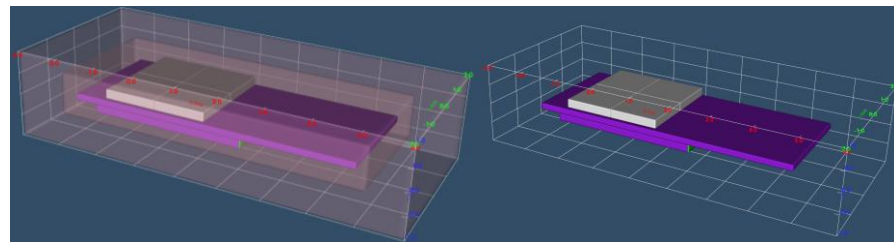
- Four detector packages
 - 2 primary and 2 backup
 - 4 MSNDs on each package
- SWaP (per DSC-domino):
 - Size: 66.8 x 27.2 x 5.75 mm
 - Weight: 15 g
 - Power: 65 mW (across all operations)
- Cost
 - 1k USD per package (RDT Student Discount)



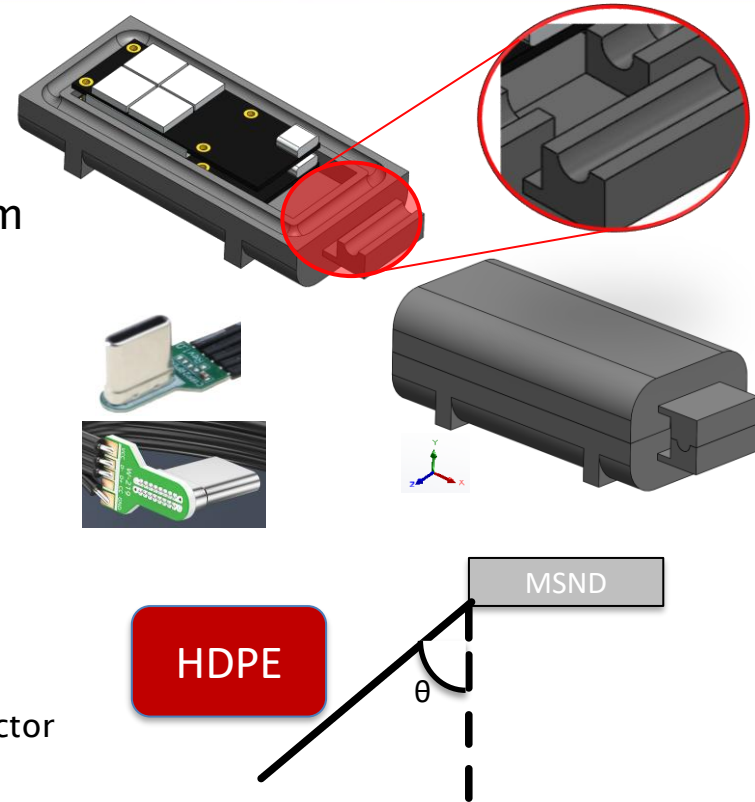
- Calibration and testing
 - Performed by supplier (RDT)
- Data validation
 - Onboard detector verification
 - Validation from previous missions
 - E.g., ISS detection data, NeRDI-1A and -1B, etc.
- Other payload specifications
 - Operating temperature: -20 °C – +55 °C
 - System interface: USB-C ports



- Two coarse energy bins
 - MSND Backfill (^6LiF and Parylene-D)
 - Moderation (HDPE)
 - 10 mm thick
 - Efficiency Peaks
 - 0.025 eV (thermal)
 - 100 keV (epithermal)
 - ~~18 MeV (fast)~~



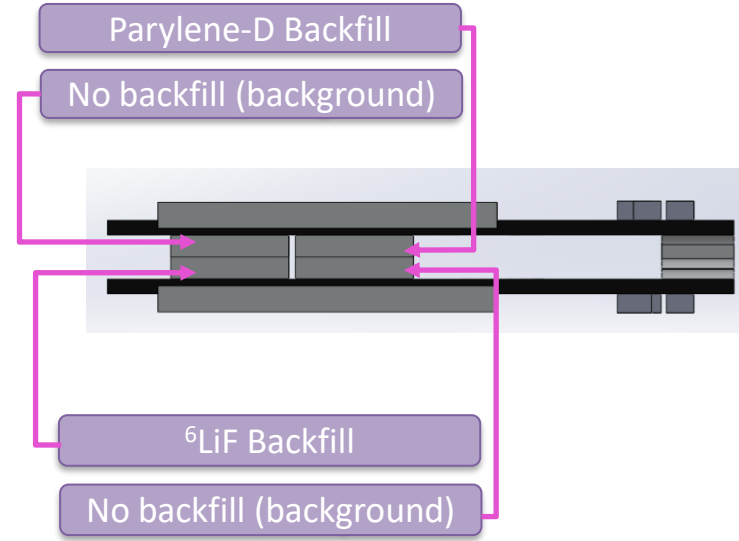
- Why?
 - Slows down epithermal neutrons allowing them to be detected in our thermal platform
- Dimensions
 - 104.8 x 33.8 x 55.2 mm, Mass of 109.6 g
 - M2 fasteners
- Requirements
 - Low outgassing (ASTM E595)
 - 10mm thick
 - Cone of exclusion
 - Exclusion cone ($\theta=70$ deg) from Nadir and zenith vector at any point on any face of unmoderated detectors



Anti-Coincidence

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- Why?
 - Allows us to confidentially say a count was a neutron or background interaction
- How
 - Stack a neutron sensitive detector on top of a non-neutron sensitive detector
 - Time poll each detector for interactions that happen in the same time window
- Issues
 - Polling frequency (Hardware limitations)
 - Only "effective" for protons above 20Mev

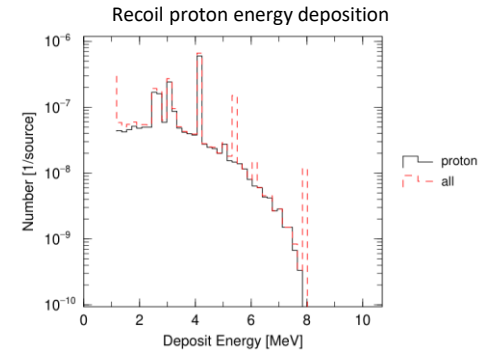
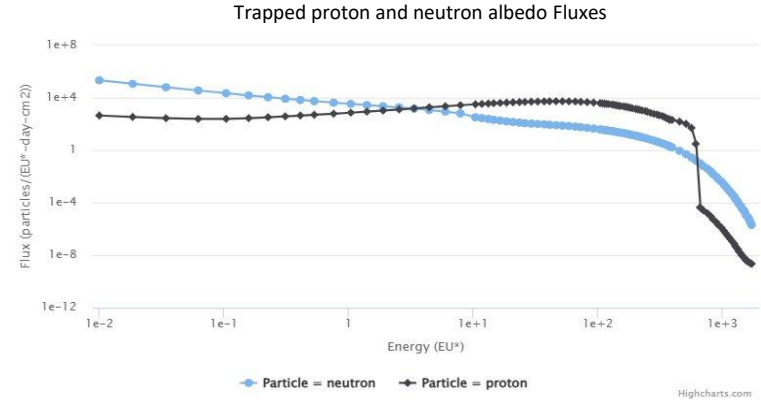


Expected Data

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Data obtained from OLTARIS orbit simulations with a 0.5cm silicon sphere to simulate MSND response

- Fast
 - Proton and Neutron flux is similar at our detector's efficiency peak
- Solution
 - Neutrons detected by recoil proton
 - From elastic collision we should see an energy peak that can be distinguished from background
- We plan to descope this from our mission due to the low efficiency

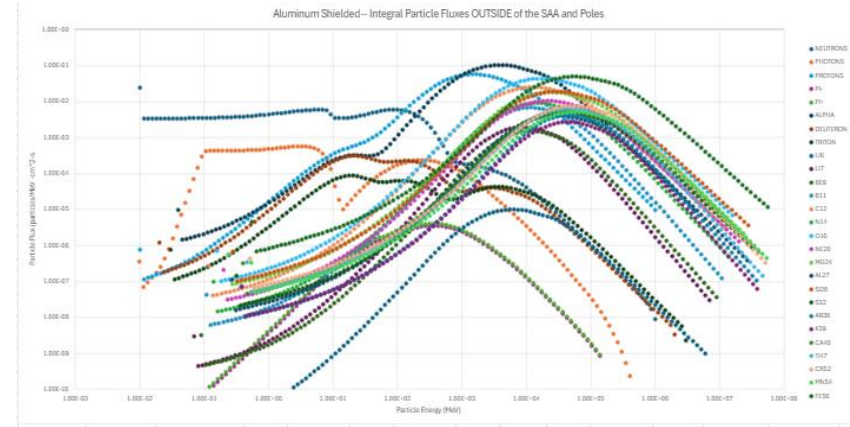


Expected Data

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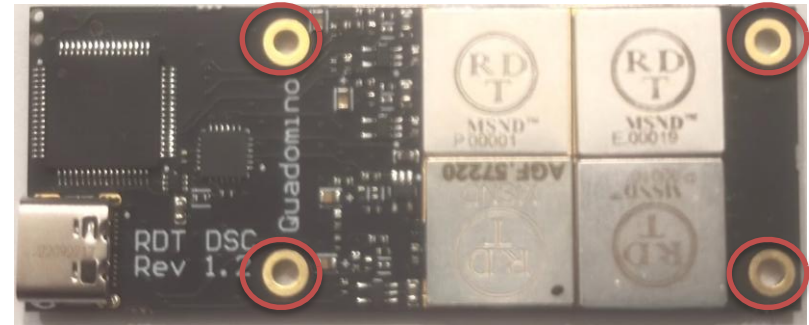
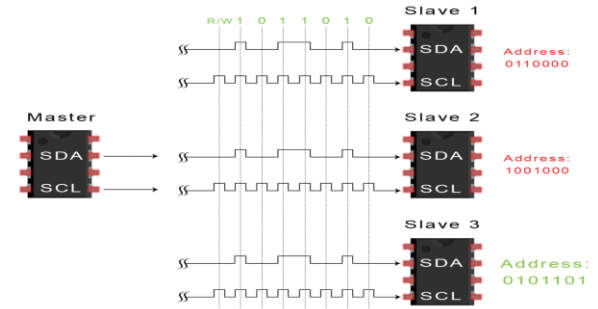
- Thermal and epithermal
 - A long detection window (70h) is needed to get statistically significant data without moderation outside of SAA/Poles
- Solution
 - Explore the feasibility of a LLD a ULD to lower the amount of background detected.
 - Research is needed into Pulse Pileup effects

Data obtained from OLTARIS orbit simulations with a 0.5cm silicon sphere to simulate MSND response

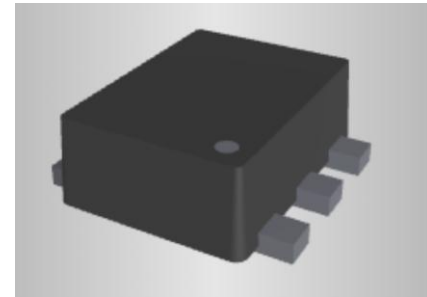


Cumulative (integrated) time to reach set percent uncertainty*		Percent Uncertainty	15%	10%	5%
Nominal	LiF backfill	Unmoderated	8.07 hours	18.35 hours	77.86 hours
		Moderated	22.08 hours	50.82 hours	232.05 hours
	Parylene backfill	Unmoderated	Unattainable – lowest is 281.79%		
		Moderated	Unattainable – lowest is 239.77%		
SAA/Poles	LiF backfill	Unmoderated	Unattainable – lowest is 160.75%		
		Moderated	223.59 hours	Unattainable – lowest is 11.68%	
	Parylene backfill	Unmoderated	Unattainable – lowest is 61756.45%		
		Moderated	Unattainable – lowest is 366.41%		

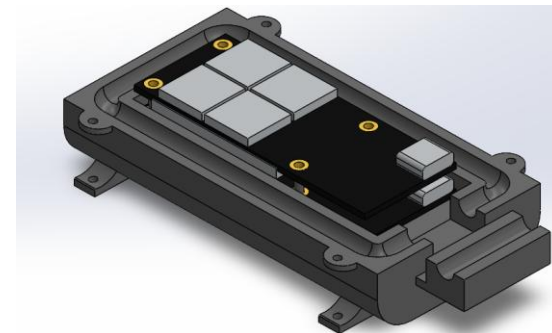
- Electrical
 - Firmware for USB-C
 - We are going to run a I2C protocol through a USB-C connector
- Mechanical
 - 4 Fasteners per package
 - M2 screw or bolt
 - Mechanical Limitations
 - Acceleration
 - Vibration
 - TBD through testing



- RDT implemented monitor
 - Readout temperature data
- Detector Platform Limitations
 - Operating temperature: -20 °C – +55 °C
 - Survival temperature: -40 °C – +100 °C
 - Heat dissipation system (If necessary)
 - Find hot points on PCB then design conduction method to radiator
 - Use the same cable maze for USB-C connector



Texas Instruments TMP112AIDRLT



Payload Outlook

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Requirement Compliance

- Non-compliant with Thermal interface, Payload accuracy, Charged particle rejection

Subsystems Compatibility

- We need adjustments done to the DSC domino for I2C compatibility
- The payload is compatible otherwise

The Payload has a Path Toward CDR

The Payload Subsystem is Capable of Supporting NuKAT Mission

In light of this, the payload should proceed toward CDR with further effort to lower statistical error.

END OF DAY 1 OF 3

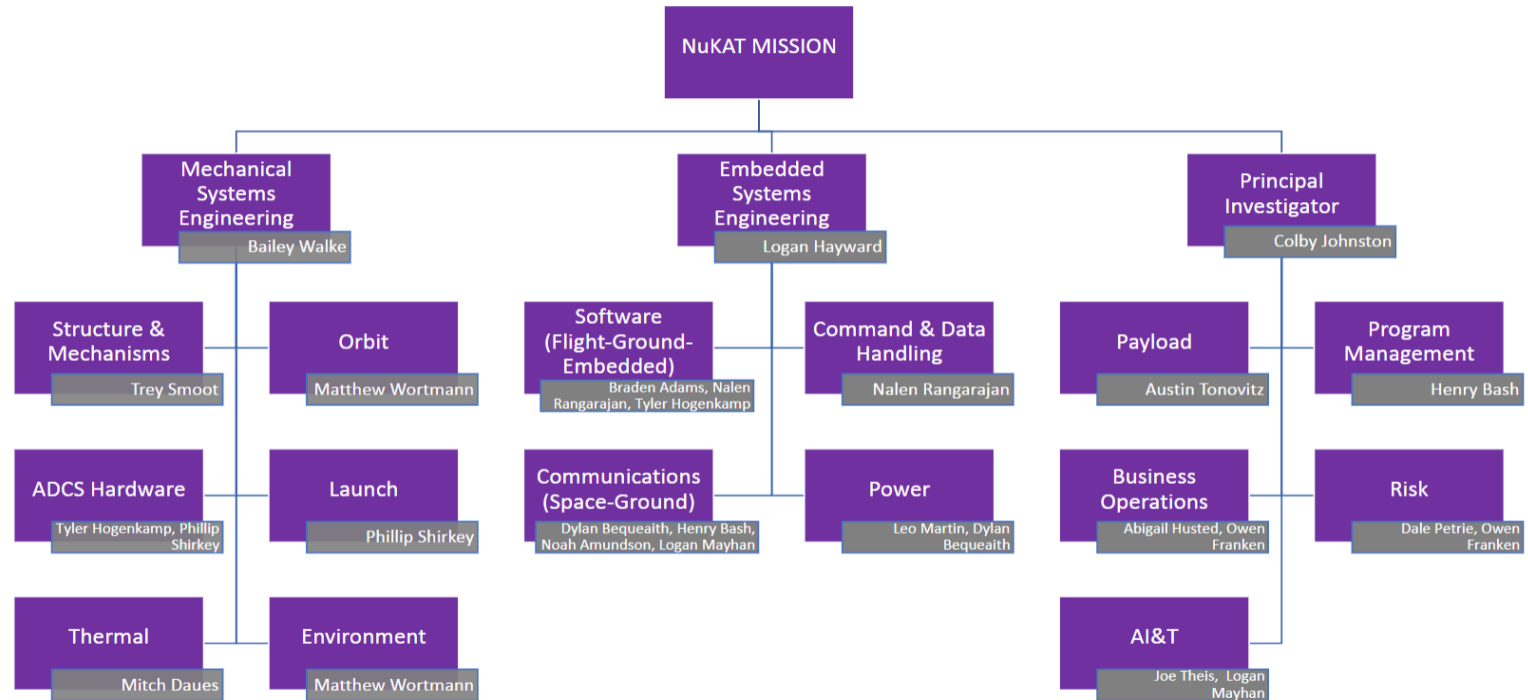
We now open the floor to questions from the audience

BACKUP SLIDES – DAY ONE



Team Structure

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Mission Requirements

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NUKAT REQUIREMENTS VERIFICATION MATRIX

Req. ID	Title	Requirement	Impact	Probability	Risk Score	Risk Handling
NuKAT LEVEL 1 REQUIREMENTS: Quantitative Mission-Level						
NU-1.01	Neutron Detector Flight Heritage	The neutron detector shall gather thermal and epithermal counts for comparison to accepted orbital readings and detector longevity qualification.	3	2	6	A - Accept
NU-1.02	Results Communication	The NuKAT mission shall write and distribute a publically-available paper detailing mission results following end of satellite operations.	2	2	4	A - Accept
NU-1.03	Operational Integration and Success	NuKAT shall have a fully operational mission (science) lifetime of at least 2 months with payload supported and integrated using a satellite bus system and ground station.	4	3	12	M - Mitigate
NU-1.04	Operational Optimization	NuKAT shall ensure volume, power budget, data accuracy, data processing, link budget, orbit, and financial budget are optimized to ensure at least minimum mission success.	5	3	15	M - Mitigate
NU-1.06	Launch Survivability	All components and systems of NuKAT shall be capable of surviving the conditions present in travel from K-State to the launch provider, while waiting integration, during integration with the launch vehicle, and during launch up until release into its final orbit.	5	2	10	M - Mitigate



Mission Requirements

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NUKAT REQUIREMENTS VERIFICATION MATRIX

Req. ID	Title	Requirement	Impact	Probability	Risk Score	Risk Handling
NuKAT LEVEL 1 REQUIREMENTS: Quantitative Mission-Level						
NU-1.07	Space Environment Survivability	All components and systems of NuKAT shall be capable of surviving the space environment at the selected orbit and altitude.	5	2	10	M - Mitigate
NU-1.08	Satellite Compliance	NuKAT shall be compliant with applicable guidelines from necessary governing bodies and launch providers.	5	1	5	M - Mitigate
NU-1.09	Satellite Guidelines	NuKAT should be designed to meet the requirements specified in Caly Poly CubeSat Design Specifications (CP-CDS-R14.1) and University NanoSat Program Satellite Development Guidelines (EXTDOC-315).	4	1	4	A - Accept
NU-1.10	Contamination and Hazardous Materials	NuKAT systems shall NOT use cadmium parts, cadmium-plated parts, zinc plating, mercury, and/or compounds containing mercury, pure tin or tin electroplate, and silicone rubber or room-temperature-vulcanizing (RTV) silicones with probability or transfer to Co-payload(s) or Launch Vehicle hardware.	4	1	4	A - Accept
NU-1.11	K-State Radio Station Use	NuKAT shall utilize the K-State radio station as a primary ground station for data downlink/uplink	3	2	6	A - Accept
NU-1.12	Deorbit Time	NuKAT shall be able to deorbit 5 years after end of science mission.	3	1	3	A - Accept



Mission Data Package

Reduced Data Package

Four data sets
(One background-corrected set from each detector configuration)

$1/60$ Hz measurement frequency

Occasional location tag attached to data pulls

Full Data Package

Eight data sets
(Two background-corrected sets from each board)

$1/30$ Hz measurement frequency

Periodic full raw data downlink
(Sixteen data sets, no background correction)

Location tag attached to each data pull

Mission Data Package

Differential Data

Small timesteps
(the frequency of data pulls –
driven by power and link budgets)

Used to observe orbital trends

Comparison with NeRDI-1A mission

High individual measurement uncertainty

Integrated Data

Large timesteps
(sum of the small timesteps)

Used to verify internal models and back out fluence
information

Performed on the ground

Acceptable uncertainty
(around 5-10%)

Mission Data Package – Measurement Times

Cumulative (integrated) time to reach set percent uncertainty*		Percent Uncertainty	15%	10%	5%
Nominal	LiF backfill	Unmoderated	8.07 hours	18.35 hours	77.86 hours
		Moderated	22.08 hours	50.82 hours	232.05 hours
	Parylene backfill	Unmoderated	Unattainable – lowest is 281.79%		
		Moderated	Unattainable – lowest is 239.77%		
SAA/Poles	LiF backfill	Unmoderated	Unattainable – lowest is 160.75%		
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	Parylene backfill	Unmoderated	Unattainable – lowest is 61756.45%		
		Moderated	Unattainable – lowest is 366.41%		

*Only protons and neutrons used for these calculations. More particle interaction efficiency simulations needed for future work.



Mission Product Calculations⁶³

Spaceborne Particle Detection Simulation: Architecture & Documentation

This notebook contains the simulation architecture for determining the expected net neutron counts and associated statistical uncertainties for spaceborne detectors in low-earth orbit. The framework accounts for energy-dependent orbital albedo distributions, dynamic satellite orientations, intrinsic detector efficiencies, and background subtraction using symbolic error propagation.

1. Dependencies

The simulation relies on the following standard Python libraries:

- **numpy** : Utilized for array operations, trigonometric functions for angular projections, and vectorized mathematical operations across the final timing arrays.
- **pandas** : Used for data ingestion and structured management of the environment fluxes and detector efficiency profiles.
- **sympy** : The core of the uncertainty propagation engine. It allows the code to symbolically derive partial derivatives for variables with associated errors, eliminating the need for hardcoded error-propagation formulas.

2. Overall Architecture

Data Management

- **File Dictionaries (`eff_files` , `flux_files`)**: The architecture uses nested dictionaries to dynamically map user-defined environmental conditions (Nominal vs. SAA, Moderated vs. Unmoderated) and detector selections (LiF vs. Parylene) to their corresponding CSV profiles.

Physics & Logic Functions

- **calculate_eff_unit_area** : Projects the detector's 2D physical area into the 3D trajectory of incoming particles, returning a degradation multiplier (0.0 to 1.0).
- **get_interpolated_efficiency** : Bridges the gap between the environment's defined flux energy bins and the detector's characterized efficiency bins using linear interpolation.
- **run_detector_simulation** : The primary aggregator. It loops through all energy bins dynamically adjusting the effective area based on the specific particle angle of that bin, calculates the CPS per bin, and returns the total gross counts, variance, and a breakdown of variance components.
- **measure_timing** : The statistical endpoint. It handles the subtraction of background arrays from the gross arrays, generates the requested timing windows, and calculates both Statistical (Poisson) and Systematic (Analytical) errors, combining them in quadrature.

Target Percent Uncertainty Engine

Instead of simulating static time intervals, the `main()` block features an analytical engine that calculates the *exact* time required to hit a user-defined target percent uncertainty. It calculates the theoretical systematic floor, rejects unattainable targets, and solves for t to output highly precise mission-planning timelines.

*Documentation compiled with the help of Google Gemini



Mission Product Calculations⁶⁴

3. Core Logic & Physical Nuances

Symbolic Error Propagation & Variance Tracking

Instead of hardcoding standard Gaussian error propagation (e.g., $\sigma_f = \sqrt{a^2 + b^2}$), the architecture uses `make_symbolic_uncertainty_function`. For any calculation, the function defines SymPy symbols, calculates the exact partial derivatives with respect to each variable, and dynamically evaluates the resulting error. Crucially, it isolates the variance components $\left(\frac{\partial f}{\partial x_i} \sigma_{x_i}\right)^2$ to determine exactly which hardware or environmental parameter is dominating the mathematical error.

Energy-Dependent Angular Topologies (Limb Brightening)

The Earth's neutron albedo is not isotropic. Thermal neutrons, having scattered extensively, peak near nadir (0°). Fast neutrons peak near the horizon/limb of the Earth due to the thicker slant-path of the atmosphere. The `run_detector_simulation` loop dynamically reads the expected angle for *each specific energy bin* and calculates a unique angular degradation multiplier for that specific bin before summing the total CPS.

Twin-Detector Subtraction Variance

To isolate the neutron signal, the architecture simulates a backfilled MSND (detecting neutrons + background) and a witness MSND (detecting background only). Because both measurements are noisy, the background error is introduced twice during subtraction. The code explicitly accounts for this by applying the statistical rule $V_{net} = V_{neutron} + 2 \cdot V_{bkg}$ as the backfilled MSND is assumed to have the same response to non-neutrons as the witness MSND.

Statistical vs. Systematic Uncertainty (Quadrature Addition)

The total uncertainty of the detector is split into two independent regimes that are combined in quadrature: $\sigma_{total} = \sqrt{\sigma_{stat}^2 + \sigma_{sys}^2}$.

- **Statistical Error (Poisson):** Governed by the stochastic nature of radiation detection. The variance of a subtracted signal ($A - B$) is $N_{total} + N_{bkg}$. This dominates at short time scales where counts are low.
- **Systematic Error (Analytical):** Governed by the uncertainties of the hardware and environment characterization (loaded from the CSVs). This scales linearly with time and dominates at long integration scales.

Mission Product Calculations⁶⁵

4. Assumed CSV Formatting

The architecture expects structured CSV inputs. The `load_flux_csv` and `load_efficiency_csv` functions include fallbacks: if the `Uncertainty` column is omitted, the code defaults to $\sqrt{x}$. If `Expected_Angle_Deg` is missing, it defaults to perfectly nadir (0.0°).

Environment Flux Files (`flux_neutron_mod.csv` , etc.)

Energy_Lower_MeV	Energy_Upper_MeV	Flux	Uncertainty	Expected_Angle_Deg
0.01	0.1	150.0	12.0	0.0
0.1	1.0	85.0	9.0	45.0
1.0	10.0	42.0	6.0	65.0

Detector Efficiency Files (`eff_neutron_LiF.csv` , etc.)

Energy_MeV	Efficiency	Uncertainty
0.05	0.85	0.02
0.55	0.10	0.05
5.0	0.01	0.01

5. Hardcoded Parameters in Execution

1. `detector_area` ($1.26^2 \approx 1.5876 \text{ cm}^2$): The absolute physical cross-sectional area of the specific MSND face.
2. `sat_angle` (20.0°): The satellite oscillation off of perfect nadir.
3. `target_percents` (within the second `main()` block): A user-defined array (e.g., `[20.0, 15.0, 10.0, 7.5, 5.0]`) fed into the Target Engine to calculate integration times.

*Documentation compiled with the help of Google Gemini

Mission Product Calculations⁶⁶

6. Hand-Calculated Verification of Simulation Logic

This section provides a step-by-step hand-calculated proof of a made up scenario to verify the underlying logic the analytical target-time solver.

The Example Problem

- **Neutron Base Signal:** 20.0 cps $\pm$ 3.2015 (Variance = 10.25)
- **Background Signal:** 10.0 cps $\pm$ 2.0 (Variance = 4.0)

Step 1: Subtraction and Gross Counts

- **Gross Rate** (Backfilled MSND): $R_{gross} = 20.0 + 10.0 = \mathbf{30.0}$ cps
- **Net Rate** (Subtracting Witness MSND from Backfilled MSND): $R_{net} = 30.0 - 10.0 = \mathbf{20.0}$ cps

Step 2: Twin-Detector Systematic Variance

Because we use two noisy detectors to isolate the signal, absolute variances add:

$$V_{net} = V_{neutron} + 2 \cdot V_{bkg}$$

$$V_{net} = 10.25 + 2(4.0) = 10.25 + 8.0 = \mathbf{18.25}$$

Systematic Rate Error (σ_{sys_rate}): $\sqrt{18.25} \approx \mathbf{4.272}$ cps

Step 3: Absolute Systematic Floor

The mathematical limit of the detector at infinite time:

$$F_{sys} = \frac{\sigma_{sys_rate}}{R_{net}} = \frac{4.272}{20.0} = 0.2136 = \mathbf{21.36\%}$$

(Any target below 21.36% will be mathematically flagged as "Unattainable" by the code).

*Documentation compiled with the help of Google Gemini

Mission Product Calculations⁶⁷

Step 4: The Timing Regimes & Quadrature Combination

The code calculates total counts, isolates the Poisson error, isolates the Systematic error, and adds them in quadrature: $\sigma_{tot} = \sqrt{\sigma_{stat}^2 + \sigma_{sys}^2}$.

Short Integration Time ($t = 1$ second):

- **Net Counts:** $20.0 \text{ cps} \cdot 1 \text{ s} = 20.0$
- **Stat Err (Poisson):** $\sqrt{(30 \cdot 1) + (10 \cdot 1)} = \sqrt{40} \approx 6.32$
- **Sys Err (Propagated):** $4.272 \text{ cps} \cdot 1 \text{ s} = 4.27$
- **Total Unc:** $\sqrt{6.32^2 + 4.27^2} = \sqrt{40 + 18.25} = \sqrt{58.25} \approx \pm 7.63 \text{ counts}$ (38.1%)

Long Integration Time ($t = 100$ seconds):

- **Net Counts:** $20.0 \text{ cps} \cdot 100 \text{ s} = 2000.0$
- **Stat Err (Poisson):** $\sqrt{(30 \cdot 100) + (10 \cdot 100)} = \sqrt{4000} \approx 63.24$
- **Sys Err (Propagated):** $4.272 \text{ cps} \cdot 100 \text{ s} = 427.20$
- **Total Unc:** $\sqrt{63.24^2 + 427.20^2} = \sqrt{4000 + 182500} = \sqrt{186500} \approx \pm 431.85 \text{ counts}$ (21.59%)

Step 5: Analytical Target-Time Solver Proof

Asking the code to find the exact time required to hit a **30.0% target** ($U = 0.30$). The code evaluates: $t = \frac{R_{gross} + R_{sig}}{R_{net}^2(U^2 - F_{sig}^2)}$

$$t = \frac{30 + 10}{20^2(0.30^2 - 0.2136^2)} = \frac{40}{400(0.09 - 0.045625)}$$

$$t = \frac{40}{400(0.044375)} = \frac{40}{17.75} \approx 2.2535 \text{ seconds}$$

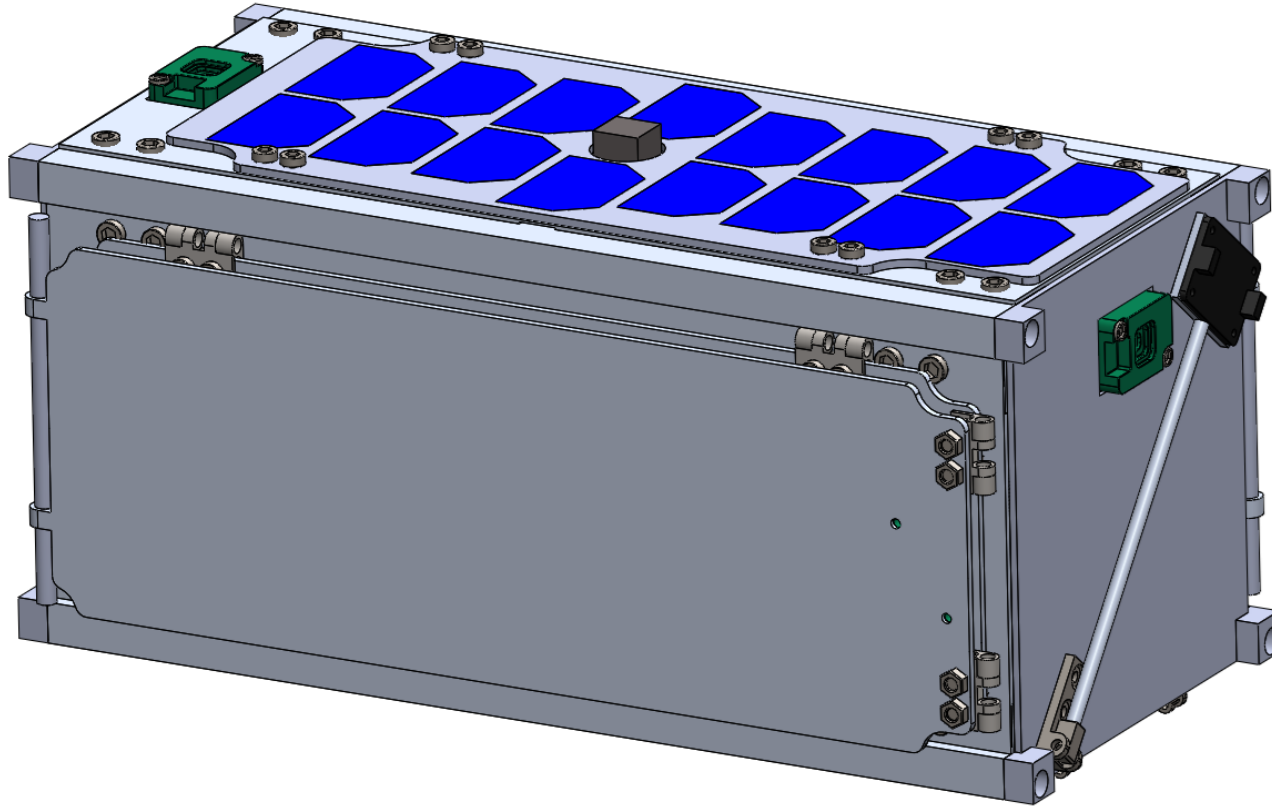
Verification at exactly $t = 2.2535$:

- **Net Counts:** $20 \cdot 2.2535 = 45.07$
- **Stat Err:** $\sqrt{40 \cdot 2.2535} = \sqrt{90.14} = 9.494$
- **Sys Err:** $4.272 \cdot 2.2535 = 9.627$
- **Total Err:** $\sqrt{9.494^2 + 9.627^2} = \sqrt{90.14 + 92.68} = \sqrt{182.82} = 13.52$
- **Final Percent Uncertainty:** $\frac{13.52}{45.07} = 0.3000$ (Exactly 30.0)

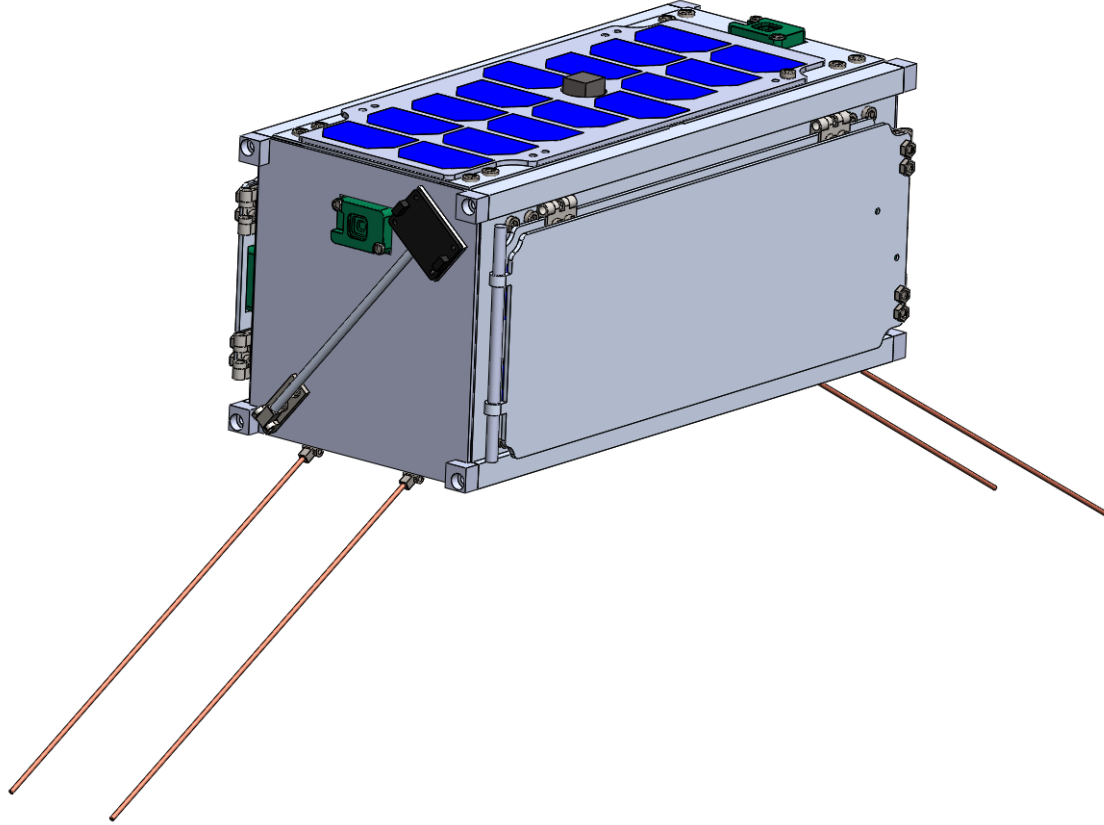
*Documentation compiled with the help of Google Gemini

Initialization:LV Deployment

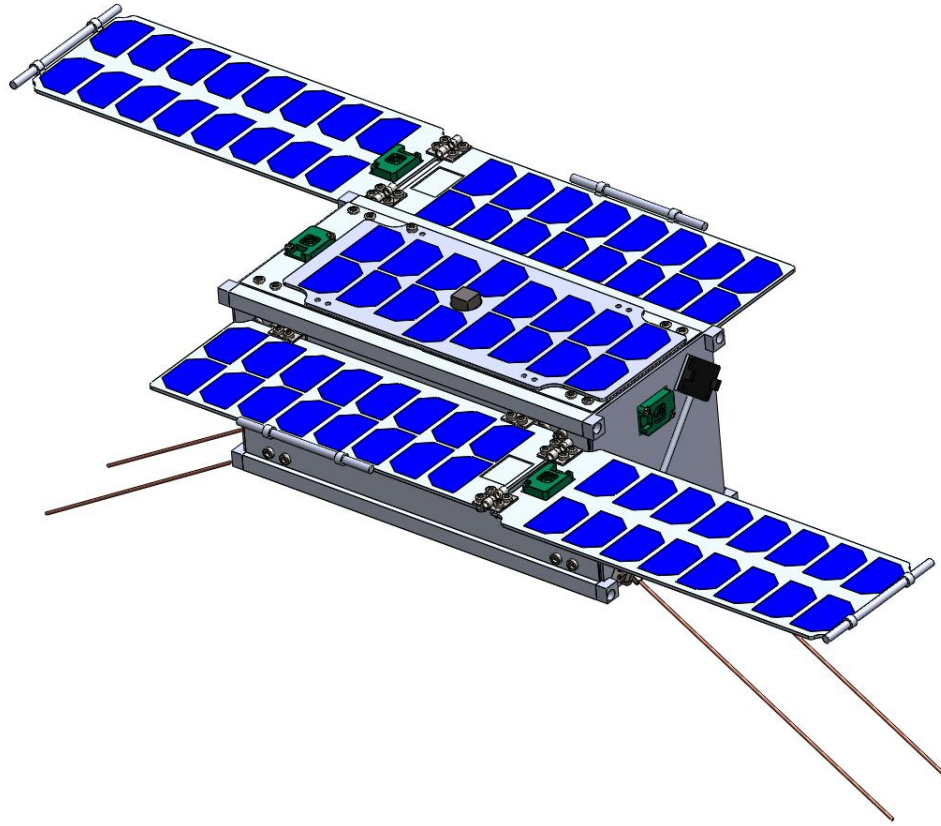
68



Initialization: Antenna Deployment⁶⁹

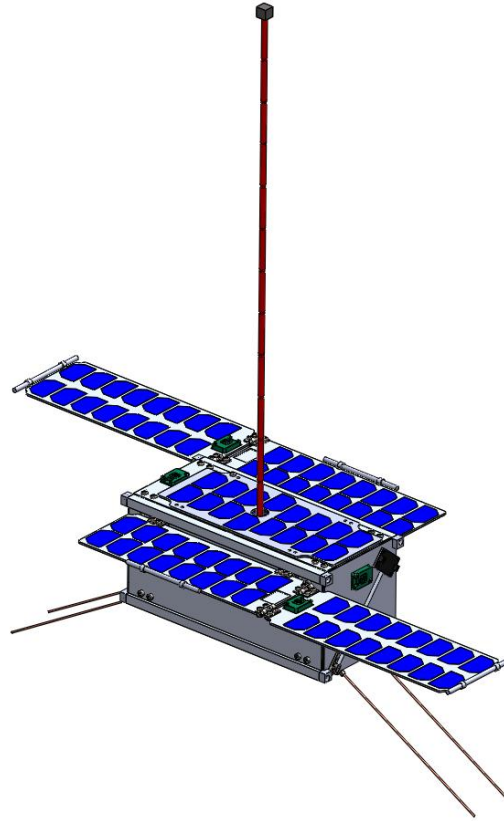


Initialization: Solar Panel Deployment ⁷⁰



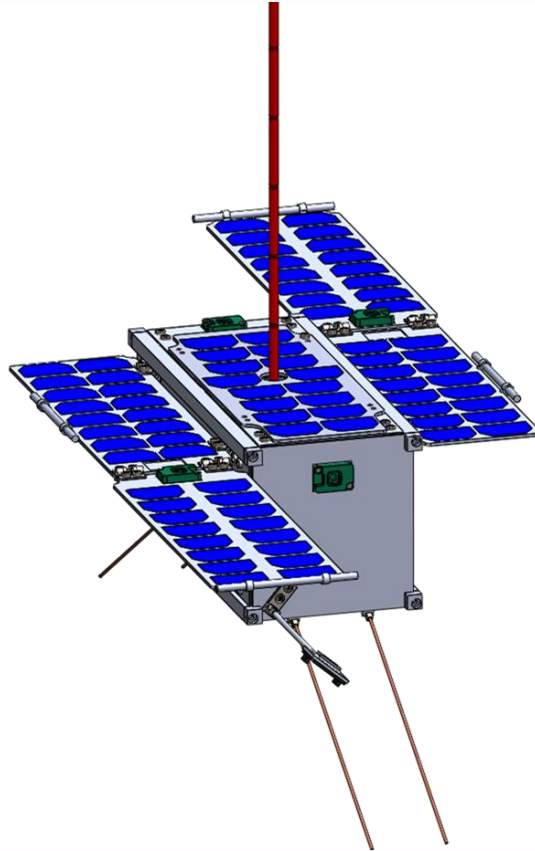
Initialization: Boom Deployment

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Initialization: Standby

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- **OLTARIS** used for radiation environment modeling
 - Simulation performed at 500 km from 1/1/2022 to 1/1/2023 using the AP8 library
 - Unmoderated system = Only the 1.5 mm aluminum paneling
 - Moderated system = Aluminum paneling + 10 mm HDPE moderator (to shift MSND efficiency spectrum into epithermal)
- **MCNP** used for neutron and proton efficiencies
 - Intrinsic efficiency calculated by dividing the radiation detected by the radiation impingent on the detector: $\epsilon_{int} = \frac{\text{F8 Tally bin of particles registering counts above LLD}}{\text{F1 Tally bin of all particles entering the detector}}$
 - F1 Tally – Surface current card (with cosine binning card allowing discrimination in favor of particle current traveling between 180 and 90-degrees of the surface normals for an RPP)
 - F8 Tally – Pulse height tally (used to determine the energy spectrum deposited by monoenergetic sources, allowing artificial LLD placement)
- **Angular dependence in calculation**
 - The satellite's oscillations and the angular nature of albedo radiation were taken into account with a multiplying factor ranging from 0 to 1

OLTARIS Environment Summary⁷⁴

Aluminum Shielded, **Outside** SAA/Poles
(GCR + Neutron Albedo)

Particles	Integral Flux (1/cm ² -s)
neutrons	3.26E-01
photons	1.91E-02
Protons	1.18E+00
pi-	7.68E-05
pi+	8.79E-05
alpha	2.20E+00
deuteron	7.85E-03
triton	2.80E-03
Li6	2.07E-04
Li7	3.45E-02
Be9	2.75E-02
B11	1.28E-01
C12	5.33E-01
N14	1.84E-01
O16	9.37E-01
Ne20	2.19E-01
Mg24	3.92E-01
Al27	8.76E-02
Si28	4.11E-01
S32	1.08E-01
Ar38	7.49E-02
K39	5.09E-02
Ca40	1.15E-01
Ti47	1.24E-01
Cr52	1.52E-01
Mn54	1.07E-01
Fe56	1.11E+00

Particles	Differential Flux (1/MeV-cm ² -s)
neutrons	1.77E+01
photons	2.62E-01
protons	1.71E-02
pi-	5.41E-06
pi+	6.34E-06
alpha	1.41E-02
deuteron	4.20E-03
triton	1.36E-03
Li6	5.01E-07
Li7	5.23E-05
Be9	3.11E-05
B11	1.60E-04
C12	6.51E-04
N14	1.88E-04
O16	8.74E-04
Ne20	1.64E-04
Mg24	2.74E-04
Al27	4.91E-05
Si28	2.44E-04
S32	4.75E-05
Ar38	2.36E-05
K39	1.83E-05
Ca40	5.41E-05
Ti47	5.01E-05
Cr52	5.41E-05
Mn54	3.67E-05
Fe56	4.22E-04

Aluminum + HDPE Shielded, **Outside** SAA/Poles
(GCR + Neutron Albedo)

Particles	Integral Flux (1/cm ² -s)
neutrons	4.09E-01
photons	8.06E-02
protons	1.20E+00
pi-	2.96E-04
pi+	3.64E-04
alpha	2.14E+00
deuteron	2.39E-02
triton	1.13E-02
Li6	1.43E-03
Li7	3.37E-02
Be9	2.76E-02
B11	1.25E-01
C12	5.06E-01
N14	1.77E-01
O16	8.68E-01
Ne20	2.02E-01
Mg24	3.55E-01
Al27	8.39E-02
Si28	3.70E-01
S32	9.93E-02
Ar38	7.18E-02
K39	5.03E-02
Ca40	9.94E-02
Ti47	1.13E-01
Cr52	1.37E-01
Mn54	9.99E-02
Fe56	9.22E-01

Particles	Differential Flux (1/MeV-cm ² -s)
neutrons	1.20E+01
photons	1.15E+00
protons	1.83E-02
pi-	1.93E-05
pi+	2.41E-05
alpha	1.08E-02
deuteron	3.83E-03
triton	8.78E-04
Li6	3.29E-06
Li7	5.22E-05
Be9	3.24E-05
B11	1.57E-04
C12	6.18E-04
N14	1.83E-04
O16	8.04E-04
Ne20	1.48E-04
Mg24	2.38E-04
Al27	4.59E-05
Si28	2.08E-04
S32	4.25E-05
Ar38	2.31E-05
K39	1.81E-05
Ca40	4.26E-05
Ti47	4.26E-05
Cr52	4.72E-05
Mn54	3.24E-05
Fe56	3.01E-04

OLTARIS Environment Summary⁷⁵

Aluminum Shielded, **Inside** SAA/Poles (GCR + Trapped Protons + Neutron Albedo)

Particles	Integral Flux (1/cm ² -s)	Particles	Differential Flux (1/MeV-cm ² -s)
neutrons	3.26E-01	neutrons	1.77E+01
photons	1.40E-01	photons	2.40E+00
protons	1.19E+01	protons	2.41E+00
pi-	9.77E-05	pi-	1.22E-05
pi+	1.13E-04	pi+	1.46E-05
alpha	2.21E+00	alpha	3.96E-02
deuteron	2.02E-02	deuteron	1.66E-02
triton	6.11E-03	triton	5.28E-03
Li6	2.07E-04	Li6	5.01E-07
Li7	3.45E-02	Li7	5.23E-05
Be9	2.75E-02	Be9	3.11E-05
B11	1.28E-01	B11	1.60E-04
C12	5.33E-01	C12	6.51E-04
N14	1.84E-01	N14	1.88E-04
O16	9.37E-01	O16	8.74E-04
Ne20	2.19E-01	Ne20	1.64E-04
Mg24	3.92E-01	Mg24	2.74E-04
Al27	8.76E-02	Al27	4.91E-05
Si28	4.11E-01	Si28	2.44E-04
S32	1.08E-01	S32	4.75E-05
Ar38	7.49E-02	Ar38	2.36E-05
K39	5.09E-02	K39	1.83E-05
Ca40	1.15E-01	Ca40	5.41E-05
Ti47	1.24E-01	Ti47	5.01E-05
Cr52	1.52E-01	Cr52	5.41E-05
Mn54	1.07E-01	Mn54	3.67E-05
Fe56	1.11E+00	Fe56	4.22E-04

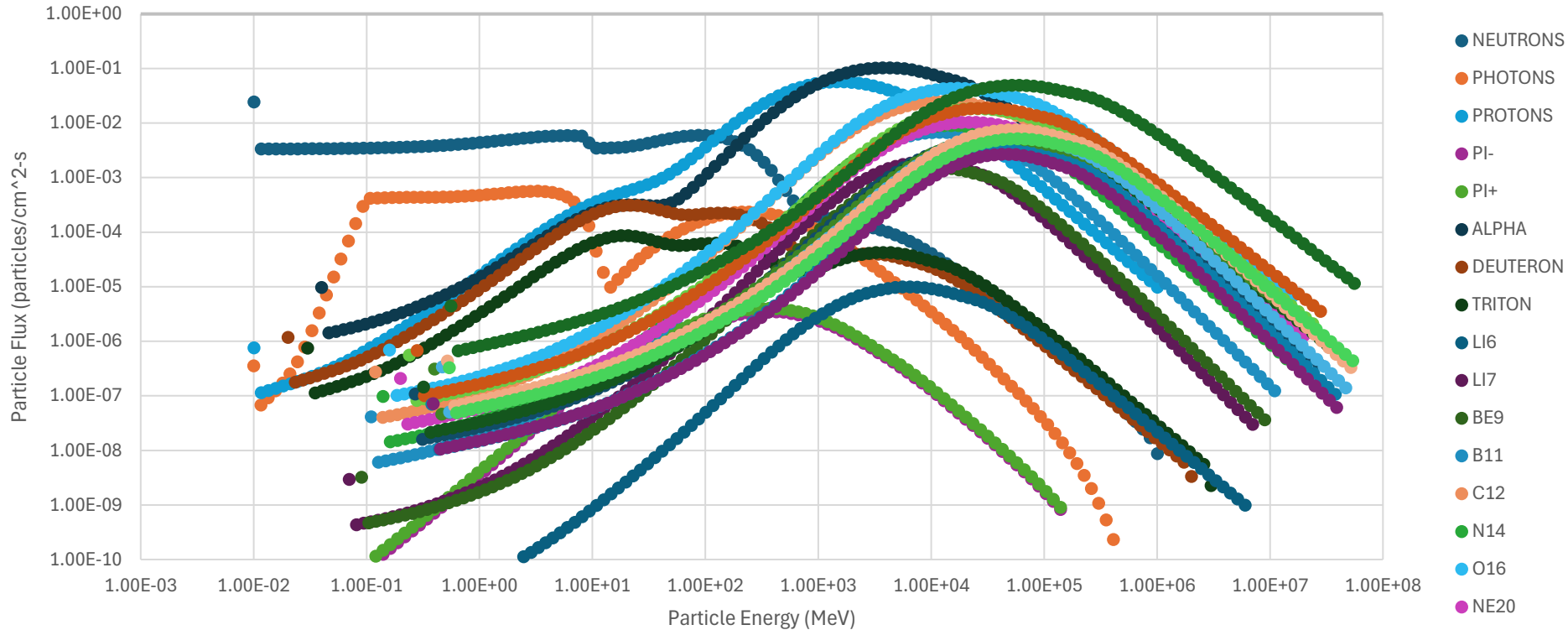
Aluminum + HDPE Shielded, **Inside** SAA/Poles (GCR + Trapped Protons + Neutron Albedo)

Particles	Integral Flux (1/cm ² -s)	Particles	Differential Flux (1/MeV-cm ² -s)
neutrons	4.09E-01	neutrons	1.20E+01
photons	4.20E-01	photons	7.38E+00
protons	1.62E+00	protons	7.45E+00
pi-	3.63E-04	pi-	4.34E-05
pi+	4.48E-04	pi+	5.38E-05
alpha	2.14E+00	alpha	1.83E-02
deuteron	6.87E-02	deuteron	1.96E-02
triton	1.86E-02	triton	4.12E-03
Li6	1.43E-03	Li6	3.29E-06
Li7	3.37E-02	Li7	5.22E-05
Be9	2.76E-02	Be9	3.24E-05
B11	1.25E-01	B11	1.57E-04
C12	5.06E-01	C12	6.18E-04
N14	1.77E-01	N14	1.83E-04
O16	8.68E-01	O16	8.04E-04
Ne20	2.02E-01	Ne20	1.48E-04
Mg24	3.55E-01	Mg24	2.38E-04
Al27	8.39E-02	Al27	4.59E-05
Si28	3.70E-01	Si28	2.08E-04
S32	9.93E-02	S32	4.25E-05
Ar38	7.18E-02	Ar38	2.31E-05
K39	5.03E-02	K39	1.81E-05
Ca40	9.94E-02	Ca40	4.26E-05
Ti47	1.13E-01	Ti47	4.26E-05
Cr52	1.37E-01	Cr52	4.72E-05
Mn54	9.99E-02	Mn54	3.24E-05
Fe56	9.22E-01	Fe56	3.01E-04

OLTARIS Environments - Nominal

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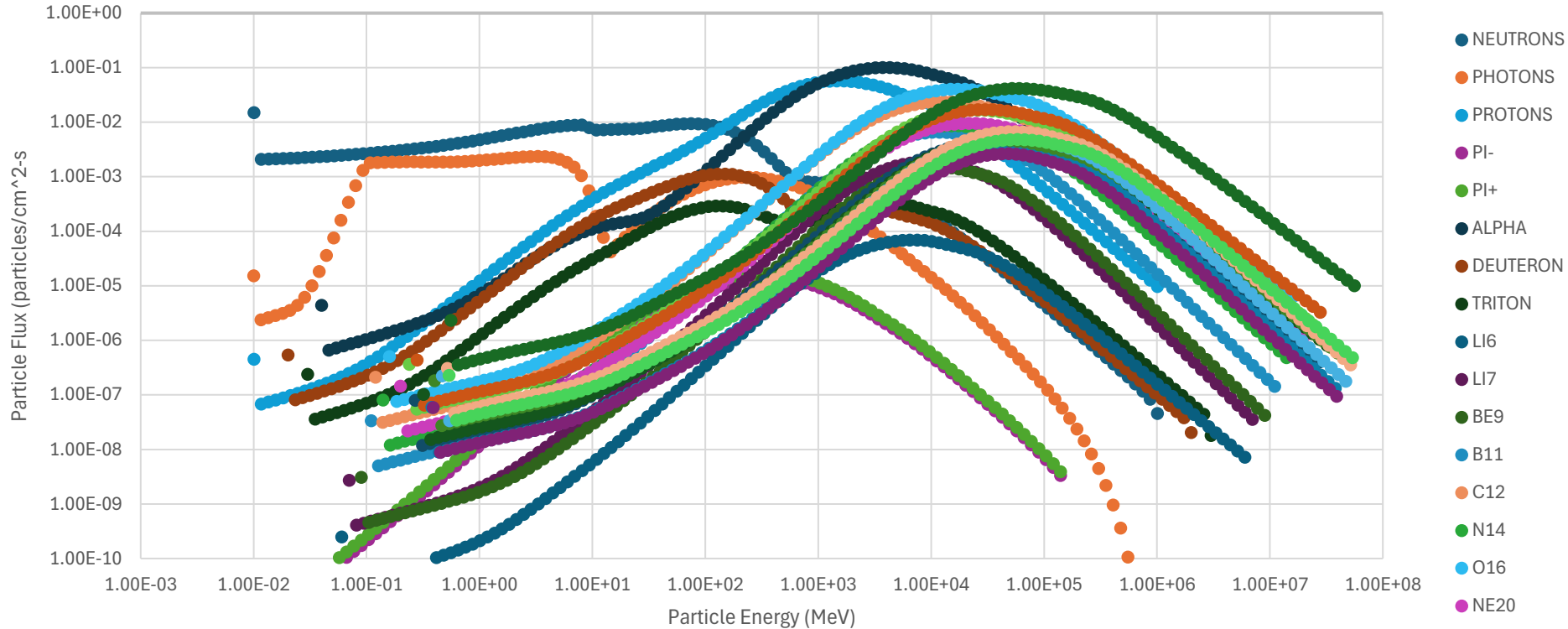
Aluminum -- **Integral** Particle Fluxes **OUTSIDE** of the SAA and Poles



OLTARIS Environments - Nominal

77

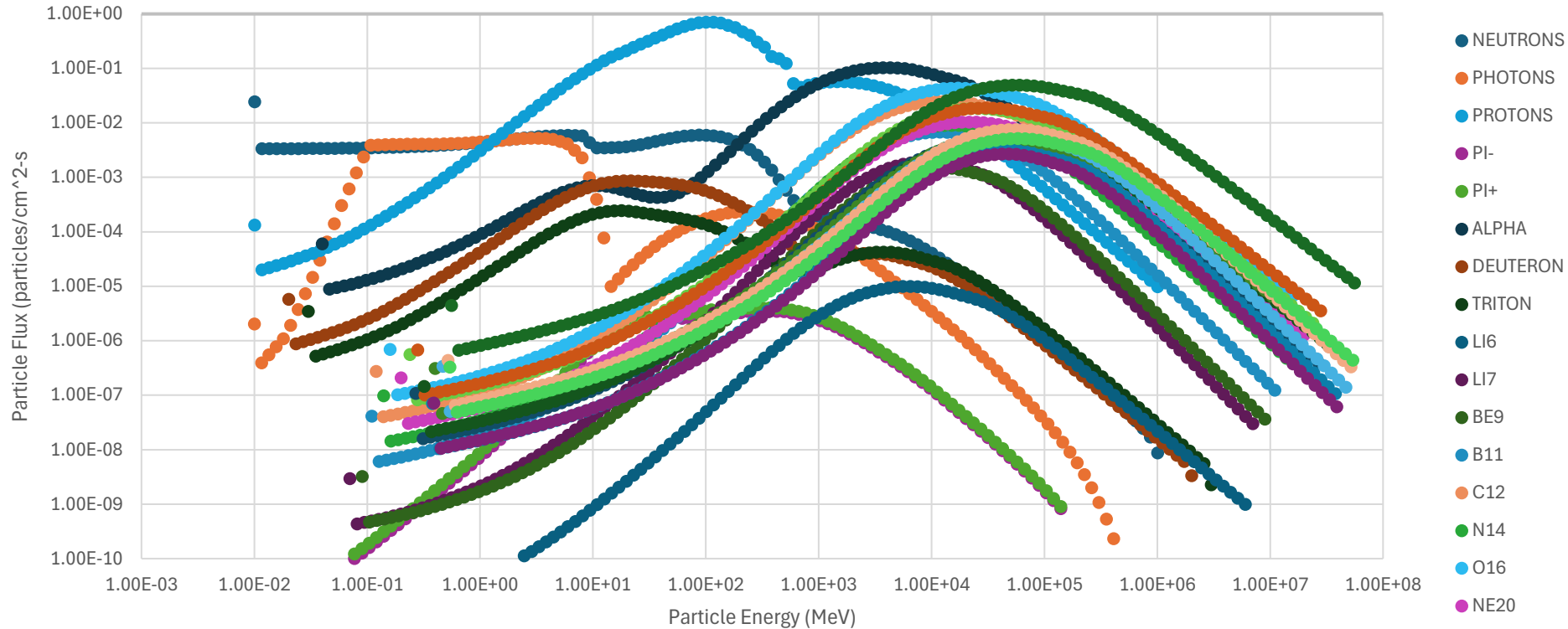
Aluminum + HDPE -- Integral Particle Fluxes **OUTSIDE** of the SAA and Poles



OLTARIS Environments - SAA

78

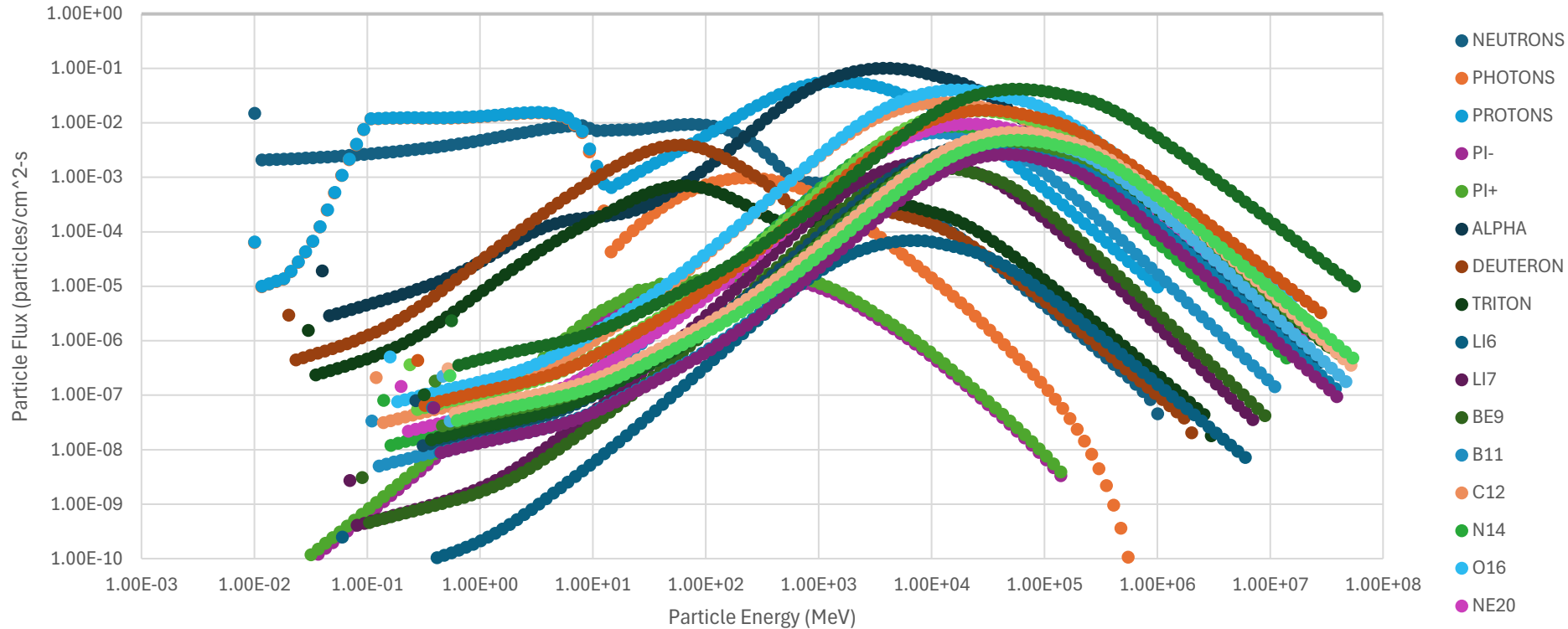
Aluminum -- **Integral** Particle Fluxes **INSIDE** of the SAA and Poles



OLTARIS Environments - SAA

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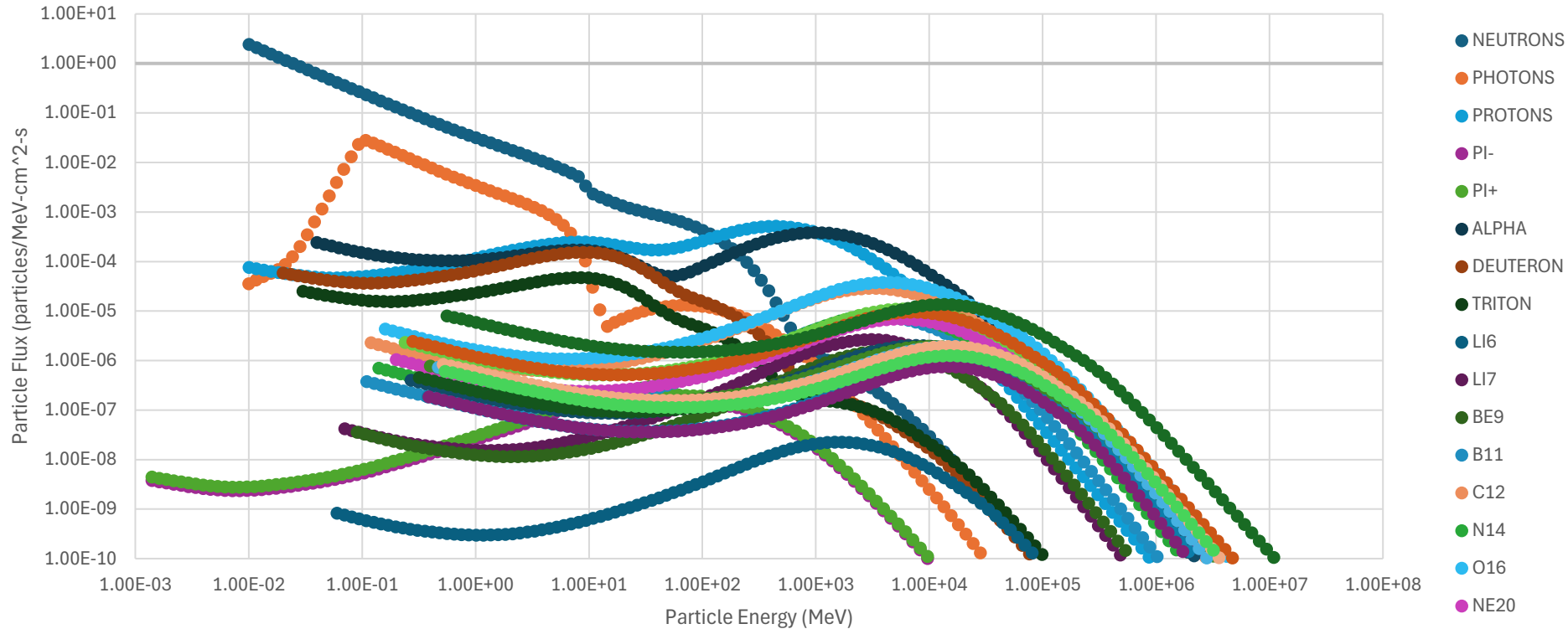
Aluminum + HDPE -- Integral Particle Fluxes **INSIDE** of the SAA and Poles



OLTARIS Environments - Nominal

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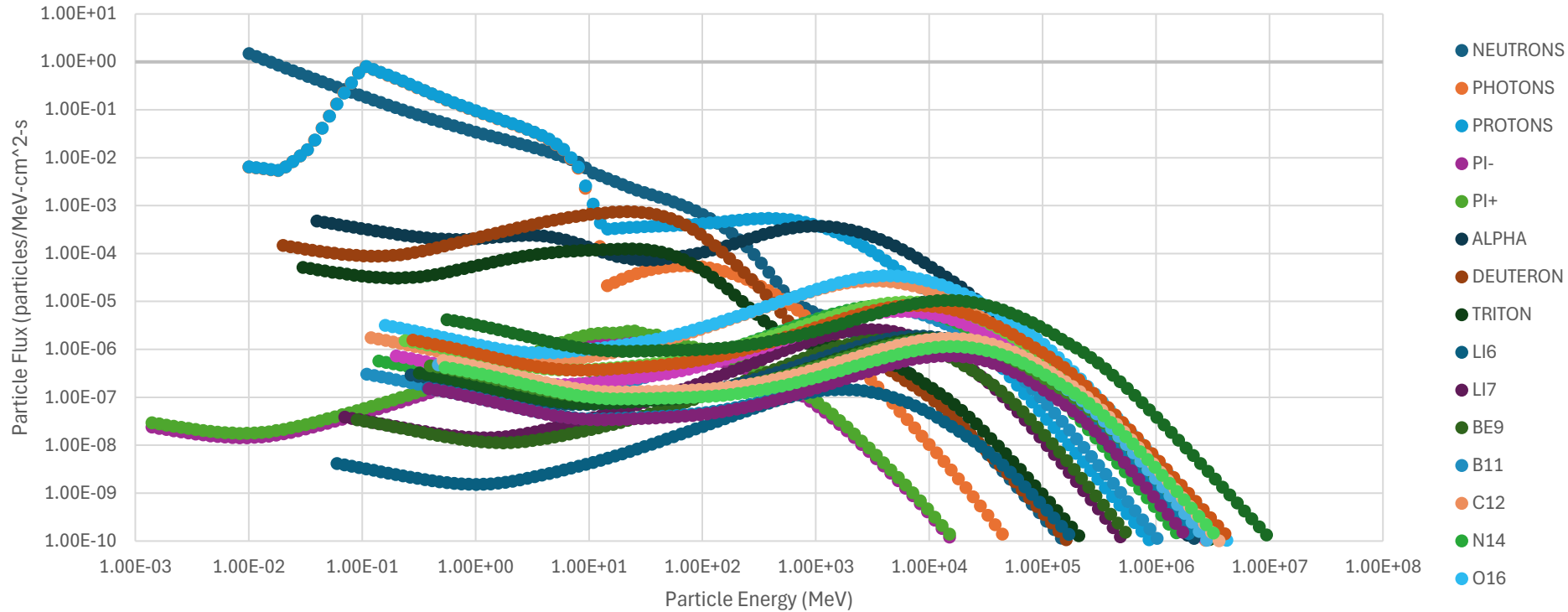
Aluminum -- **Differential** Particle Fluxes **OUTSIDE** of the SAA and Poles



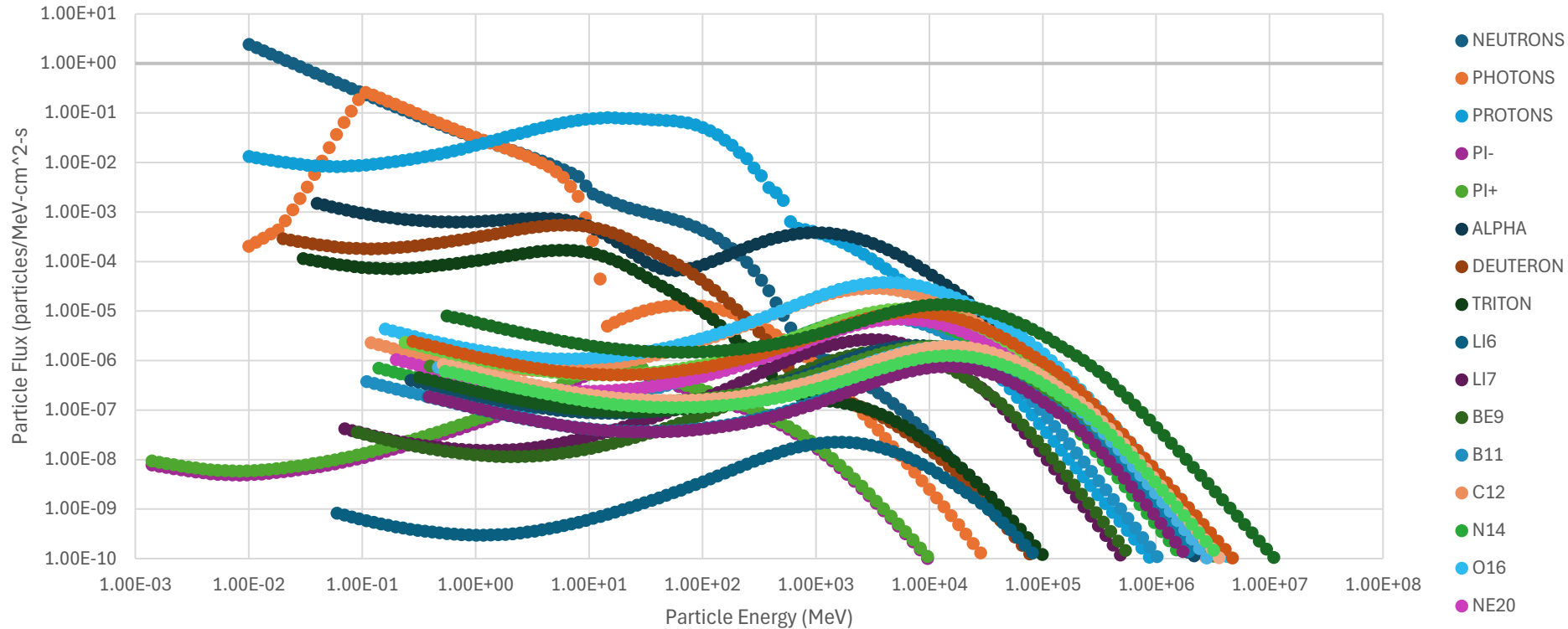
OLTARIS Environments - Nominal

81

Aluminum + HDPE -- Differential Particle Fluxes **INSIDE** of the SAA and Poles



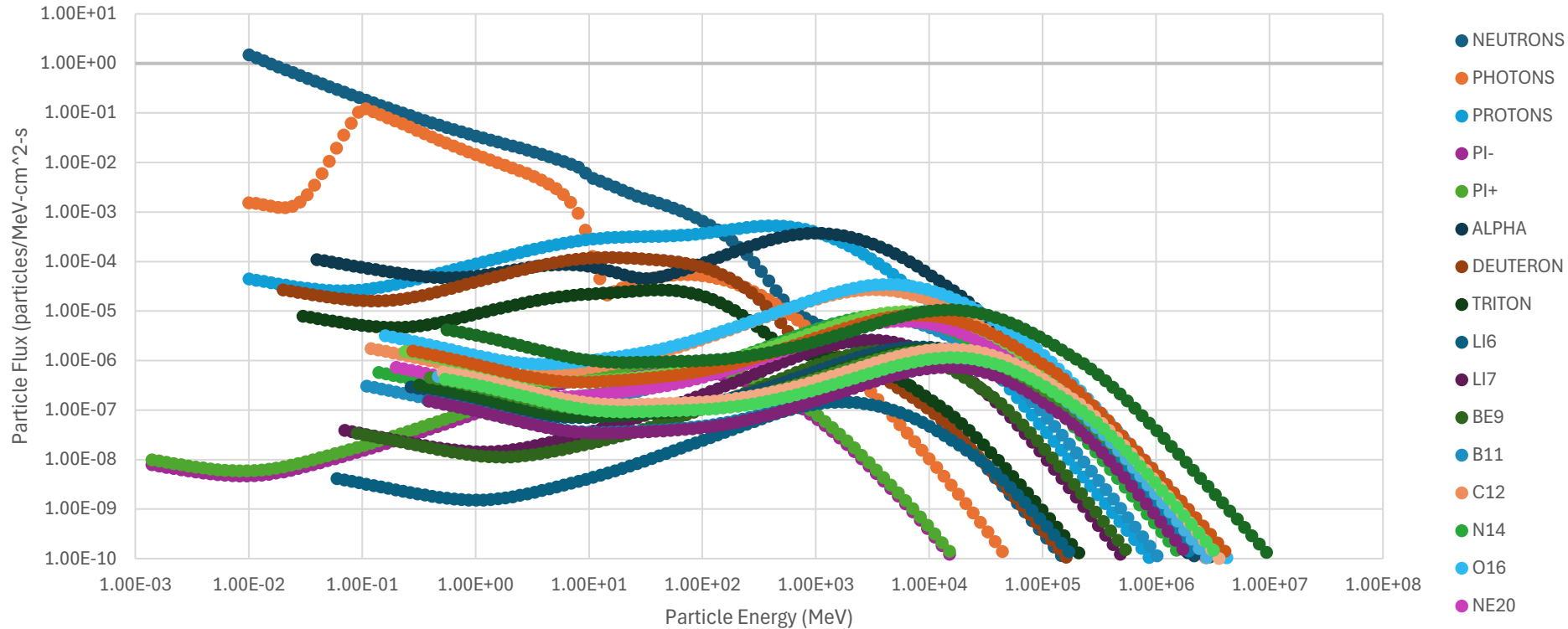
Aluminum -- **Differential** Particle Fluxes **INSIDE** of the SAA and Poles



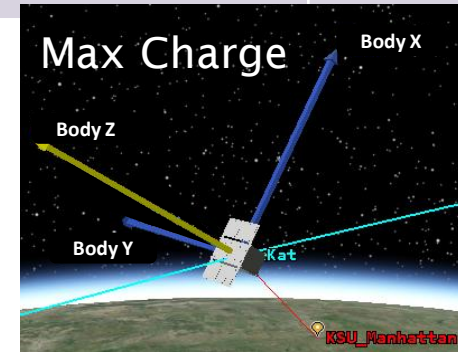
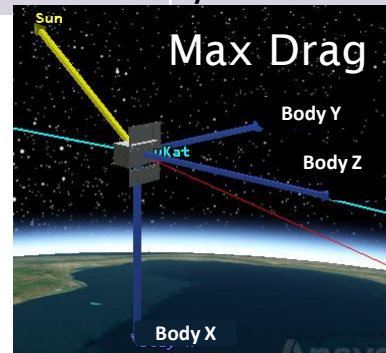
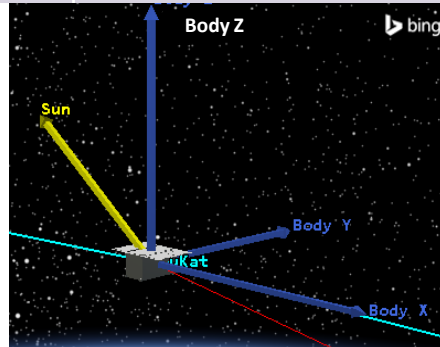
OLTARIS Environments - SAA

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Aluminum + HDPE -- Differential Particle Fluxes **INSIDE** of the SAA and Poles



Mode	Attitude/Orientation	Mission Life from 500km	Mission Life from 545km	Mission Life from 600km
Science	2U Face Forward	1-5 years	1-5 years	1-5 years
Max Drag	Solar Panel Face Forward	+ 0.9-0.3 years	+ 4.6-2.8 years	+ 10.7-10.3 years
Max Charge	Solar Panel Face Sun Normal	+ 1.3-0.4 years	+ 8.1-5.1 years	+ 12.0-11.6 years



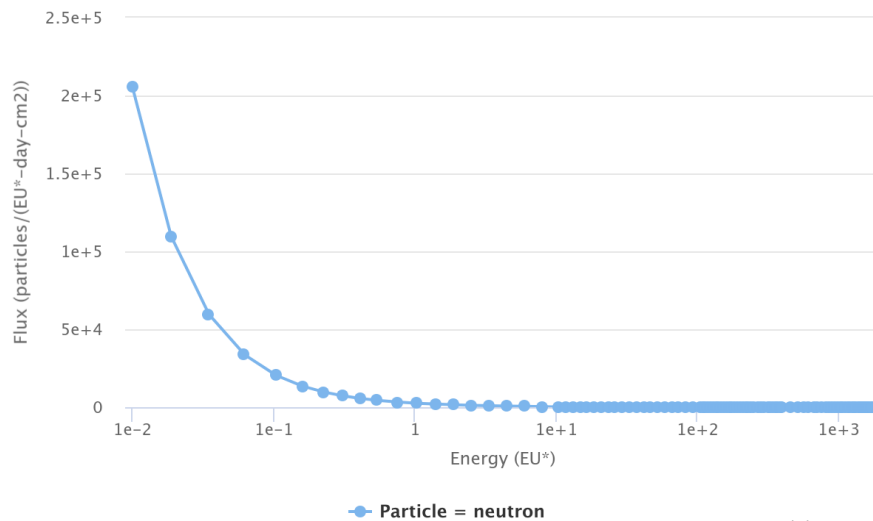
Left Figure: Attitude with 2U Face Forward / Science

Middle Figure: Attitude with Solar Panel Face Forward / Max Drag

Right Figure: Attitude with Solar Panel Face Sun Normal / Max Charge

Reentry < 5 years
after loss-of-comms

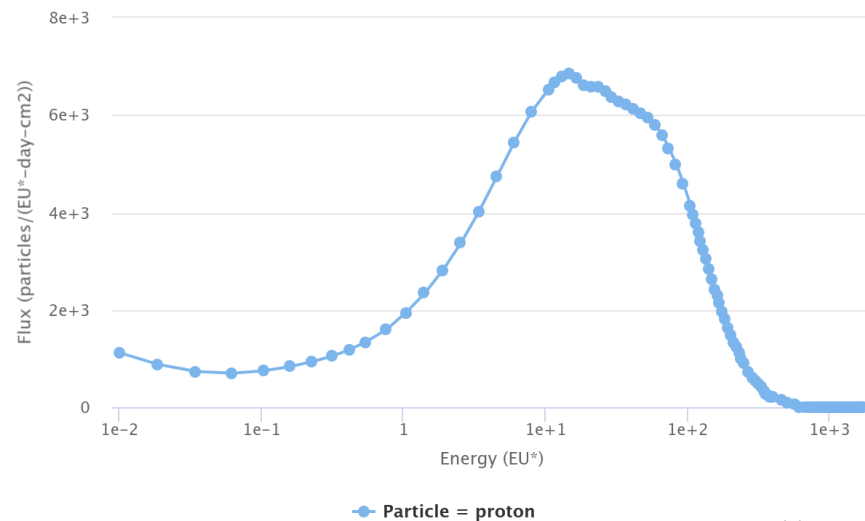
Flux after Transport



Highcharts.com

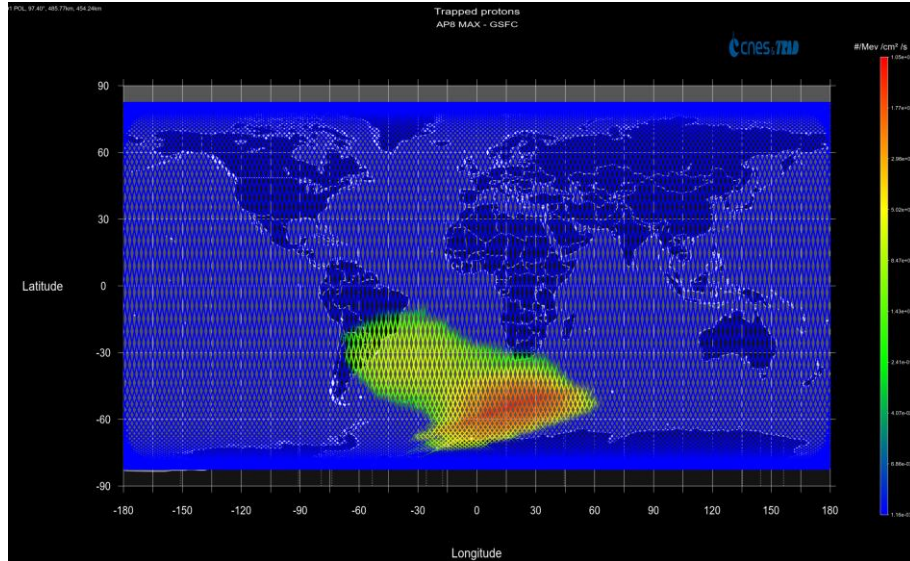
Albedo Neutrons – Avg over mission life

Flux after Transport

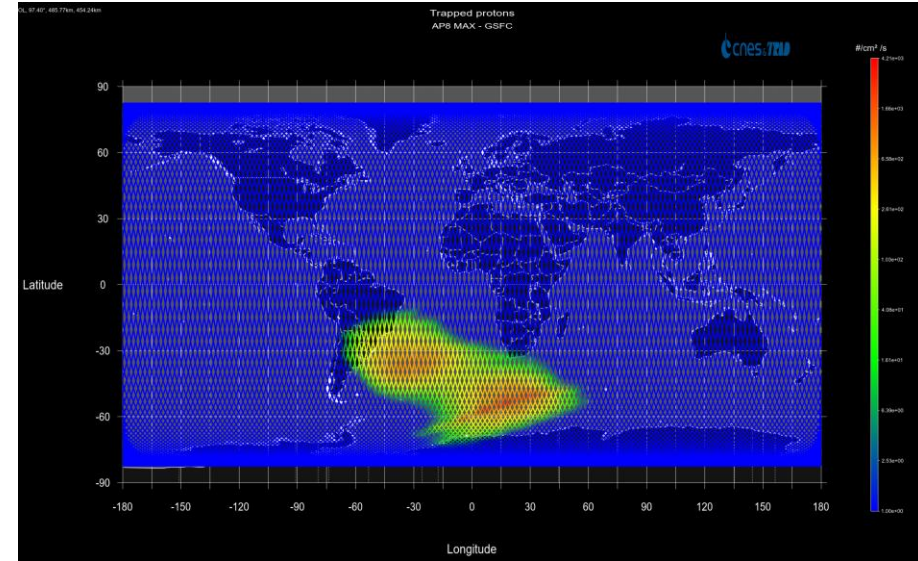


Highcharts.com

Trapped Protons – Avg over mission life



Trapped Protons Differential Flux Map

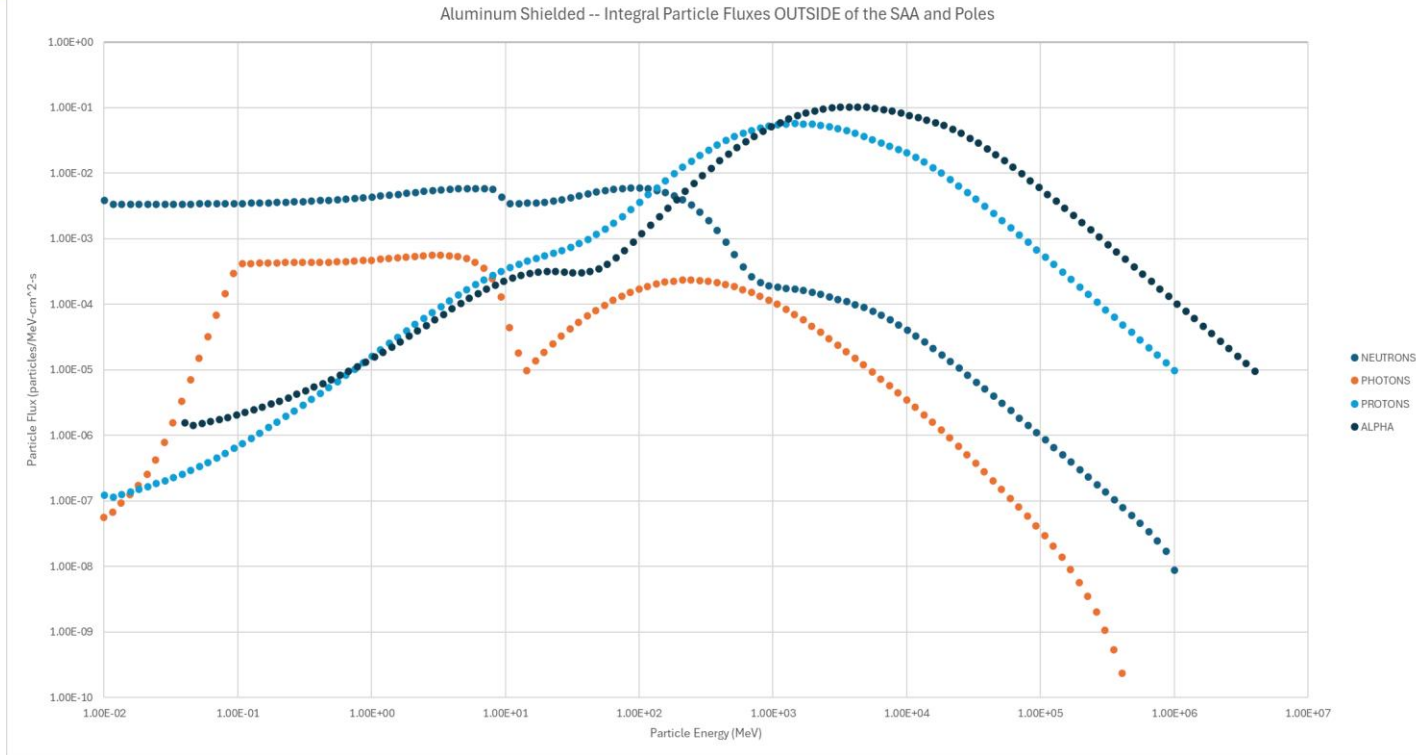


Trapped Protons Integral Flux Map

Simulation Parameters: AP8 Max GSFC 12/66 at 500 km orbit

Environment

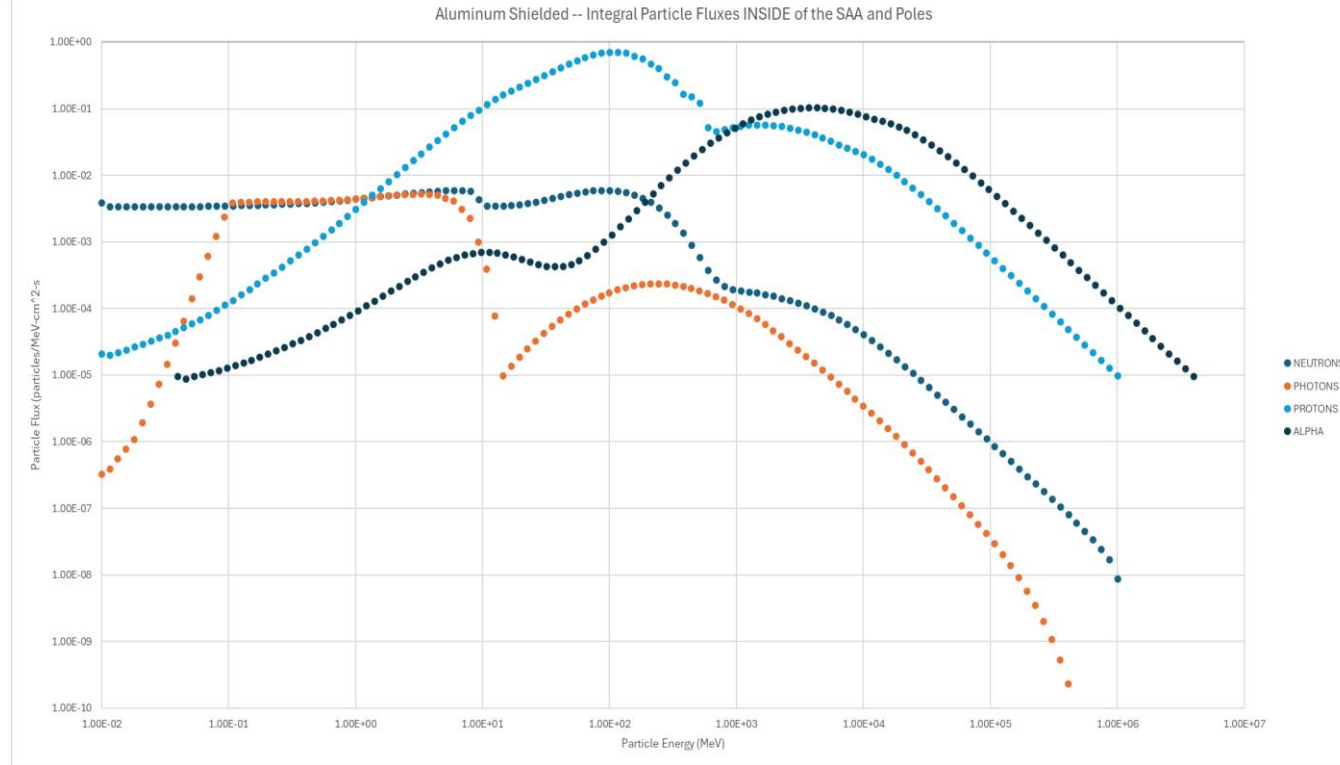
87



Unmoderated flux of highest contributing particles outside of the SAA and Poles

Environment

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Unmoderated flux of highest contributing particles
inside of the SAA and Poles

Launch Environment

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- Launch Vehicle – Falcon 9, 10-2027
- Source: Falcon 9 User's Guide
 - <https://spacex.com/pl/files/2017-10/falcon-9-users-guide-rev-2.0.pdf?4f8d2248dc>
 - Table and Figure labels from this document
- All values are design nominal, and will be checked by coupled load analysis and physical testing
- Load Environment
 - Fundamental Vibrational Frequencies assumed for this environment
 - Primary bending mode > 10Hz
 - Primary axial mode > 25Hz
 - Acceleration Range in Figure 4-2
 - Axial load from +8.5g to -4g
 - Lateral load from +3g to -3g
 - Shock
 - Maximum load and its frequency, at moment of deployment, in Table 4-5

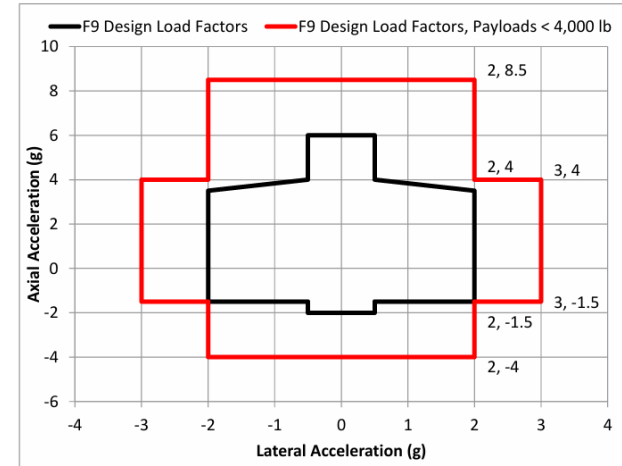


Figure 4-2: Falcon 9 payload design load factors, Light mass (under 4,000 lb)

Table 4-5: Payload adapter-induced shock at the spacecraft separation plan

Frequency (Hz)	SRS (g)
100	30
1000	1,000
10000	1,000

- Acoustic Environment
 - Pressure added due to sound
 - Maximum values at liftoff and transonic flight

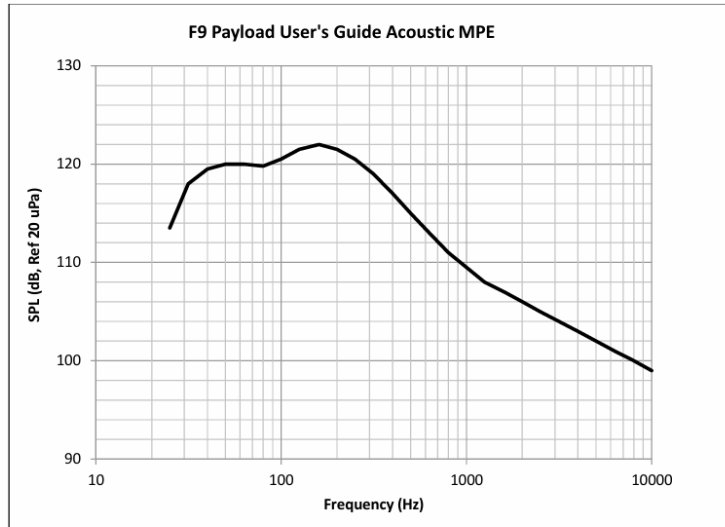


Figure 4-5: Falcon 9 maximum predicted acoustic environment (P95/50), 60% fill-factor, 131.4 dB OASPL (1/3 octave)

Table 4-3: Falcon 9 maximum predicted acoustic environment (P95/50), 60% fill-factor, 131.4 dB OASPL (1/3 octave)

Frequency (Hz)	Acoustic Limit Levels (P95/50), 60% Fill-Factor (1/3 Octave)
25	113.5
31.5	118
40	119.5
50	120
63	120
80	119.8
100	120.5
125	121.5
160	122
200	121.5
250	120.5
315	119
400	117
500	115
630	113
800	111
1000	109.5
1250	108
1600	107
2000	106
2500	105
3150	104
4000	103
5000	102
6300	101
8000	100
10000	99
OASPL (dB)	131.4

- Acoustic Environment
 - Pressure added due to sound
 - Maximum values at liftoff and transonic flight

Table 4-4: Falcon 9 maximum predicted acoustic environment (P95/50), 60% fill-factor, 131.4 dB OASPL (full octave)

Frequency (Hz)	Acoustic Limit Levels (P95/50), 60% Fill-Factor, (Full Octave)
31.5	122.4
63	124.7
125	126.1
250	125.2
500	120.1
1000	114.4
2000	110.8
4000	107.8
8000	104.8
OASPL (dB)	131.4

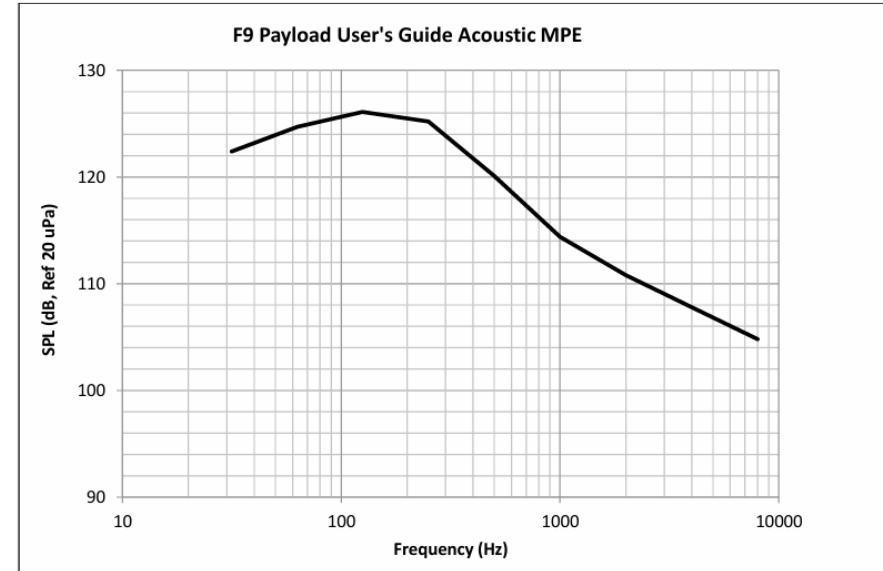


Figure 4-6: Falcon 9 maximum predicted acoustic environment (P95/50), 60% fill-factor, 131.4 dB OASPL (full octave)

- Thermal
 - Expect some heating (less than $1,135 \text{ W/m}^2$) after deployment
 - Fairing temperature reaches 180°F at deployment
- Ascent Venting
 - Payload outgassing and bus depressurization up to - 4.5kPa/sec on ascent ($\sim 200\text{s}$ total)
 - Sensors must be oriented away from debris exit path

Payload Requirements

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Level 2 Requirements:

Req. ID	Title	Requirement	Rationale	Risk Score	Status
PL-2.01	Payload System Scope	The Payload System shall consist of 2-4 neutron detector platforms (minimum of 1 moderated, 1 unmoderated with goal of 2 in each moderator configuration).	This is the definition of our payload scope.	3	Compliant
PL-2.02	Neutron Detector Interface	PL shall interface to the bus mechanically, electrically and thermally.	This hosted payload must fit within the SWaP allocation not used by bus/imager.	10	Compliant
PL-2.03a	Detector Orientation	Neutron detector shall have the MSNDs side of the DSC-Domino within ± 20 -degrees of nadir while pointing in orbit and while primarily focusing on counting (e.g., not in a survival charging mode).	While in orbit, allowing for emergency exceptions (i.e., energy depletion), the detector must remain close to perpendicular to nadir for accurate fast counts	6	Compliant
PL-2.03b	Detector Orientation	The detectors, there shall be no moderator from the adjacent detectors within a cone of view as described by 70° from Nadir at any point on any face.		12	Partial Compliance
PL-2.04	Mission Lifetime	PL shall operate to gather science data for a minimum of 2 months with a goal of 1 year.	PL mission life is determined by stakeholder needs.	9	N/A
PL-2.05	Moderated and Unmoderated Detector Spacing	The unmoderated detector(s) shall be placed a minimum lateral distance of [TBD] mm from moderated detector package(s).	To ensure there are no statistically significant moderated neutrons that are contributing to the unmoderated detector counts (both fast and thermal)	8	Partial Compliance
PL-2.06	DSC-Domino keep-out zone	The two unmoderated platforms shall have their nadir- and zenith-faces unobstructed by the HDPE moderator within a cone of 70-degrees relative to zenith/nadir	To ensure that the moderator is not impacting the unmoderated detectors, they should be located in the 1U allocated for the bus systems where there is no HDPE directly above or below the detector. Ideally, they're on the opposite side of the craft from the moderator	16	Partial Compliance
PL-2.07	Neutron Detector Redundancy	Each main neutron detector platform shall have a backup detector for failure detection and space anomaly verification resulting in four total DSC Domino boards	To ensure anomalies are correctly observed and distinguishable from detector problems	3	Compliant

Payload Requirements

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Level 3 Requirements:

Req. ID	Title	Requirement	Rationale	Risk Score	Status
PL-3.01	Neutron Detector Payload Accuracy	The neutron detector payload shall detect neutrons within 10% of the accepted orbital neutron profile (after proper efficiency corrections from moderator choice/thickness and detector efficiency).	To determine the accuracy of the detector, it must be compared to the accepted measurements	12	Not Compliant
PL-3.03	Charged Particle Rejection (proton, beta, GCRs, gamma, etc.)	The non-backfilled MSNDs shall enable non-neutron particle rejection within "x"% accuracy	To ensure only neutrons are being counted, a rejection technique must be employed (coincidence detection, temporal discrimination, etc.)	8	Not Compliant
PL-3.04	Neutron Detector Payload Power Consumption	The neutron detector payload shall have a power consumption of 65 mW or less per detector platform, with a 5 V voltage requirement, respectively.	Max power during detector operation using the DSC-Domino from RDT	4	Compliant
PL-3.05	Neutron Detector Payload Mass	The neutron detector payload shall have a mass of 0.31 kg or less (4 x 14.25 g for the detector platforms and 246 g for the neutron moderator).	Mass of the detector and the neutron moderator	2	Compliant
PL-3.06	Neutron Detector Size	The neutron detector assembly (including moderator) shall occupy 0.3 U or less	Total space allocated for the detector and neutron moderator	16	Compliant
PL-3.07	Neutron Detector Thermal Limits	The neutron detector payload shall be kept between -20 °C and +55 °C during operation and 40°C to +100°C during survival	The optimal operating temperature is 30 °C, but must be in these bounds for operation.	12	Partial Compliance



Payload Requirements

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Level 3 Requirements:

Req. ID	Title	Requirement	Rationale	Risk Score	Status
PL-3.08	DSC-Domino power interface	The dominos shall use a usb-c cable to receive power	The platform comes with a built-in USB-C port for all power needs	9	Compliant
PL-3.09	Neutron Detector Vibration Limits	The neutron detector shall be integrated such that it does not exceed [70%] its [TBD vibration amplitude and frequency] limit	The detector can only handle this maximum vibration load, so mounting, integration, and other factors must ensure an acceptable factor of safety is maintained	16	Compliant
PL-3.10	Neutron Detector Acceleration Limits	The neutron detector shall be integrated such it can accommodate [30g] of acceleration with its nominal [x g] limit	The integration of the detector should allow detector survival of the expected launch and deployment accelerations	12	Compliant
PL-3.11	Detector Efficiency	Neutron detector shall have peak detection efficiencies at 0.025 eV, 0.1 MeV, and 18 MeV	The shifted spectrums from the moderator will allow relative detection efficiencies at the energies we are interested in, possessing a certain efficiency that can be used to back out neutronic properties at the location (i.e., flux, dose, etc.)	3	N/A
PL-3.12	DSC-Domino upset flagging	The data gathered from the backup dominos shall be compared to the primary dominos and any difference of {5%} or more over an average of {20 } worth of data collection will be flagged as potential detector error	To ensure detectors are functioning as expected and no issues are occurring, the backup platform in each moderation condition will check the measurements of the primary platform	6	N/A
PL-3.13	DSC-Domino data frequency	While in science mode, the detectors shall have their counts pulled at a frequency of at least {0.0333 Hz} for nominal operation and {0.01667 Hz} for minimum success and their temperatures pulled at a frequency of at least {0.01667 Hz}	To get quality data the minimum count frequency will allow data points around every kilometer of distance traveled and the temperature pull will ensure adequate temperature regulation and reaction	3	Compliant

Payload Requirements

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Level 3 Requirements:

Req. ID	Title	Requirement	Rationale	Risk Score	Status
PL-3.14	Neutron Energy Bins	Neutron detector moderation shall shift detection bins into thermal (0 - 1 eV), epithermal (1 eV - 0.5 MeV), fast (0.5 MeV+)	The moderator will allow us to bin energies in each detector to obtain a certain separation in detected neutron energies	2	Compliant
PL-3.15	Detector Mounting	Neutron detector shall be <u>mounted</u> with the MSNDs facing parallel to the Nadir datum face within 5-degrees.	In assembling/modeling the satellite package, the detector will be mounted perpendicular to nadir (or parallel to the nadir datum face)	6	Partial Compliance
PL-3.16	DSC-Domino fastening	The dominos shall be securely fastened to either the moderator or the structure using {insert size} bolt using the four mounting holes on each platform	There are four mounting holes on each DSC-Domino package that should be used to secure the domino inside the moderator and to the frame for the two different moderation conditions	8	Compliant
PL-3.17	DSC-Domino data interface	The dominos shall use a usb-c cable to receive commands and relay data	The platform comes with a built-in USB-C port for all communication needs	9	Partial Compliance
PL-3.18	Detector power distribution {if needed for separate power and data interfaces}	There shall be a 5 Volt rail capable of supplying 0.012 amps to each detector (62 mW to each detector, or 0.248 W for the system)	In the case of compartmentalized power and data interfaces with the detector, a 5 V rail capable of supplying enough power for all four detectors will be needed for the subsystem	16	Compliant
PL-3.19	DSC-Domino thermal interface	The dominos shall have thermal couples to the heater(s) and radiator(s) to ensure they are kept in thermal limits (-20 °C and 55 °C) during operation	The detectors must be kept within thermal limits when operating to keep thermal noise to a minimum (from high temperature) and ensure hardware functions (from low temperature)	12	Not Compliant

Payload Requirements

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Level 4 requirements:

Req. ID	Title	Requirement	Rationale	Risk Score	Status
PL-4.01	Detector Moderator	HDPE shall be used as the neutron moderator	To produce the desired detector efficiency energy ranges, a neutron moderator must be used	3	Compliant
PL-4.02	Neutron Moderator Outgassing	The neutron detector moderator shall have a total mass loss coefficient below 1.0%. (or be encased such that the loss coefficient is below 1.0%)	To avoid interfering with other sensors/equipment, the chosen moderator must not have outgassing that would inhibit the overall mission	3	Compliant
PL-4.03	Neutron Moderator Size	The neutron moderator shall encase 2 of the detectors with a thickness of 10 mm on the "top" side of the board and 10 mm on the bottom side of the board.	Two of the detectors shall be encased in the moderator to shift the energy spectrum as needed	3	Partial Compliance
PL-4.04	Moderator fastening	The moderator shall be securely fastened to the satellite structure using {M2} bolts	The moderator, being the housing for two of the detectors, must be secured to the satellite frame to ensure payload safety and accurate pointing	8	Compliant

NuKAT Preliminary Design Review (PDR) – Day Two

May 6th, 2026

This presentation has been approved for public release.
Limited distribution as approved by NuKAT team.



PDR Panelist Introduction

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- **Dr. Amir Bahadori**
 - Professor and Program Director, Kansas State University Nuclear Engineering
 - Former NASA civil servant in the Space Radiation Analysis Group (SRAG)
- **Chris Brandt**
 - President, K-State's Research Foundation
- **Dave Copeland**
 - Space Communications System Engineer, John Hopkins University Applied Physics Laboratory
 - Co-Founder, Silversat
- **Glen Fountain**
 - Project Manager, John Hopkins University Applied Physics Laboratory – Retired
 - Former Industry Advisor, Mike Wiegers Department of Electrical and Computer Engineering, Kansas State University



PDR Panelist Introduction

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- **Don Golding**
 - Chief Avionics Architect, Nextage
- **Dr. John Hatcliff**
 - Professor, Kansas State University
Computer Science
- **Chris Hersman**
 - Mission Systems Engineer, John
Hopkins University Applied Physics
Laboratory
- **Zach Metzinger**
 - Engineer, METECS, Lunar Gateway
Mission
 - Member, AMSAT Board of Directors
 - Volunteer, AMSAT GOLF Program
- **Dr. Matt O'Keefe**
 - Professor, Kansas State University
Chemical Engineering



PDR Panelist Introduction

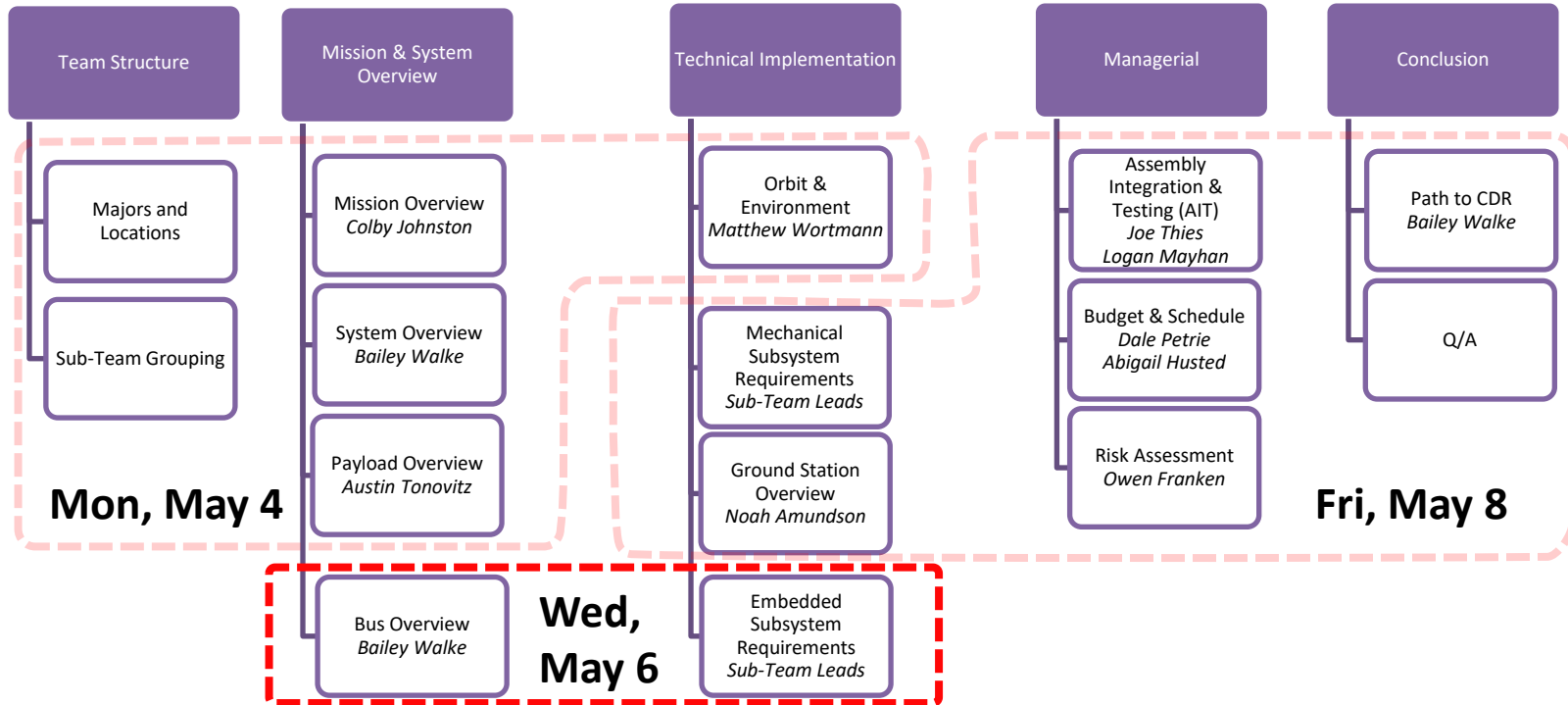
101

- **Jeffrey Sheldon**
 - Director, K-State's Sunderland Foundation Innovation Lab
 - FCC Trustee, K-State's Amateur Radio Club (W0QQQ)
- **Dr. Tim Steffensmeier**
 - Assistant Vice President and Director of Engagements and Outreach, Kansas State University
- **Kyle Tidball**
 - Engineer Specialist, Electronics Development Group, Textron Aviation
- **Tim Trenkle**
 - Engineer (Retired), NASA Goddard Space Flight Center



Agenda

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Presenters

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Bailey Walke	Chief Mechanical Systems Engineer
Leo Martin	EPDS Lead
Tyler Hogenkamp	ADCS Lead
Dylan Bequeaith	Communications Lead
Henry Bash	Program Manager
Logan Hayward	Chief Embedded Systems Engineer
Nalen Rangarajan	C&DH Lead
Braden Adams	FSW Lead



BUS OVERVIEW

Bailey Walke – NuKAT Chief Mechanical Systems Engineer

Mechanical Engineering Masters | Spring 2026



Bus Requirements

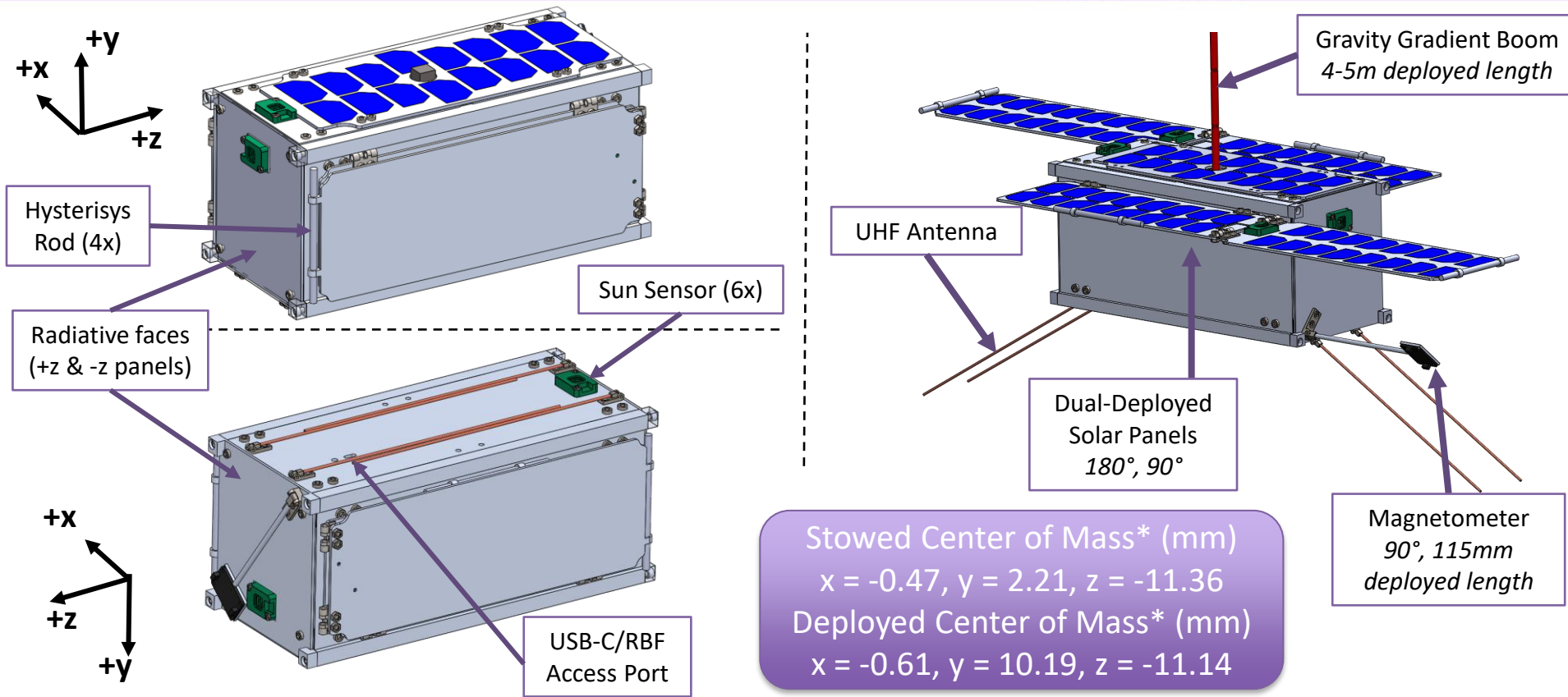
105

Req. ID	Title	Requirement	Risk Score	Status
2.3 Bus Requirements				
BUS-2.01	Bus System Purpose	NuKAT bus systems shall support payload needs, and integrate payload systems and bus subsystems.	5	Compliant
BUS-2.02	Bus System Size	BUS subsystems shall take up no more than 1.75U of the satellite.	12	Compliant
BUS-2.03	Bus System Weight	BUS must weigh less than 3.5 kg.	5	Compliant
BUS-2.04	Bus System Power	BUS shall provide at least 8W of power.	16	Partial Compliance
BUS-2.05	Bus System Thermal Range	BUS must be able to operate between -20 and +55 degrees Celsius.	8	Compliant
BUS-2.06	Bus System Radiation Hardness	BUS must be able to endure 7.25 krad of radiation over the mission lifetime.	15	Partial Compliance
BUS-2.09	Bus-Payload Integration	BUS shall provide data handling, ground communication (4800 kbps), attitude control, structure, and flight software to support payload needs.	10	Partial Compliance
BUS-2.11	Space-Rated Components	BUS shall consist of components rated for space environments and proven to be space worthy for at least 24 months.	5	Compliant
BUS-2.13	Satellite Communication	BUS shall be able to communicate with the ground station. Communication includes receiving uplinked commands and downlinking relevant science data and health beacons.	10	Partial Compliance
BUS-2.15	Deployable/Dispenser Independence	Deployables shall be constrained by the CubeSat, not the dispenser.	5	Compliant
BUS-2.16	Deployable Inhibitors	NuKAT shall have at least three independent inhibits to prohibit the inadvertent release of any deployable structures.	10	N/A
BUS-2.17	Deployable Before Ejection	All parts shall remain attached to NuKAT during launch, ejection, and operation.	5	N/A
BUS-2.22	Acceleration Loading	BUS shall be designed for quasi-static loading of 30g acceleration applied independently in each primary axis.	5	N/A
BUS-2.23	Shock Loading	BUS should be designed to withstand the launch vehicle vibroacoustic environment of 15 GRMS for one minute in each axis.	5	N/A
BUS-2.24	Access Port	BUS shall provide a USB-C and RBF pin access port on one 2U panel, accessible without disassembly, per launch provider requirements.	4	Compliant



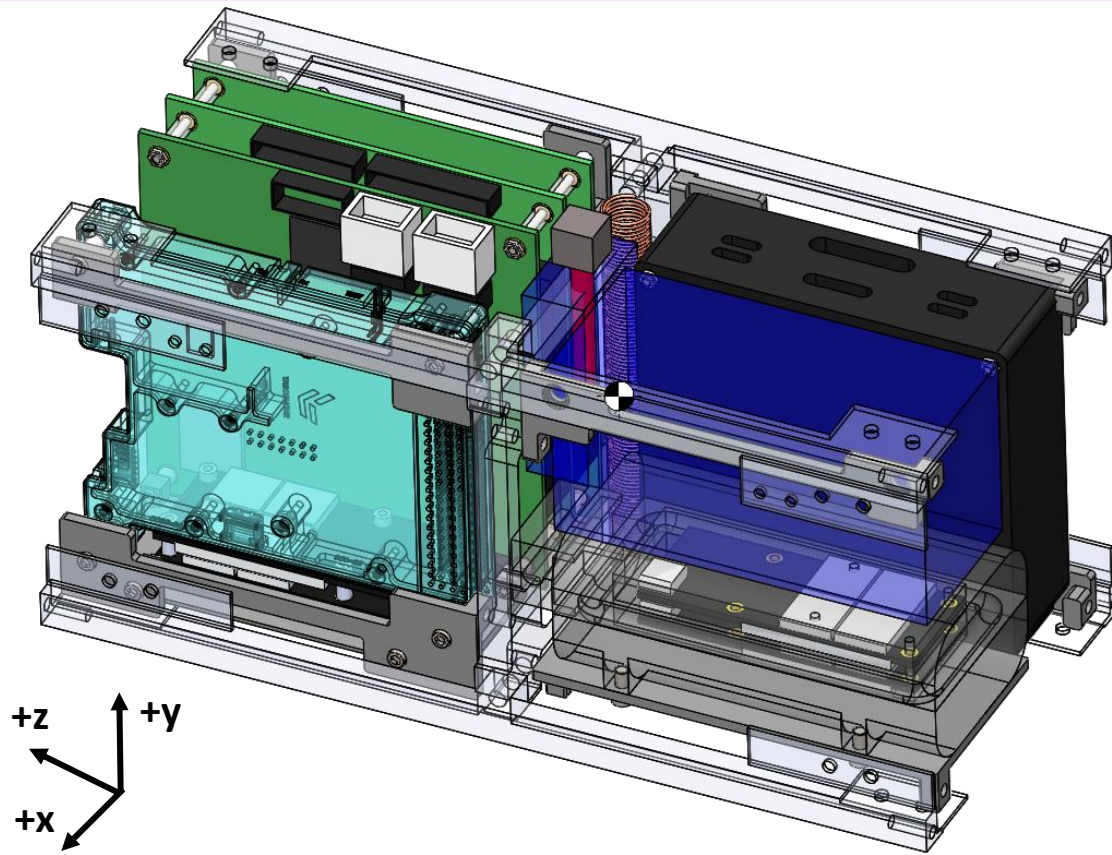
Satellite Overview (External)

106



Satellite Overview (Internal)

107



PAYLOAD

- 4x RTD DSC Domino; 2x moderated, 2x unmoderated

ATTITUDE CONTROL

- *Passive attitude control*: Gravity gradient boom
- *Active attitude control*: Electromagnet

COMMUNICATIONS

- *Science and telemetry downlink, command uplink*: UHF amateur bands (4.8kbps)

POWER

- 50Whr battery
- PowerDrive PDCU

COMMAND & DATA HANDLING

- NextAge NMC2 onboard computer

STRUCTURE

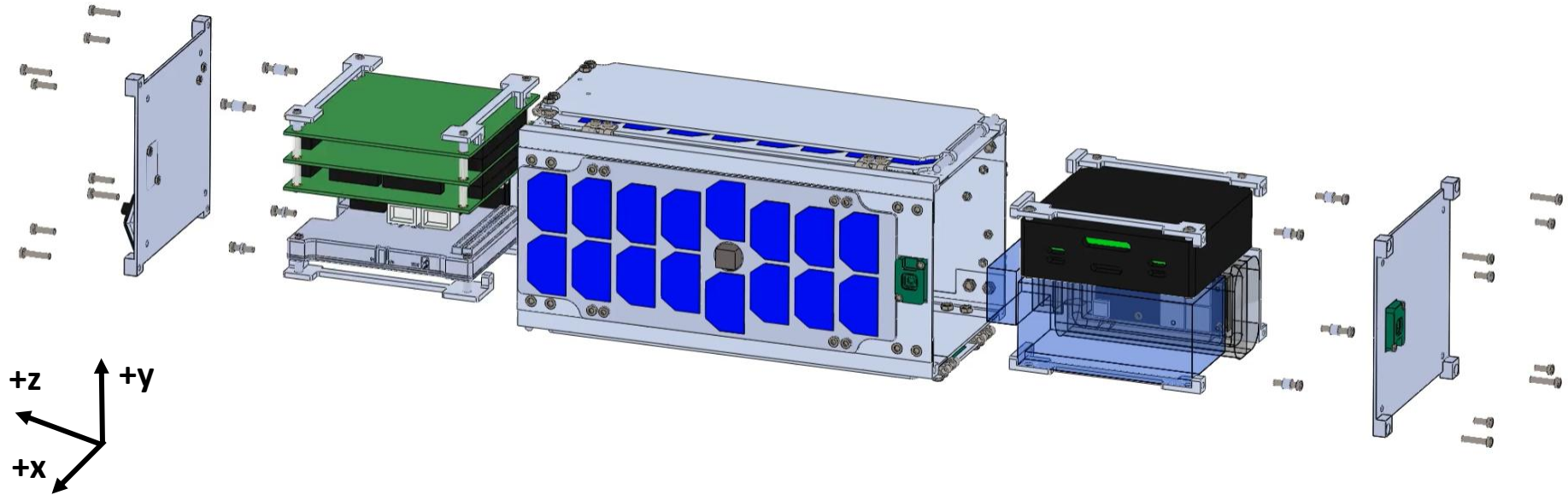
- E-Stack brackets, 6061
- 18-8 fasteners
- Delrin spacers
- Rail design compliant with CalPoly spec

THERMAL (not modeled)

- Conductive heat strips
- Radiator
- Heater

E-Stack Removal & Assembly

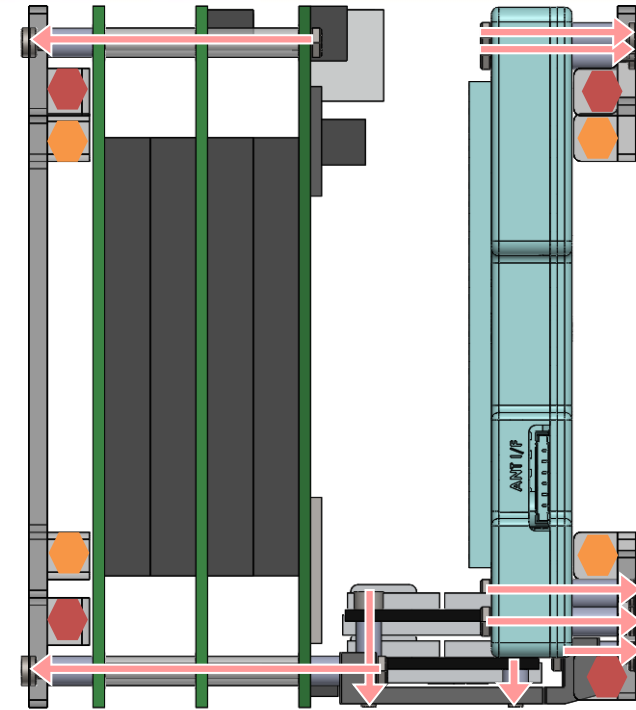
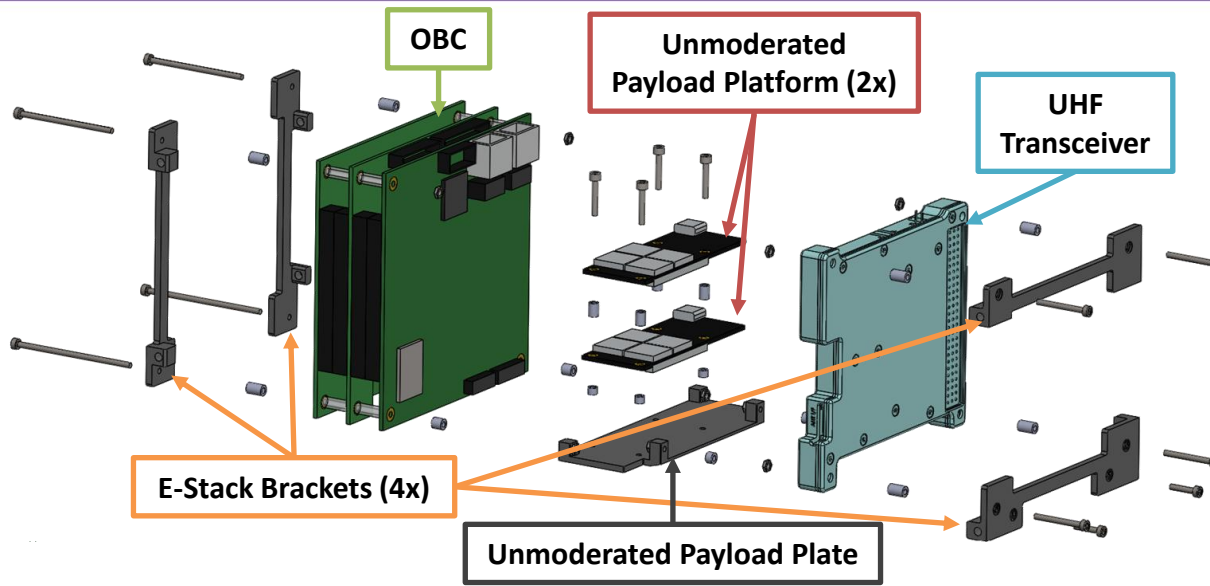
108



E-Stacks are removable with clearance through “z” panels without disassembly of chassis or x/y panels

“+z” E-Stack

109

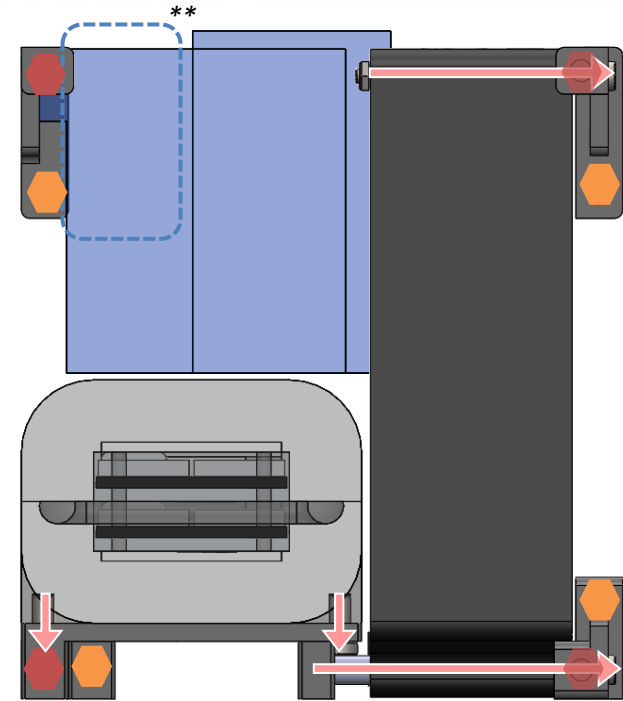
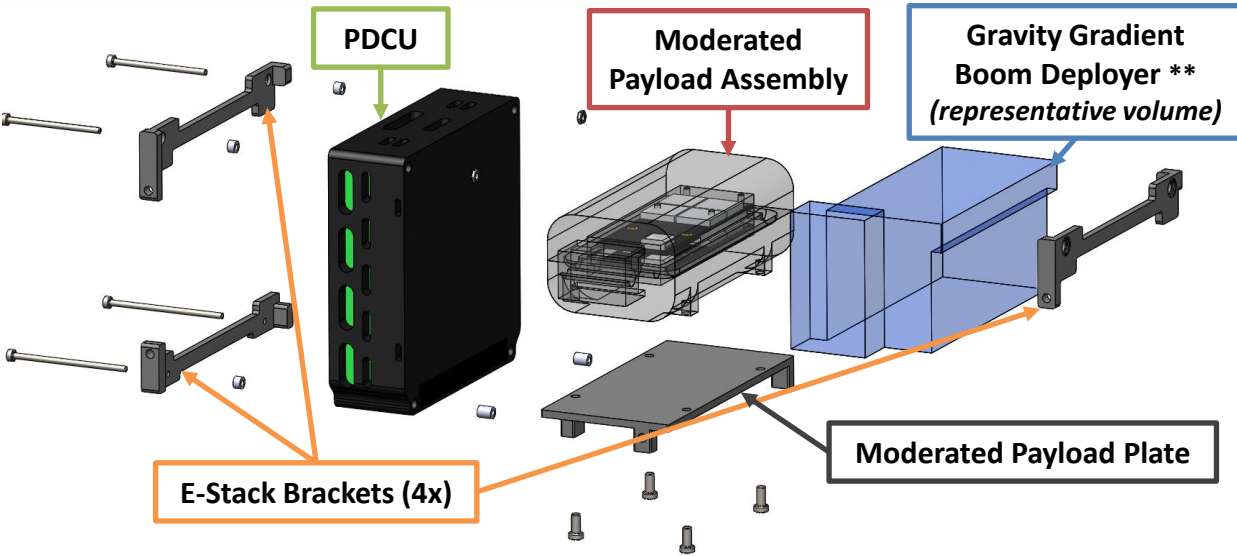


- Retention ring contact (into page)
- +z panel contact (out of page)
- E-stack : Component contact and load direction

- 96mm x 86mm envelope
- All components constrained with 4 points of contact
- E-Stack constrained with 8 points of contact to chassis/panels
- M4 18-8 bolts, nuts, and threaded inserts
- M4 Delrin spacers

“-z” E-Stack

110



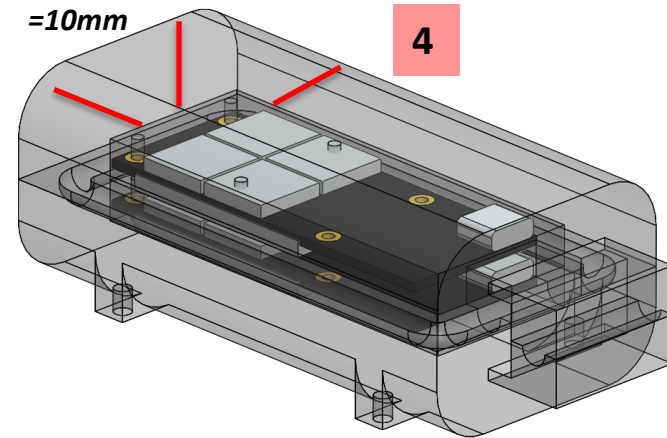
- Retention ring contact (into page)
- +z panel contact (out of page)
- E-stack : Component contact and load direction

- 96mm x 86mm envelope
- All components constrained with 4 points of contact
- E-Stack constrained with 8 points of contact to chassis/panels
- M4 18-8 bolts, nuts, and threaded inserts
- M4 Delrin spacers

** Not designed due to unknown geometry

PDCU: power distribution and control unit

111

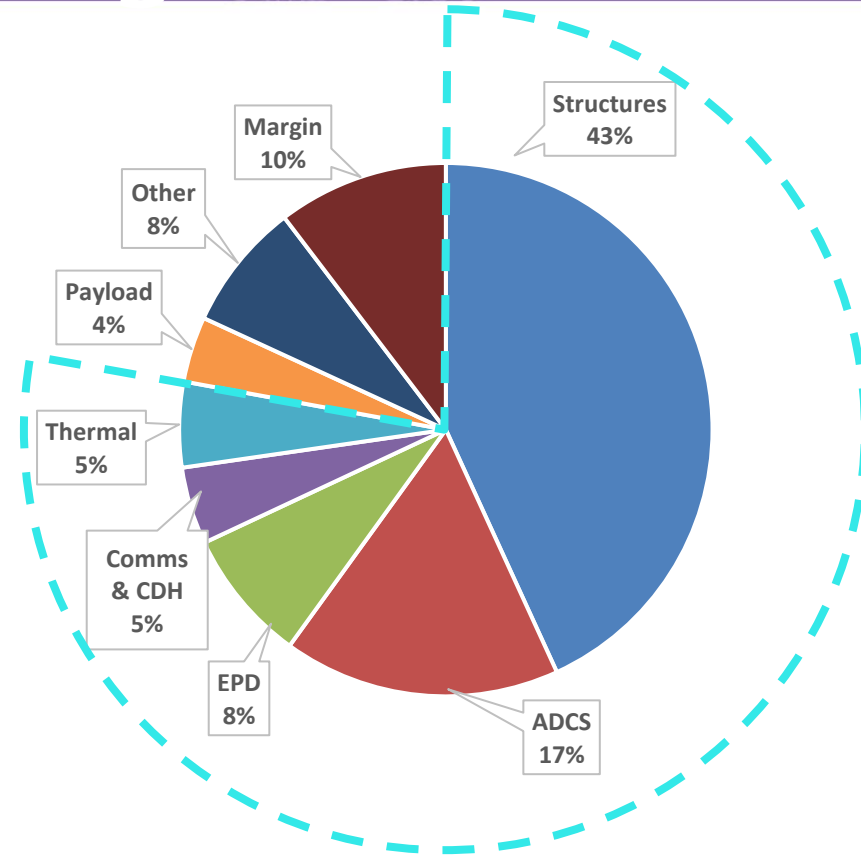


1. Platform normal are in line with earth normal (y)
2. Moderated and unmoderated platforms must not overlap in z (or x) direction
3. Unmoderated platform must have a 70° cone with respect to the nadir and zenith directions where no HDPE is present
4. Moderator must provide 10mm of HDPE material on every side of the platforms
5. Each platform must interface with OBC via USB-C

Mass Budget

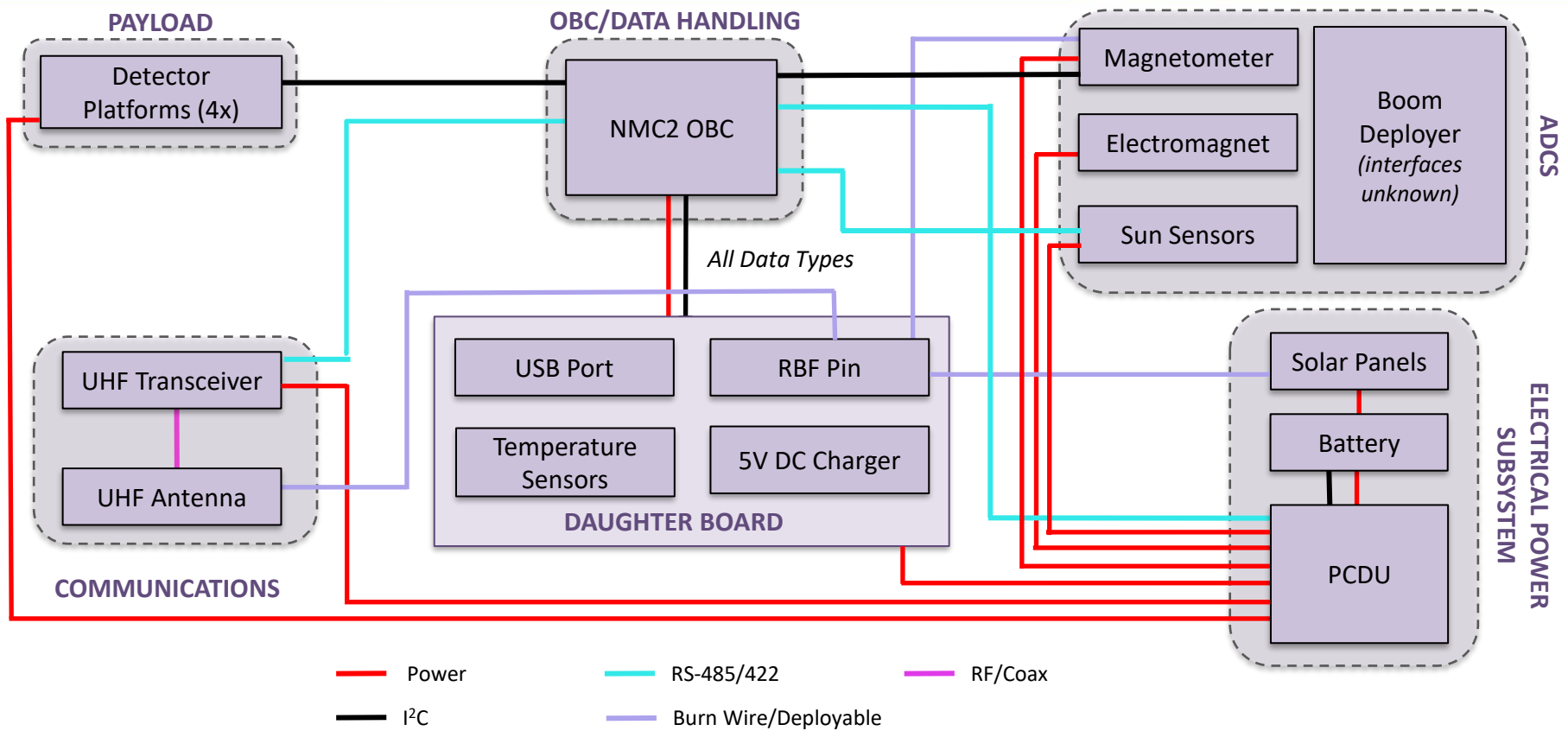
112

Subsystem	Mass (g)	Percentage
Structures & Mechanisms	1667.12	43.16%
Attitude Determination & Control	649	16.80%
Electrical Power & Distribution	311.52	8.07%
Thermal	200	5.18%
Communications, Commands and Data Handling	181.04	4.69%
Payload	153.64	3.98%
Other (wiring, coatings)	300	7.77%
Margin	400	10.36%
Total	3862.32	



System Block Diagram

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Requirement Compliance

- Most requirements that are able to be addressed at this time are compliant
- Environmental and launch load survivability requirements need further analysis and testing to determine compliance

Subsystems Compatibility

- ADC system design
- UHF antenna
- Power budget
- Thermal design

Bus has a Path Toward CDR

Bus is Capable of Supporting NuKAT Mission

In light of this, Bus should proceed toward CDR with further effort to develop detailed designs of subsystems, ensure power budget closure, conduct environmental and launch load survivability analysis, and test subsystem interfacing

EMBEDDED SATELLITE SUBSYSTEMS

Leo Martin, Tyler Hogenkamp, Phillip Shirkey, Dylan Bequeaith, Henry Bash, Nalen Rangarajan, Logan Hayward, Braden Adams



ELECTRICAL POWER DISTRIBUTION SUBSYSTEM

Leo Martin – EPDS Lead

Electrical Engineering | Spring 2027



Supplier change

- DMV -> TakeMe2Space

Closer to closing power budget

EPDS – Subsystem Requirements

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Level 3 Requirements

NUKAT REQUIREMENTS VERIFICATION MATRIX				
Req. ID	Title	Requirement	Risk Score	Status
3.5 Electrical Power Distribution Requirements				
EPD-3.01	External Charging	The EPDS shall have a means of charging its battery that does not require solar power.	3	Compliant
EPD-3.02	External Power Supply	The EPDS shall have a means of externally powering the spacecraft that does not require battery or solar power.	3	Compliant
EPD-3.03	Battery Level	The battery level shall remain above 50% during science/nominal mode, while remaining above 30% when upset mode occurs.	9	Not Compliant
EPD-3.04	Idle Mode Power	EPD shall be capable of powering all components at the calculated worst-case peak power draws during Idle Mode.	8	Compliant
EPD-3.05	Active Mode Power	EPD shall be capable of powering all components at the calculated worst-case peak power draws during Active Mode.	6	Compliant
EPD-3.06	Downlink Mode Power	EPD shall be capable of powering all components at the calculated worst-case peak power draws during Downlink Mode.	6	Compliant
EPD-3.07	Distribution and Control	EPD shall safely distribute and control all of the power generated.	10	Compliant
EPD-3.08	Power Conversion	EPD shall provide a 3.3V rail.	8	Compliant
EPD-3.09	Power Conversion	EPD shall provide a 5V rail.	8	Compliant
EPD-3.10	Switched outputs	EPD shall provide at least four separately-switched outputs.	8	Compliant
EPD-3.11	Bus Isolation	EPD shall provide bus isolation between subsystem loads.	12	Compliant
EPD-3.12	Bus Monitoring	EPD shall provide bus monitoring electronics such that each load may be accurately monitored for fault and efficiency.	3	Compliant
EPD-3.13	Protection	EPD shall protect itself from EMI, transients, bus faults, and load faults.	15	Compliant
EPD-3.14	Lifetime	Component's expected life-span should exceed mission life expectancy of '4' months.	5	Compliant

EPDS – Subsystem Requirements

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Level 4 Requirements:

4.5 Electrical Power Distribution Requirements				
EPD-4.01	EPS	The EPS shall be the TakeMe2Space PowerDrive		
EPD-4.01.1	Power Consumption	The EPS shall not draw more than (2) Watts of power when operating.	6	Partial Compliance
EPD-4.01.2	Voltage Rails	The EPS shall maintain the necessary voltage levels for the busses.	4	Compliant
EPD-4.01.3	Current	The EPS shall be able to provide the necessary currents on all voltage rails	4	Compliant
EPD-4.01.4	Solar Power	The EPS shall accept the full power of the solar array	4	Compliant
EPD-4.02	Battery	The battery shall be the TakeMe2Space PowerBank		
EPD-4.02.1	Battery Size	The battery shall be of adequate (at least 40Wh) size to provide power to systems while vehicle is in Earth's shadow .	10	Compliant
EPD-4.03	Solar Array	The solar array shall be 10U of TakeMe2Space PowerPlane		
EPD-4.03.1	Solar Array Size	The solar arrays shall conform to a 2U vehicle size.	3	Compliant
EPD-4.03.2	Solar Array Power	The solar array shall provide an orbit average of at least 8W.	4	Compliant
EPD-4.03.2	Solar Array Configuration	The array shall consist of 20 cells, in a S/P configuration to maximize voltage without going over input limit	1	Partial Compliance
EPD-4.04	Errata			
EPD-4.04.1	Solar Safe/Arm Plug	A solar array arming plug shall be used to prevent the solar array from powering the spacecraft.	9	Compliant
EPD-4.04.2	Battery Safe/Arm Plug	A battery arming plug shall be used to connect battery power to the spacecraft.	9	Compliant
EPD-4.04.3	Wiring / Harnesses	Wiring shall be procured to connect EPS to all components (USB, MOLEX, etc.)	4	N/A
EPD-4.04.4	Power wiring gauge	All wiring carrying 100mA or greater shall be 22awg or thicker	8	N/A
EPD-4.05	Total EPD System			
EPD-4.05.1	Total Mass	The EPDS shall be no more than 800g	4	Partial Compliance
EPD-4.05.2	Total Cost	The EPDS shall cost no more than \$15k	12	Partial Compliance

EPDS – External Links

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Mechanical

Mounting	PC104
Exterior Solar Panels	Custom-designed mount
Ground Station Accessibility	Battery Solar Panel Connection

Thermal

Conduction	EPS Battery
Solar Panel Heat Radiation	
Battery Heaters	

Command & Software

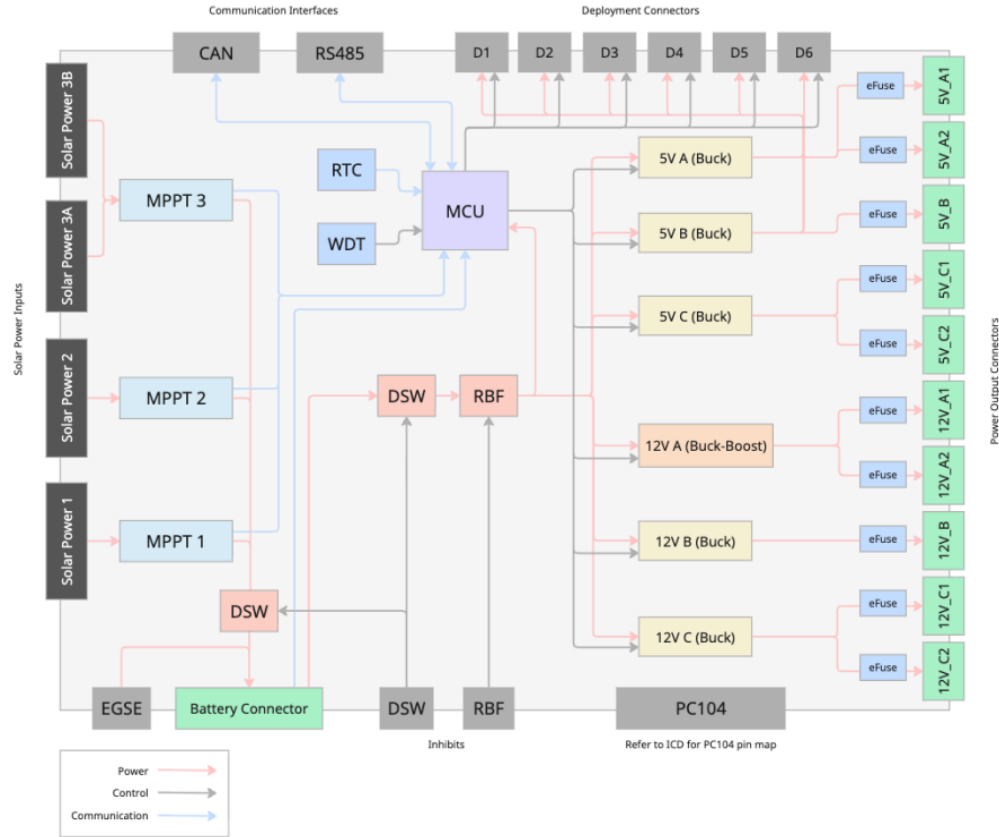
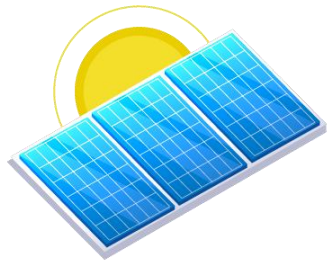
Command Protocol	CAN, RS-485 PC104 Bus
Enable/Disable	
Telemetry Outputs	Voltage, Current, Temperature & lots more

Electrical

Voltage Rails	Configurable 3.3V - 24V
---------------	-------------------------

EPDS – Block Diagram

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- Solar Panel - TakeMe2Space PowerPlane
 - Potential to Include
 - Sun Sensor
 - Spectral Sensor
 - Temperature Sensor
 - Gyroscope
 - Magnetometer
- Reported Characteristics
 - ~2W/U
 - 28.5% Efficiency



- EPS – TakeMe2Space PowerDrive
 - Provides 3V3, 5V, & Configurable Voltage
 - MPPT Solar inputs
 - 35V Max
 - Telemetry
 - Input voltage/current
 - MPPT voltage
 - Output voltage/current
 - Temperature
 - Battery Telemetry
 - Single-core ARM CPU
 - Investigating possibilities



- TakeMe2Space PowerBank
 - Configuration options
 - 4S1P, 2S2P
 - Nominal capacity: 50.4Wh
 - 5,000 cycles @ 50% depth-of-discharge
 - 80% capacity = 40Wh
 - With 90min orbit we hit this within a year
 - Potential for additional decrease due to aging
 - Integrated heaters
 - Active Cell Balancing
 - Telemetry including battery health



EPDS – Preliminary Budget

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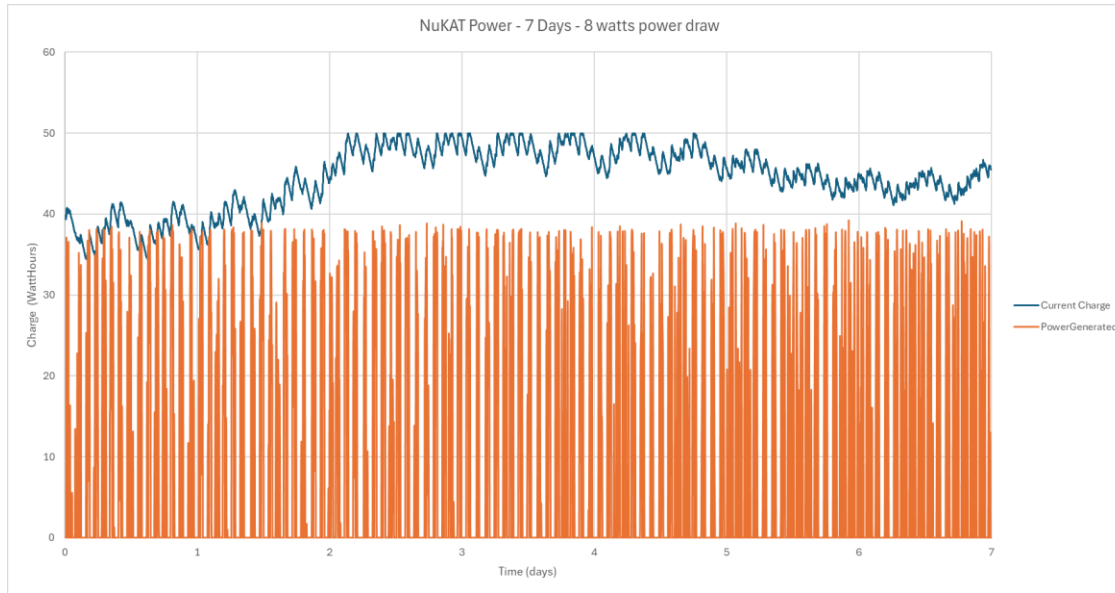
Subsystem	Payload	OBC	ADCS	XCVR	Thermal	EPS	Structure	Total
Orbit-Average Power (W)	0.4	4	0	0.5	1	2	0	7.9
Peak Power (W)	0.4	4	0	3	2	2	0	11.4



EPDS - Components

Electrical Power & Distribution

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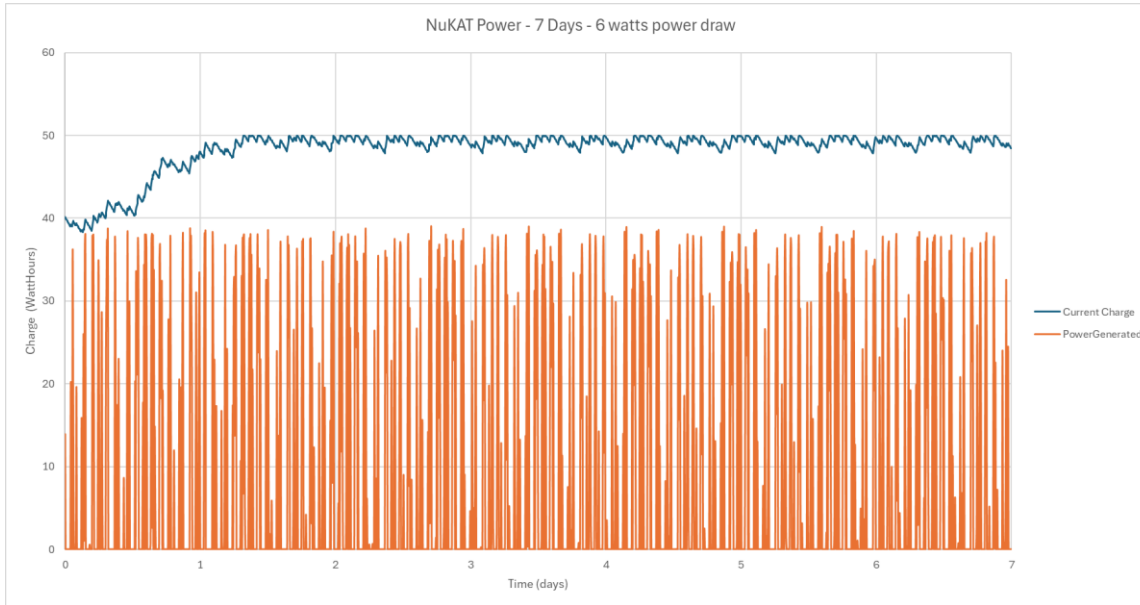


- Assuming **8W** average power
 - We are barely power-positive
 - Battery stays charged but does not reach 100% every orbit
 - Assuming 40Wh initial charge, never drops below 35Wh
 - Not ideal due to lack of safety factor

EPDS - Components

Electrical Power & Distribution

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- Assuming **6W** average power
 - Power-positive with margin
 - Goal consumption
 - We have extra power for
 - Efficiency diminishing
 - Upset events
 - Additional Rx/Tx

Requirement Compliance

- Battery Level
- EPS/Battery Efficiency
- Wiring

Subsystems Compatibility

- No compatibility issues

EPDS has a path toward CDR

EPDS will be capable of supporting NuKAT Mission

In light of this, EPDS should proceed toward CDR with further effort to reduce power consumption.

ADCS SUBSYSTEM

Tyler Hogenkamp – ADCS Lead

Mechanical Engineering | Spring 2027



Active vs. Passive Control

- Closed trade study
- Moving to passive control (gravity gradient boom)

Satellite Orientation

- Finalized orientation and coordinate axes

Stability Calculation

- Determined approximate length of boom
- Determined approximate mass of boom end mass

ADCS Requirements

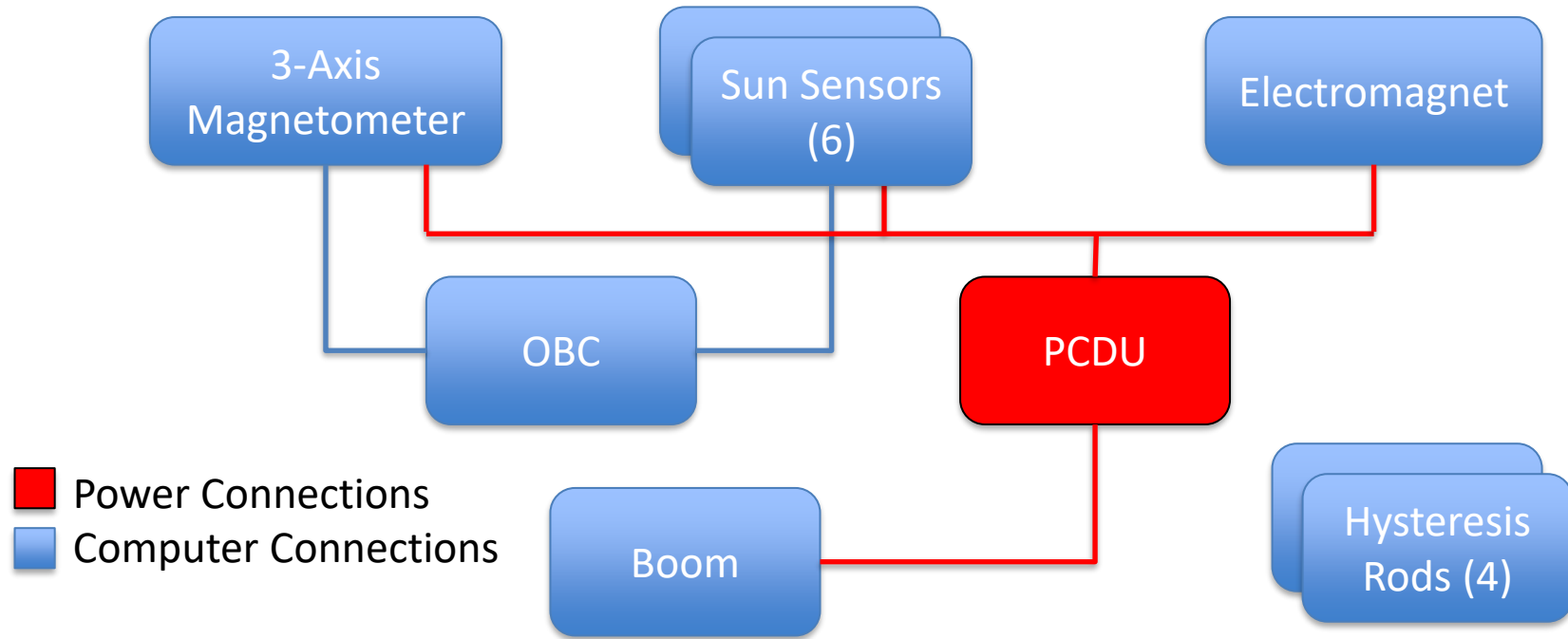
Level 3 Requirements

Req. ID	Title	Requirement	Risk Score	Status
ADC-3.04a	GS Knowledge	ADC shall determine angle w.r.t. Ground Station, within "x" deg.	4	Partial Compliance
ADC-3.04b	GS Pointing	ADC shall allow pointing to Ground Station, within "x" deg.	4	Not Compliant
ADC-3.12	Computing Budget	ADC shall require less than "x" bit/s of onboard data processing for telemetry and flight ops.	4	Partial Compliance

Level 4 Requirements

Req. ID	Title	Requirement	Risk Score	Status
ADC-4.05	Magnetorquer Angular Accuracy	ADC shall use Magnetorquers to make maneuvers accurate to less than 5 degrees.	4	Partial Compliance

Block Diagram



Interfaces

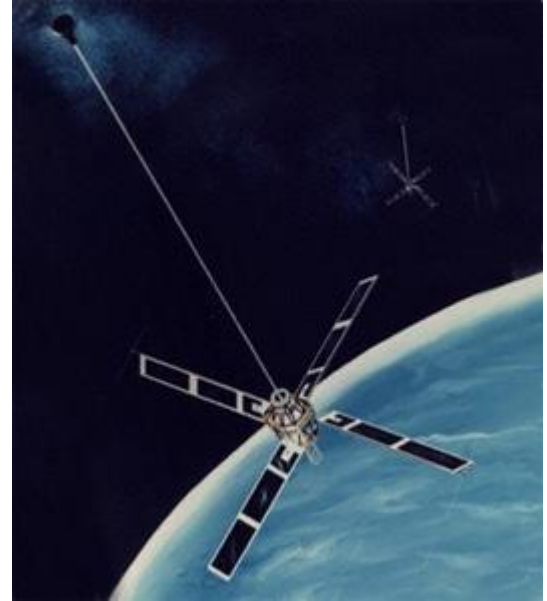
- Sun Sensors
 - RS-485 serial communication
- 3-Axis Magnetometer
 - I2C communication through SDA pin
 - Requires installation of a library to code
- Electromagnet
 - Molex Pico – Blade 2 pin

Initialization of Boom

1. Separation from launch vehicle
2. Waiting period to allow detumble via hysteresis rods
3. Activate electromagnet at one of the poles
4. Deploy boom arm

Boom and Mass

- Size: 4 – 5 m
- Weight: 0.5 – 0.8 kg
- Power: N/A
- Cost: more analysis is required



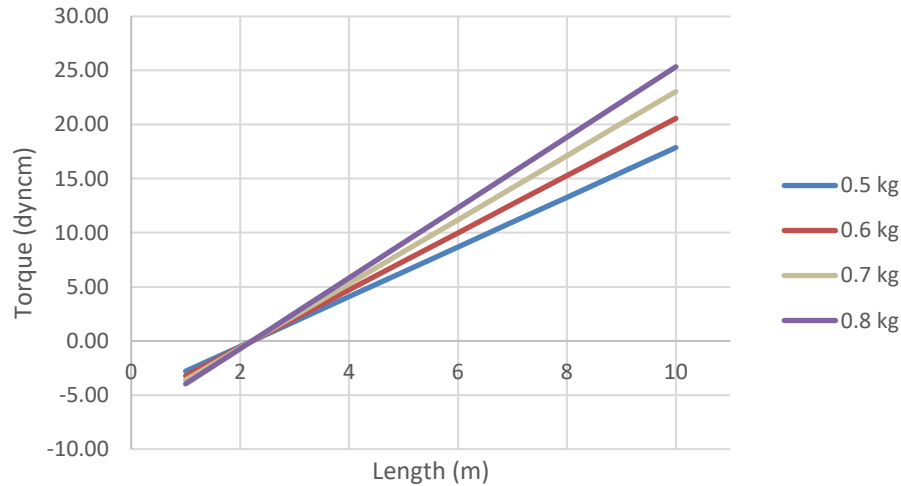
Boom and Mass

$$\tau = mL + b$$

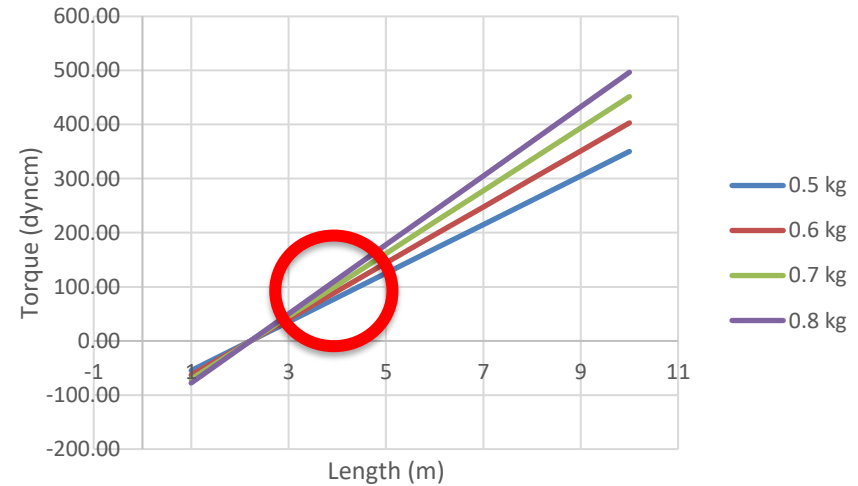
Length: 4 – 5m

Weight: Further considerations needed

Calculated Torque Generated at 1°



Calculated Torque Generated at 20°



Work to Go

- Create a plan to manufacture a boom and mass
- Contact suppliers to get quotes
- Determine disturbance torques on the spacecraft (aerodynamic, solar pressure, and torques due to residual magnetism in the spacecraft)
- Run simulations of deployment of the boom and stabilization
- Compute natural frequencies of pitch and roll axis



Requirement Compliance

- Update the requirements to reflect a passive control system

Subsystems Compatibility

- No Sun pointing for power
- Can stay within +/- 20 of nadir for data collection and comms

ADCS has a Path Toward CDR

The ADCS Subsystem is Capable of Supporting NuKAT Mission

In light of this, ADCS should proceed toward CDR with further effort to perform deployment and stability simulations.

COMMUNICATIONS SUBSYSTEM

Dylan Bequeaith – Communications Lead

Electrical Engineering | Spring 2027

Henry Bash – Project Manager & Comms Member

Electrical Engineering | Spring 2027



Link Budget Established

- Substantially more accurate link budget established using AMSAT-IARU spreadsheet

Antenna Selection

- Two courses of action for antenna selection

Comms – Level 3 Requirements 141

NUKAT REQUIREMENTS VERIFICATION MATRIX				
Req. ID	Title	Requirement	Risk Score	Status
3.4 Communications Requirements				
COM-3.01	UHF Operating Frequency	The COM UHF subsystem shall operate between (420-450 MHz) range.	4	Compliant
COM-3.02	Link Margin	The COM subsystem will establish a link margin of at least 3dB to ensure stable and robust communication.	10	Compliant
COM-3.03	Modulation Scheme	The COM subsystem will Gaussian Minimum Shift Keying (GMSK) Modulation for all communications	4	Compliant
COM-3.04	Bit Error Rate (BER)	The COM subsystem will define a maximum allowable Bit Error Rate (BER) of 10^{-5} to ensure accurate communication.	6	N/A
COM-3.06	Uplink Data Rate	The COM subsystem shall accept and process uplink data at a rate of at least 4800 bps.	3	Compliant
COM-3.07	Downlink Data Rate	The COM subsystem shall be capable of downlinking at a rate of 4800 bps.	3	Compliant

Comms – Level 4 Requirements

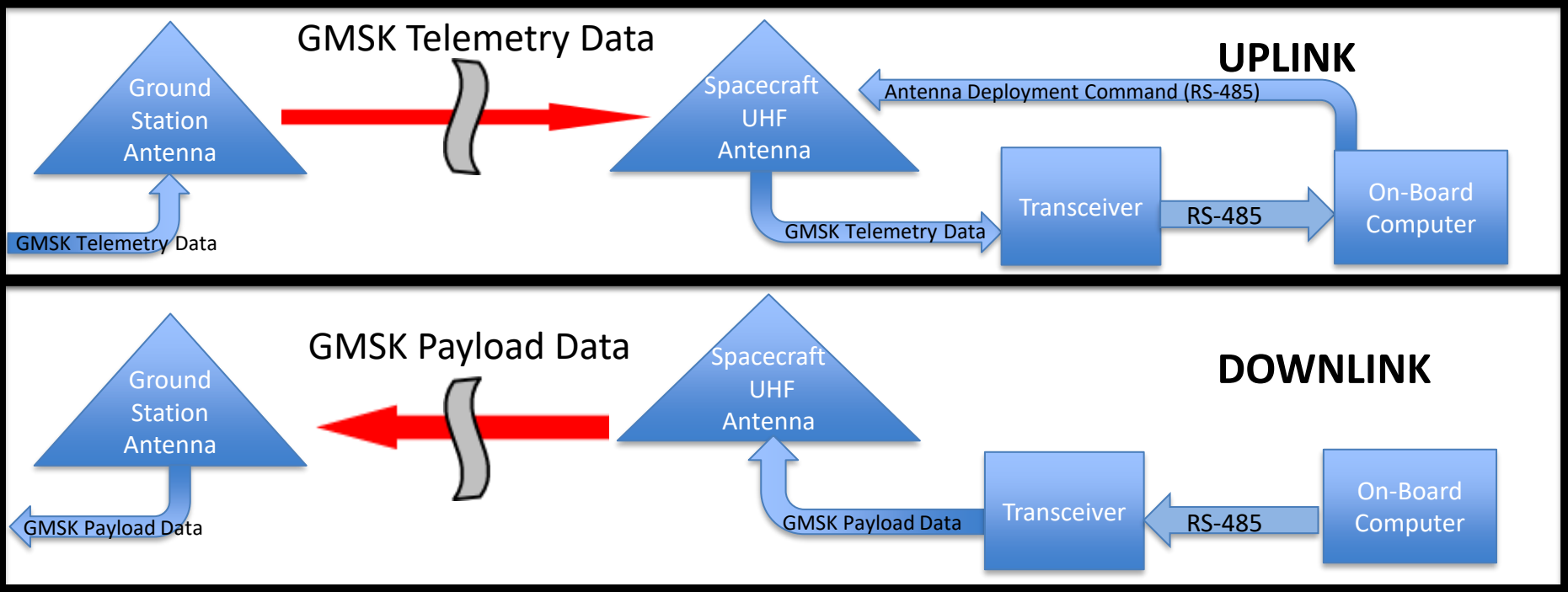
142

NUKAT REQUIREMENTS VERIFICATION MATRIX

Req. ID	Title	Requirement	Risk Score	Status
4.9 Communication Requirements				
COM-4.01	Satellite Antenna Beamwidth Gain	On-board satellite antenna will have a minimum gain of -3 dBi across its operational beamwidth.	3	N/A
COM-4.02	Satellite Antenna Worst-Case Gain	On-board satellite antenna will have a minimum gain of "X" across all attitudes	4	Not Compliant
COM-4.03	Satellite Antenna Beamwidth	On-board satellite antenna will have an operational beamwidth of 120 degrees.	3	Not Compliant
COM-4.04	Satellite Antenna Polarization	On-board satellite antenna will be linearly polarized for simplicity and redundancy	2	Compliant
COM-4.05	Forward Error Correction	The flight software will implement Reed-Solomon Error Correction to the data that is sent and received.	2	Not Compliant
COM-4.06a	Satellite Antenna Size	Antenna shall attach to the AZIMUTH face of the spacecraft and fit within 1U of space.	3	Compliant
COM-4.06b	Satellite Antenna Weight/Mass	Antenna shall require less than "X" kg of mass.	3	Compliant
COM-4.06c	Satellite Antenna Cost	Antenna hardware shall cost under "X" dollars to purchase.	3	Compliant
COM-4.07a	Transceiver Size	Transceiver dimensions shall be less than or equal to "X by X by X" mm.	3	Compliant
COM-4.07b	Transceiver Weight/Mass	Transceiver shall require less than "X" kg of mass	3	Compliant
COM-4.07c	Transceiver Power	Transceiver shall require at most 3 W of instantaneous power, but average no more than 0.5W across operational life.	3	Compliant
COM-4.07d	Transceiver Cost	Transceiver hardware shall cost under "X" dollars to purchase.	3	Compliant
COM-4.08a	Transceiver Data Rate Cap (Tx)	The transceiver downlink data rate shall be bounded to 9600 bps.	6	Compliant
COM-4.08b	Transceiver Data Rate Cap (Rx)	The transceiver uplink data rate shall be bounded to 9600 bps.	6	Compliant



Subsystem Block Diagram



Endurosat UHF Transceiver II

- Frequency: 430-440 MHz
- Power Draw: 1W (nominal) 3W (max) 0.5W (min)
- Sensitivity: -121 dBm
- Dimensions: 89 x 95 x 23.2 mm



Figure: EnduroSat UHF Transceiver II

V-Dipole Antenna (Theory)

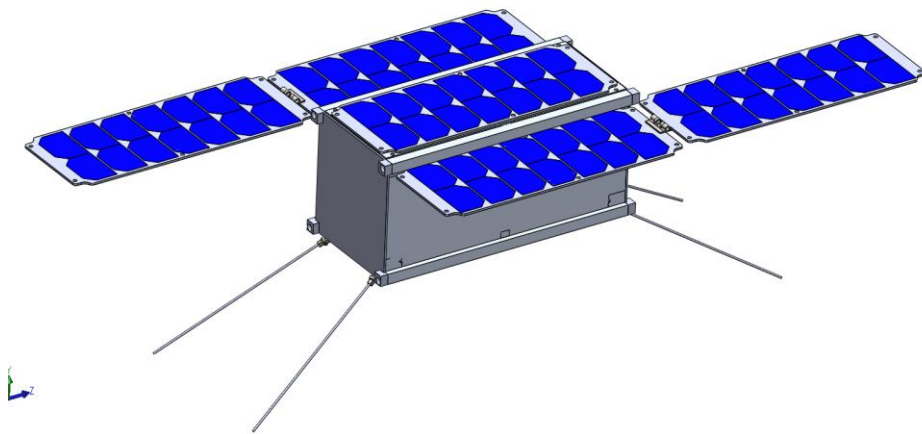
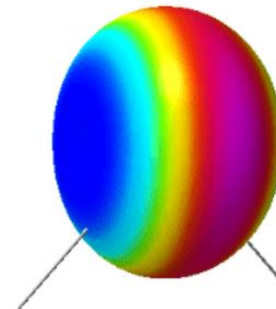


Figure: Proposed Custom Antenna

Half-Wave Resonant Inverted-V Dipole Antenna

Compiled by - Basu VU2NSB - <https://vu2nsb.com/>



Antenna in Free Space

Power Gain (TX)	:	Approx. 1.7 dBi
RDF (RX Gain)	:	1.75 dBi
Radiation Efficiency	:	Approx. 98%

Figure: V-Dipole Radiation Pattern

Antenna Course of Action #1: Fully Custom Antenna

- Utilize Electrical and Computer Engineering (ECE) Senior Design Class to design and build a custom antenna through the next Fall and Spring semester
- Complete design and plan complete by Fall 2026, hardware implementation by Spring 2027

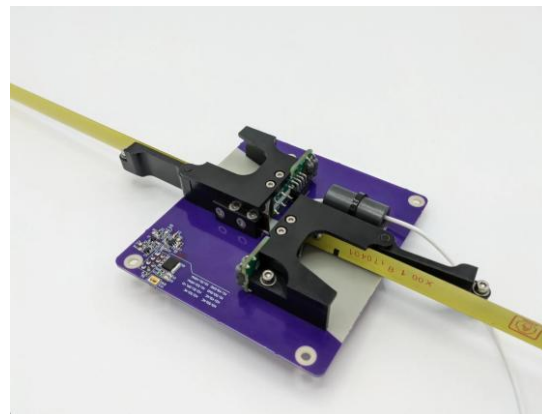


Figure: OrcaSat Custom
CubeSat Antenna

Antenna Course of Action #2: Purchase and Modify

- C3S Antenna System
 - Center frequency: 401 MHz (with +/- 15 MHz Bandwidth)
 - Polarization: Linear
 - Input port: Coaxial
 - Antenna main beam gain: 3.8 dBi
 - Deployable

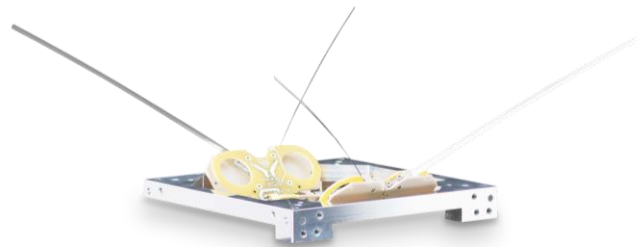


Figure: C3S V-Dipole Antenna

- Will require modification to work with 435-438 MHz

Comms – Downlink Link Budget 148

NOTE: This is calculated before FEC implementation. FEC will increase margin.

Parameter:	Value:	Units:	Comments:
Spacecraft:			
Spacecraft Transmitter Power Output:	1.0 watts		This value is transferred from "Transmitters" W/S, Cell [E50]
In dBW:	0.0	dBW	Transmitter power expressed in dB above one watt
In dBm:	30.0	dBm	Transmitter power expressed in dB above one milliwatt
Spacecraft Total Transmission Line Losses:	2.0 dB		This value is transferred from "Transmitters" W/S, Cell [I68]
Spacecraft Antenna Gain:	2.2 dBi		This value is selected at "Antenna Gain" W/S, Cell [E41]
Spacecraft EIRP:	0.2	dBW	Spacecraft Effective Isotropic Radiated Power (EIRP) [EIRP=Pt x Ltl x Ga]
Downlink Path:			
Spacecraft Antenna Pointing Loss:	0.3 dB		This value is calculated in the "Antenna Pointing Losses" W/S, and transferred from Cell [K85]
S/C-to-Ground Antenna Polarization Loss:	0.1 dB		This value is calculated in the "Polarization Loss" W/S and is transferred from Cell [F60].
Path Loss:	149.9 dB		$L_p = 22 + 20\log(D/\lambda)$; Transferred from "Frequency" W/S
Atmospheric Loss:	2.1 dB		This value is transferred from "Atmos. & Ionos. Losses" W/S, Cell [D23]
Ionospheric Loss:	0.4 dB		This value is transferred from "Atmos. & Ionos. Losses" W/S, Cell [D47:D50]
Rain Loss:	0.0 dB		This value should be estimated by the link model operator and place into Cell [B18]
Isotropic Signal Level at Ground Station:	-152.5	dBW	This is the signal level received at the Earth in the vicinity of the ground station using an omnidirectional antenna.
Ground Station (EbNo Method):			
----- Eb/No Method -----			
Ground Station Antenna Pointing Loss:	0.3 dB		This value is transferred from "Antenna Pointing Losses" W/S, Cell [K102]
Ground Station Antenna Gain:	15.5 dBi		This value is selected at "Antenna Gain" W/S, Cell [E58]
Ground Station Total Transmission Line Losses:	2.0 dB		This value is transferred from the "Receivers" W/S, Cell [J123]
Ground Station Effective Noise Temperature:	510 K		This value is calculated in the "Receivers" W/S and Transferred from Cell [J138]
Ground Station Figure of Merit (G/T):	-13.6 dB/K		$G/T = G_a - L_{ti} - 10\log(T_s)$. This is the ultimate measure of the receiver's performance.
G.S. Signal-to-Noise Power Density (S/No):	92.2	dBHz	Boltzman's Constant: -228.6 dBW/K/Hz
System Desired Data Rate:	4800 bps		Operator selects this value. Be Careful! This is the data rate, not the symbol rate.
In dBHz:	38.8	dBHz	This is simply $= 10\log(R)$; R = data rate
Telemetry System Eb/No for the Downlink:	25.4	dB	
Demodulation Method Selected:	GMSK		Values selected in "Modulation-Demodulation W/S, Cell [E30]
Forward Error Correction Coding Used:	None		Value selected in "Modulation-Demodulation" W/S, also Cell [E30]
System Allowed or Specified Bit-Error-Rate:	1.0E-05		The selected value is transferred from the "Modulation-Demodulation W/S, Cells [E33:E50]
Demodulator Implementation Loss:	0	dB	This value is transferred from the "Modulation-Demodulation W/S, Cell [E52]
Telemetry System Required Eb/No:	9.6	dB	The selected value is transferred from the "Modulation-Demodulation W/S, Cells [F33:F50]
Eb/No Threshold:	9.6	dB	This is the result of the "Modulation-Demodulation" W/S and is transferred from Cell [H32]
System Link Margin:	15.8	dB	



Comms – Uplink Link Budget

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NOTE: This is calculated before FEC implementation. FEC will increase margin.

Parameter:	Value:	Units:	Comments:
Ground Station:			
Ground Station Transmitter Power Output:	10.0	watts	This value is transferred from "Transmitters" W/S, Cell [E15].
In dBW:	10.0	dBW	Transmitter power expressed in dB above one watt
In dBm:	40.0	dBm	Transmitter power expressed in dB above one milliwatt
Ground Stn. Total Transmission Line Losses:	2.4	dB	This value is transferred from "Transmitters" W/S, Cell [I33]
Antenna Gain:	15.5	dBi	This value is selected at "Antenna Gain" W/S, Cell [E11]
Ground Station EIRP:	23.1	dBW	Ground Station Effective Isotropic Radiated Power (EIRP) [EIRP=Pt x Ltl x Ga]
Uplink Path:			
Ground Station Antenna Pointing Loss:	0.3	dB	This value is calculated in the "Antenna Pointing Losses" W/S, and transferred from Cell [K43]
Gnd-to-S/C Antenna Polarization Losses:	0.1	dB	This value is calculated in the "Polarization Loss" W/S and is transferred from Cell [F40].
Path Loss:	149.9	dB	Lp = 22 + 20LOG(D/R); Transferred from "Frequency" W/S
Atmospheric Losses:	2.1	dB	This value is transferred from "Atmos. & Ionos. Losses" W/S, Cell [D23]
Ionospheric Losses:	0.4	dB	This value is transferred from "Atmos. & Ionos. Losses" W/S, Cell [D47:D50]
Rain Losses:	0.0	dB	This value should be estimated by the link model operator and place into Cell [B18]
Isotropic Signal Level at Spacecraft:	-129.6	dBW	This is the signal level received in space in the vicinity of the spacecraft using an omnidirectional antenna.
Spacecraft (Eb/No Method):			
----- Eb/No Method -----			
Spacecraft Antenna Pointing Loss:	0.3	dB	This value is transferred from "Antenna Pointing Losses" W/S, Cell [K63]
Spacecraft Antenna Gain:	2.2	dBi	This value is selected at "Antenna Gain" W/S, Cell [E24]
Spacecraft Total Transmission Line Losses:	2.0	dB	This value is transferred from the "Receivers" W/S, Cell [J52]
Spacecraft Effective Noise Temperature:	269	K	This value is calculated in the "Receivers" W/S and Transferred from Cell [J67]
Spacecraft Figure of Merit (G/T):	-24.1	dB/K	G/T = Ga-Lti-10log(Ts). This is the ultimate measure of the receiver's performance.
S/C Signal-to-Noise Power Density (S/No):	74.6	dBHz	Boltzman's Constant: -228.6 dBW/K/Hz
System Desired Data Rate:	4800	bps	Operator selects this value. Be Careful! This is the data rate, not the symbol rate.
In dBHz:	36.8	dBHz	This is simply = 10log(R); R= data rate
Command System Eb/No:	37.8	dB	
Demodulation Method Selected:	GMSK		Values selected in "Modulation-Demodulation W/S, Cell [E3]
Forward Error Correction Coding Used:	None		Value selected in "Modulation-Demodulation" W/S, also Cell [E3]
System Allowed or Specified Bit-Error-Rate:	1.0E-05		The selected value is transferred from the "Modulation-Demodulation W/S, Cells [E6:E23]
Demodulator Implementation Loss:	1.0	dB	This value is transferred from the "Modulation-Demodulation W/S, Cell[E25]
Telemetry System Required Eb/No:	9.6	dB	The selected value is transferred from the "Modulation-Demodulation W/S, Cells [F6:F23]
Eb/No Threshold:	10.6	dB	This is the result of the "Modulation-Demodulation" W/S and is transferred from Cell [H32]
System Link Margin:	27.2	dB	



Requirement Compliance

- The communication subsystem is non-compliant with COM 4.02 (Worst-case Gain), COM 4.03 (Beamwidth), and COM 4.05 (Reed Solomon)

Subsystem Compatibility

- The communication subsystem is compatible with other systems

Communications has a Path Toward CDR

Communication Subsystem is Capable of Supporting NuKAT Mission

In light of this, communication should proceed toward CDR with further effort to complete antenna design and implementation.

COMMAND & DATA HANDLING SUBSYSTEM

Logan Hayward – Chief Embedded Systems Engineer

Computer Science | Spring 2026

Nalen Rangarajan – C&DH Lead

Computer Science | Spring 2026



Investigating On-Board Computer

- Trade Study pending

Improvements to Data Budget

- Addition of FEC to Data Budget estimates
- Investigating need to compress data

Interfacing with other subsystems

C&DH – Level 3 Requirements

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NUKAT REQUIREMENTS VERIFICATION MATRIX				
Req. ID	Title	Requirement	Risk Score	Status
3.9 Command and Data Handling Requirements				
CDH-3.01	C&DH OBC Ownership	The C&DH subsystem shall own and specify the On-Board Computer (OBC).	5	Partial Compliance
CDH-3.02	C&DH + Comms Subsystem Interface	The C&DH subsystem shall interface with the Communications (Comms) subsystem.	5	Compliant
CDH-3.03	C&DH + EPD Subsystem Interface	The C&DH subsystem shall interface with the Electrical and Power Distribution (EPD) subsystem.	5	Compliant
CDH-3.04	C&DH + Payload Subsystem Interface	The C&DH subsystem shall interface with the Payload.	5	Partial Compliance
CDH-3.05	C&DH + ADCS Subsystem Interface	The C&DH subsystem shall interface with the Attitude Determination and Controls (ADCS) subsystem.	5	Compliant
CDH-3.06	C&DH + Thermal Subsystem Interface	The C&DH subsystem shall interface with the Thermal subsystem.	5	Compliant
CDH-3.07	External Data Framing	The C&DH subsystem shall support CCSDS for transmitting messages over RF up and down.	2	Compliant
CDH-3.08	Data Encoding	The C&DH subsystem shall support [TBD protocol] for encrypting payload data.	1	Partial Compliance
CDH-3.09	Data Compression	The C&DH subsystem shall support [TBD compression algorithm] for packaging data for RF communications.	3	Partial Compliance
CDH-3.10	Internal Data Framing	The C&DH subsystem shall define internal data framing protocols for any subsystem without Interface Control Documents.	3	Partial Compliance
CDH-3.11	Data Storage	The C&DH subsystem shall be capable of storing up to [2 x (48 hour data collection amount - TBD)] at a time in non-volatile memory over the mission lifetime.	12	Compliant

C&DH – Level 4 Requirements

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NUKAT REQUIREMENTS VERIFICATION MATRIX				
Req. ID	Title	Requirement	Risk Score	Status
4.4 Command and Data Handling Requirements				
CDH-4.01.1	OBC Size	The OBC shall require less than "x" U of space.	5	Compliant
CDH-4.01.2	OBC Weight	The OBC shall require less than "x" kg of mass.	5	Compliant
CDH-4.01.3	OBC Power	The OBC shall use between 0-4 Watts of power.	25	Not Compliant
CDH-4.02	OBC + UHF TxRx Interface	The OBC shall interface with the Comms UHF transceiver via RS-485.	5	Compliant
CDH-4.02	OBC + EPD Network Interface	The OBC shall interface with the EPD PowerDrive via RS-485.	5	Compliant
CDH-4.03	OBC + DSC Domino Payload Network Interface	The OBC shall interface with the DSC Domino Payload via [I2C or SPI ideally, currently supports USB-C].	25	Not Compliant
CDH-4.04.1	OBC + Sun Sensors Interface	The OBC shall interface with the ADCS Sun Sensors via RS-485.	2	Compliant
CDH-4.04.2	OBC+ Magnetometer Interface	The OBC shall interface with the ADCS Magnetometer via I2C.	4	Compliant
CDH-4.04.3	OBC + Electromagnet Interface	The OBC shall interface with the ADCS Electromagnet via 2 pin PWM.	5	Compliant
CDH-4.05.1	OBC + Thermistor Interface	The OBC shall interface with the Thermal Thermistors via an ADC pin using a ["x" pins] MUX.	4	Compliant
CDH-4.05.2	OBC + Thermal Network Interface	The OBC shall interface with the Thermal Heaters via 2 pin PWM.	5	Compliant

Component/Subsystem	Hardware Interface
UHF Transceiver	RS-485
EPD Computer	RS-485
DSC Domino	Ideally I2C or SPI / Currently USB-C
ADCS Subsystem	RS-485, I2C, PWM
Thermal Subsystem	ADC + MUX, PWM

C&DH – OBC Decision Matrix

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Criteria	Weighting 1 to 10 (Criteria Importance)	Nextage NMC2	Raspberry Pi Zero 2	Endurosat OBC	C3S OBC	Steel MOBC
Power	10	 -1	1	0	1	0
Cost	10	1	1	-1	 0	0
Company Heritage	8	1	1	1	1	1
Rad Hardening	8	1	-1	1	 0	1
Engagement	7	1	1	1	1	0
Engineering Support	7	1	-1	1	 0	0
Export Controls	5	1	1	1	0	1
Flight Heritage	4	-1	1	1	1	1
Availability	4	1	1	1	 0	0
Size	1	1	1	1	1	1
Weight	1	1	1	1	1	1
Total		37	35	35	31	27
Ranking		1st	2nd	2nd	4th	5th

Rank:



1 (Pro)



0 (Neutral)



-1 (Con)



More Research Necessary



KANSAS STATE
UNIVERSITY

Carl R. Ice
College of Engineering

Logan Hayward

Criteria	Nextage NMC2
Power	-1
Cost	1
Company Heritage	1
Rad Hardening	1
Engagement	1
Engineering Support	1
Export Controls	1
Flight Heritage	-1
Availability	1
Size	1
Weight	1
Total	37
Ranking	1st

Pros:

- Great company heritage with 20+ space missions (JPSS, Landsat 9, DART, and more)
- Supplied for free with Nextage engineering support

Cons:

- No flight heritage for new product
- Large power draw
 - 4W (Nominal) 10W (Max Compute)

Other:

- Can be viewed as a secondary tech-demo payload
- NMC2 undergoing redesign to reduce power draw
- Exploring collaboration agreement



Nextage NMC2
(+3 daughter boards)

Command and Data Handling



Nextage NMC2
(+3 daughter boards)

- Designed with radiation best practices
 - Automotive grade parts
 - Component placement
 - Rigorous testing
- Operating Temperature: -40C - +85C
- Processor Information
 - 1 FPGA
 - 4 Risc V Application Cores
 - 1 Risc V Monitor Core

} Programmable via TriCortex and Verilog
} Programmable via C/C++ and Python
- 400+ GPIO Pins
- 500 GB SSD → 320 GB w/ ECC
- 16 GB DDR4 RAM

Criteria	Raspberry Pi Zero 2
Power	1
Cost	1
Company Heritage	1
Rad Hardening	-1
Engagement	1
Engineering Support	-1
Export Controls	1
Flight Heritage	1
Availability	1
Size	1
Weight	1
Total	35
Ranking	2nd

Pros:

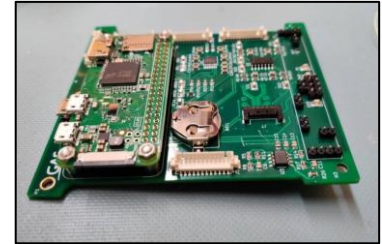
- Flight heritage on NASA CSLI + Utah State 1U GASPACS
- Small SWaP-C factors
 - 0.5W (Nominal) 3W (Max Compute)
- Leaves budget margins to design around risks

Cons:

- Must design own integration board
- Must design for radiation tolerance
- Need to determine support for interfaces

Other:

- More research needs to go into this board before decision is made



Raspberry Pi Zero 2
(+ Custom Integration Board)

Data Per Domino Sample			
Field	Type	Bits	Bytes
Background Detector 1	uint32	32.00	4.00
Background Detector 2	uint32	32.00	4.00
Fast Detector	uint32	32.00	4.00
Thermal Detector	uint32	32.00	4.00
Temperature	float (signed)	32.00	4.00
Total Data Per Sample	---	160.00	20.00

Data Per Sample Cycle			
Field	Type	Bits	Bytes
Latitude	float (signed)	32.00	4.00
Longitude	float (signed)	32.00	4.00
Time (UNIX)	uint32	32.00	4.00
Altitude	float (signed)	32.00	4.00
Earth Normal Angle	int8	8.00	1.00
Total Data Per Sample	---	136.00	17.00

4 Detectors -> 776 bits or 97 bytes of data for each sample cycle

Data Framing Standard

Field	Data Type	Bits	Bytes
Version Number	3-bit uint	3	0.375
Packet Type	1-bit bool	1	0.125
Secondary Header Flag	1-bit bool	1	0.125
APID	11-bit uint	11	1.375
Sequence Flags	2-bit enum	2	0.25
Sequence Count	14-bit uint	14	1.75
Packet Data Length	16-bit uint / ushort	16	2
Payload Data	---	$136 + 160 * \text{Detector Count}$	$17 + 20 * \text{DC}$
Checksum	32-bit uint	32	4
Header Total	---	80	10

- Data Framing Standard
 - Based on CCSDS Standard
- 1 packet per payload sample
- 10% margin when running 4 detectors

C&DH – Data Budget

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4 Detectors, 1/2 FEC Rate

Frequency (Hz)	Sample Length (s)	4 Dominos (bits / day)	4 Dominos (KB / day)	4 Dominos (KB / day * 2.6 mrg)	4800 bps Compliant?	Time to Downlink Data (secs)	Number of times a full downlink can be attempted in a day (12 min)	Number of Missed Passes Before Needing Compression	9600 bps Compliant?	Time to Downlink Data (secs)	Number of times a full downlink can be attempted in a day (12 min)	Number of Missed Passes Before Needing Compression
0.000555556	1800	41,088	5.14	13.35	Yes	22.256	32.35	32	Yes	11.128	64.70	65
0.001111111	900	82,176	10.27	26.71	Yes	44.512	16.18	16	Yes	22.256	32.35	32
0.001666667	600	123,264	15.41	40.06	Yes	66.768	10.78	11	Yes	33.384	21.57	22
0.003333333	300	246,528	30.82	80.12	Yes	133.536	5.39	5	Yes	66.768	10.78	11
0.008333333	120	616,320	77.04	200.30	Yes	333.84	2.16	2	Yes	166.92	4.31	4
0.017	60	1,232,640	154.08	400.61	Yes	667.68	1.08	1	Yes	333.84	2.16	2
0.033	30	2,465,280	308.16	801.22	No	1335.36	0.54	1	Yes	667.68	1.08	1
0.05	20	3,697,920	462.24	1,201.82	No	2003.04	0.36	0	No	1001.52	0.72	1
0.1	10	7,395,840	924.48	2,403.65	No	4006.08	0.18	0	No	2003.04	0.36	0
0.2	5	14,791,680	1,848.96	4,807.30	No	8012.16	0.09	0	No	4006.08	0.18	0
0.5	2	36,979,200	4,622.40	12,018.24	No	20030.4	0.04	0	No	10015.2	0.07	0
1	1	73,958,400	9,244.80	24,036.48	No	40060.8	0.02	0	No	20030.4	0.04	0
2	0.5	147,916,800	18,489.60	48,072.96	No	80121.6	0.01	0	No	40060.8	0.02	0
5	0.2	369,792,000	46,224.00	120,182.40	No	200304	0.00	0	No	100152	0.01	0
10	0.1	739,584,000	92,448.00	240,364.80	No	400608	0.00	0	No	200304	0.00	0

30% general margin + 2.0 encoding margin multiplier -> *2.6 margin



Assuming 12 minute link time	
Downlinkable bits per day (9600 bps)	6,912,000
Downlinkable KB per day (9600 bps)	864.00
Downlinkable bits per day (4800 bps)	3,456,000
Downlinkable KB per day (4800 bps)	432.00

- Daily Downlink Capacity is used to justify certain data schemes
- Adaptive Compression for missed passes
 - Ex: Assume 4 detectors, 60s sample length, 4800 bps, and 1/2 FEC rate
 - Missing 2 consecutive passes would cause the system to compress data until we reach a point at which all data can be downlinked in one complete pass

Requirement Compliance

- Non-compliant with C&DH 4.01.3 (OBC Power Budget).
- Non-compliant with C&DH 4.03 (OBC + Payload Interface)

Subsystems Compatibility

- The Command and Data Handling subsystem is compatible pending the closure of C&DH 4.01.3 (OBC Power Budget) and C&DH 4.03 (OBC + Payload Interface).
- Other partially compliant requirements are pending review, need further investigation, and are expected to close.

Command and Data Handling has a Path Toward CDR

- C&DH 4.01.3 (OBC Power Budget) is pending updated Nextage information and OBC trade study and selection.
- C&DH 4.03 (OBC + Payload Interface) is pending I2C/SPI response from RDT. Alternative solutions are possible.

The Command and Data Handling Subsystem is Capable of Supporting NuKAT Mission

In light of this, the command and data handling subsystem should proceed toward CDR with further effort to solidify our OBC trade.

FLIGHT SOFTWARE SUBSYSTEM

Braden Adams – Flight Software Lead

Computer Science | Spring 2026



Software Architecture

- Utilizing FPGA Technology
- Defining core Flight System applications

Testing Plan

- Needs more work but progress was made

Hardware Interactions

- Interfacing with subsystems

FSW – Level 3 Requirements

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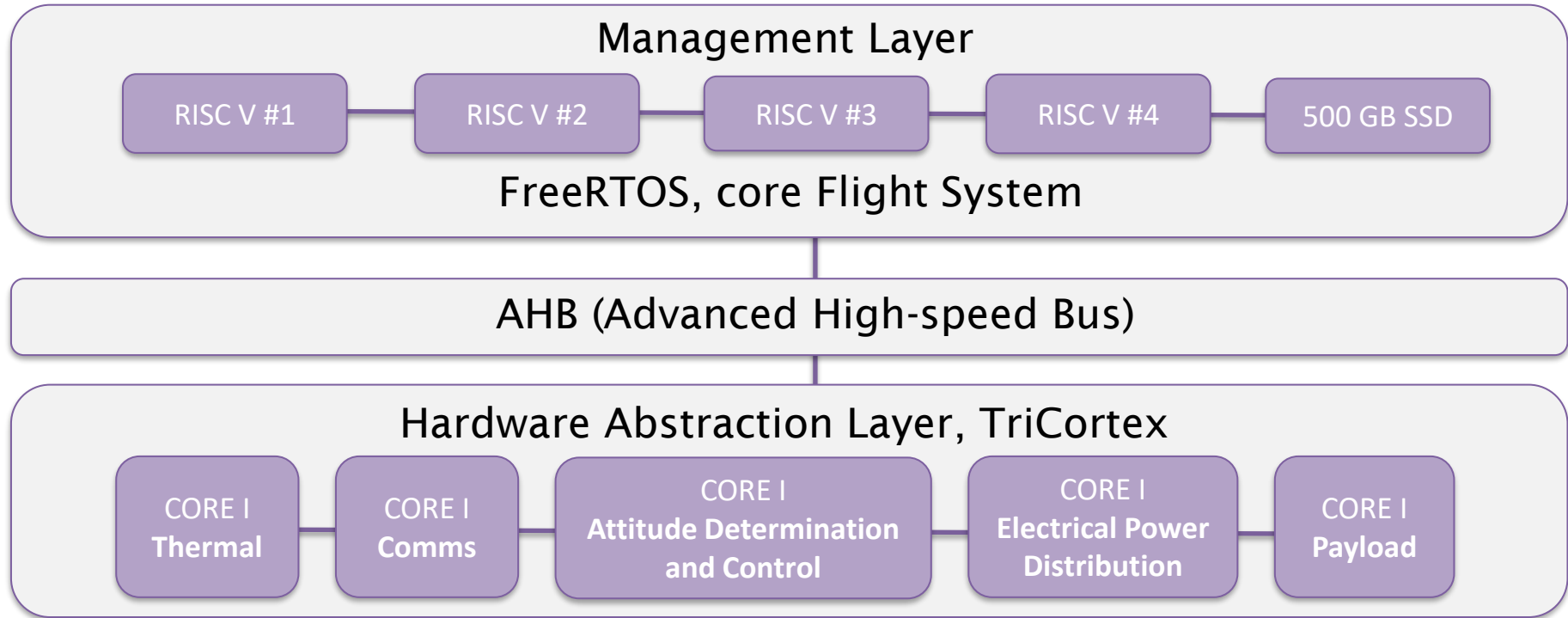
Req. ID	Title	Requirement	Risk Score	Status
3.6 Flight Software Requirements				
FSW-3.01	Telemetry Collection	FSW shall collect telemetry from all hardware onboard NuKAT.	10	Compliant
FSW-3.02	Event Logging	FSW shall maintain a log of software events related to command execution, resets, and software errors.	6	Compliant
FSW-3.03	Ground Software Interface	FSW shall be able to receive, decrypt, and interpret commands from the Ground Station Software.	15	Compliant
FSW-3.04	Power Reset Actions	FSW shall immediately command NuKAT back to a normal state following a power reset.	10	Compliant
FSW-3.05	Health Data Transmission	FSW shall relay the instantaneous health status of NuKAT to the ground upon requesting command.	6	Compliant
FSW-3.06	Neutron Count Collection Time	FSW shall be capable of tracking what time and date each neutron count reading is collected.	3	Compliant
FSW-3.07	Location Tracking	FSW shall be capable of tracking NuKAT's physical position in orbit to appropriately associate neutron count readings.	6	Compliant
FSW-3.08	Operational Modes	FSW shall be capable of manipulating and tracking the operational modes for NuKAT and change between modes when necessary.	12	Compliant
FSW-3.09	Error Checking/Correction	FSW shall be capable of detecting and correcting errors caused by SEUs and lossy data transmission	9	Compliant

FSW – Level 4 Requirements

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Req. ID	Title	Requirement	Risk Score	Status
4.6 Flight Software Requirements				
FSW-4.01	Onboard Computer Utilization	The flight software shall require an average utilization of under 50% of the OBC with peak utilization under 80%.	8	Compliant
FSW-4.02	Software Code Count	The flight software shall not exceed "X" lines of custom code.	3	Compliant
FSW-4.03	Decryption Algorithm	The flight software shall implement AES-256 decryption on all uplinked commands using the mbedTLS open-source library with keys stored on SSD.	3	Compliant
FSW-4.04	Software Package	The flight software shall be implemented using the C, C++, Python, and TriCortex real-time AI languages.	1	Compliant
FSW-4.05	Software Structure	The flight software shall utilize NASA's core Flight System (cFS) framework.	4	Compliant

- **RISC-V Cores (× 4)**
 - Physical processors, built into chip
 - Run Linux and FreeRTOS and core Flight System
- **CORE I Processors (× 100+) (Tricortex Computer, FPGA)**
 - Software-defined processors, programmed into the FPGA fabric
 - Each dedicated to one hardware device
 - All run simultaneously in parallel
- **AHB Bus**
 - Internal connection between RISC-V and CORE I
 - CORE I writes sensor data, RISC-V reads it



NASA core Flight System (cFS)

- cFS enables easy reuse, rapid development, and portability
 - Dynamic runtime environment
 - Layered architecture
 - Component-based design
- 3 main components
 - Platform Support Package
 - Operating System Abstraction Layer
 - Core Flight Executive
- Used by 40+ small to large missions, including the Roman Space Telescope.

Executive Services (ES)

Manages app startup,
shutdown, and resets

Software Bus (SB)

Routes messages
between apps

Event Services (EVS)

Logs and reports system
events and faults

Table Services (TBL)

Manages unloadable
configuration tables

Time Services (TIME)

Manages satellite clock
and time sync

FSW - Stock Apps

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CCSDS File Delivery Protocol App (CF)

Checksum App (CS)

File Data Store App (DS)

File Manager App (FM)

Housekeeping App (HK)

Health and Safety App (HS)

Limit Checker App (LC)

Memory Dwell App (MD)

Memory Manager (MM)

Scheduler Lab App (sch_lab)

Stored Command App (SC)

Mode Manager

Determines and tracks operational modes

Thermal

Monitors temperatures and controls heaters

EPD

Monitors battery and charge

Payload

Collect neutron data, package for downlink, perform any needed on-board processing/data cleanup

ADC

Interfaces with and monitors ADC integrated software

Comms

Monitors and controls communication system (uplinks and downlinks)

- cFS handles hardware interaction without FPGA
 - HWLIB (Hardware Library)

- Integration testing
 - Verify that NuKAT's Flight Software subsystems communicate and behave correctly
 - Meet NASA testing standards
- Test environment
 - "Flat Sat"
 - cFS running natively on PolarFire SoC

Requirement Compliance

- All requirements are compliant

Subsystems Compatibility

- FSW Design will be slightly altered with a change in On-Board Computer selection.
- Otherwise, FSW is compatible with other subsystems.

Flight Software has a Path Toward CDR

Flight Software Subsystem is Capable of Supporting NuKAT Mission

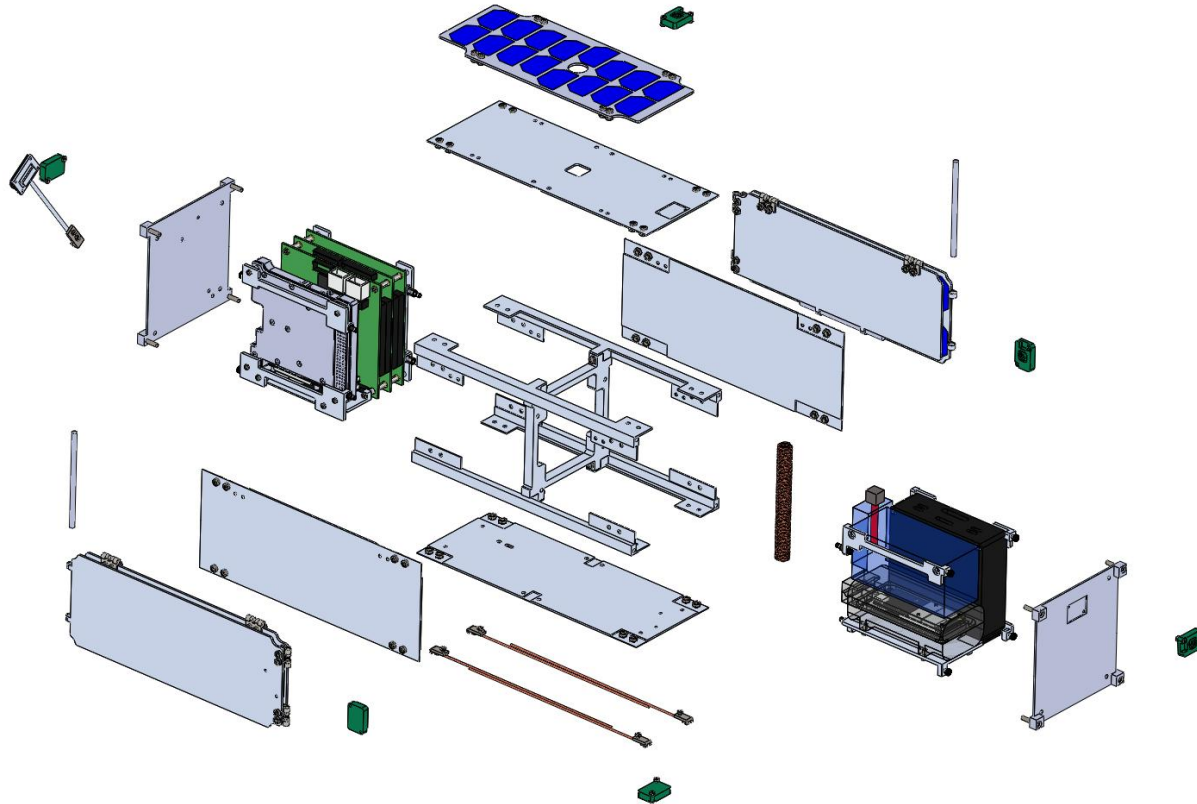
In light of this, Flight Software should proceed toward CDR with further effort to solidify testing plan and confirm design in accordance with OBC selection.

BACKUP SLIDES – DAY TWO



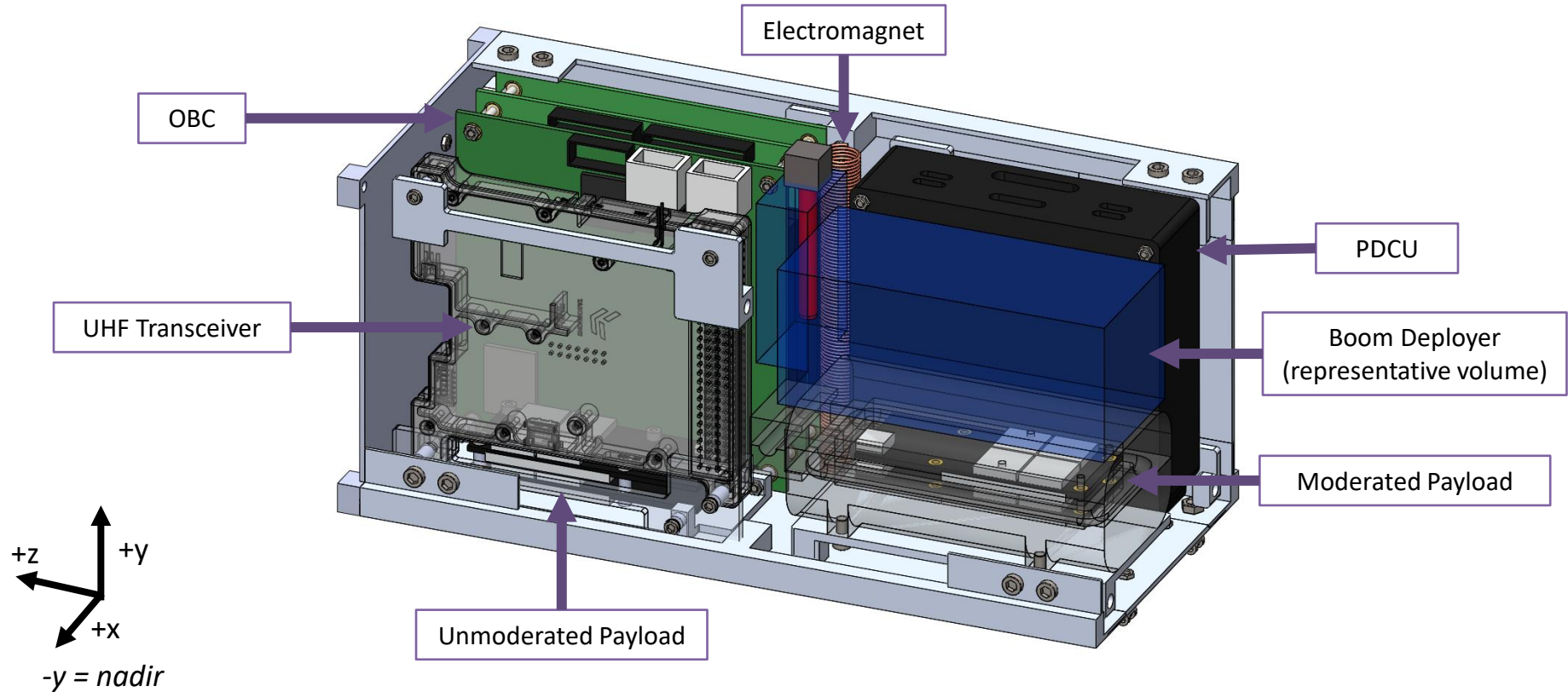
Final Assembly: Exploded

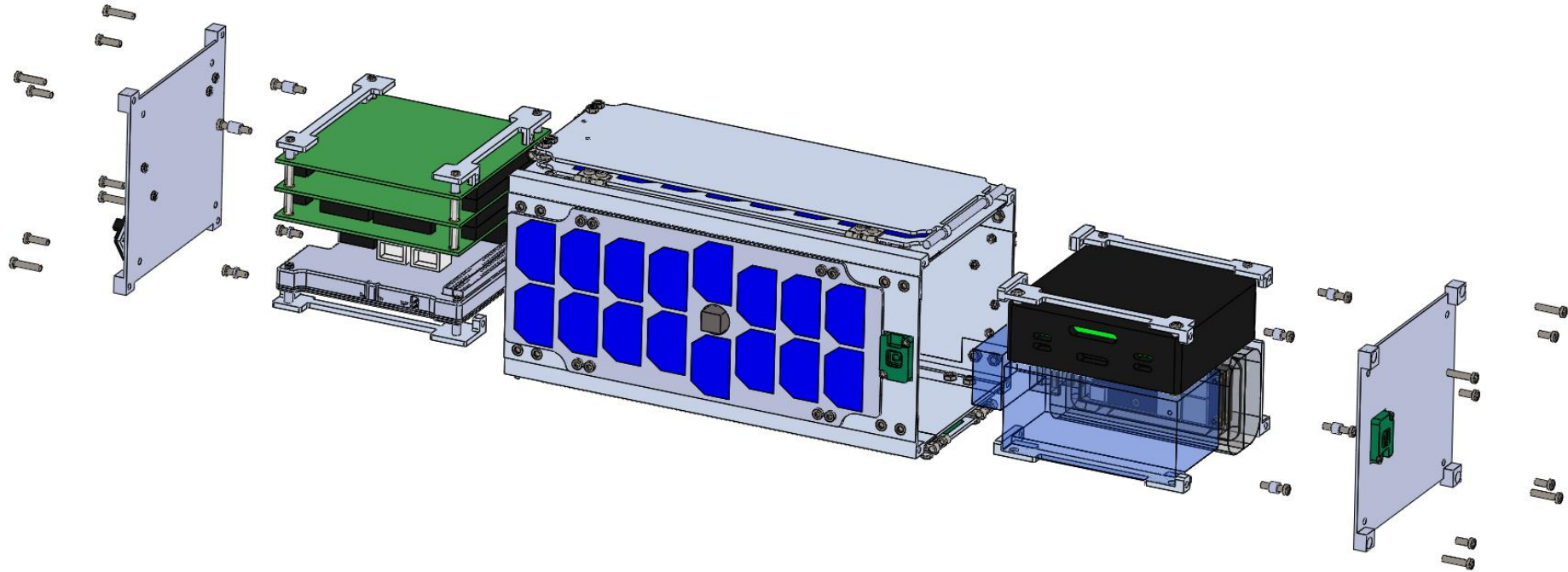
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Internal Hardware Layout

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Electrical Power & Distribution

COTS Testing Standards:

- PowerDrive/PowerBank

TVAC	2 Cycles with temperature range of +60°C to -25°C
Vibration	Random vibration acceptance test levels 20 - 2000 Hz at 4.47grms for 60 seconds
Shock	100 Hz at 30g 1000 Hz at 1500g 5000 Hz at 1500g
Mission Life Expectancy	Up to 5 Years in LEO

- PowerPlane

ETL: TVAC, EMI/EMC, Vibration and Shock Tolerance	ETL complete for ISRO PSLV profile for solar cells. Panel can be tested based on customer requirements.
	Tested for 13g (vibration), 1500g (shock) ISRO PSLV profile
Mission Life Expectancy	5 Years

Mass Moments of Inertia

186

Reported using Solidworks Mass Properties evaluation on most updated CAD model.

STOWED

Principal axes of inertia and principal moments of inertia ($\text{kg}\cdot\text{mm}^2$) at CM:

$$\begin{aligned} I_x &= (-0.02, 0.01, 1.00) & P_x &= 5292.92 \\ I_y &= (1.00, 0.01, 0.02) & P_y &= 12014.32 \\ I_z &= (-0.01, 1.00, -0.01) & P_z &= 12481.08 \end{aligned}$$

Moments of inertia($\text{kg}\cdot\text{mm}^2$) at CM (Using positive tensor notation):

$$\begin{aligned} I_{xx} &= 12012.45 & I_{xy} &= 1.74 & I_{xz} &= -112.44 \\ I_{yx} &= 1.74 & I_{yy} &= 12480.86 & I_{yz} &= 38.89 \\ I_{zx} &= -112.44 & I_{zy} &= 38.89 & I_{zz} &= 5295.01 \end{aligned}$$

ALL APPENGEGES DEPLOYED (excluding GG boom)

Principal axes of inertia and principal moments of inertia ($\text{kg}\cdot\text{mm}^2$) at CM:

$$\begin{aligned} I_x &= (-0.28, 0.01, 0.96) & P_x &= 7052.02 \\ I_y &= (0.96, 0.04, 0.28) & P_y &= 24202.79 \\ I_z &= (-0.04, 1.00, -0.02) & P_z &= 25671.45 \end{aligned}$$

Moments of inertia($\text{kg}\cdot\text{mm}^2$) at CM (Using positive tensor notation):

$$\begin{aligned} I_{xx} &= 22865.25 & I_{xy} &= 15.86 & I_{xz} &= -4603.80 \\ I_{yx} &= 15.86 & I_{yy} &= 25667.01 & I_{yz} &= 179.64 \\ I_{zx} &= -4603.80 & I_{zy} &= 179.64 & I_{zz} &= 8394.00 \end{aligned}$$

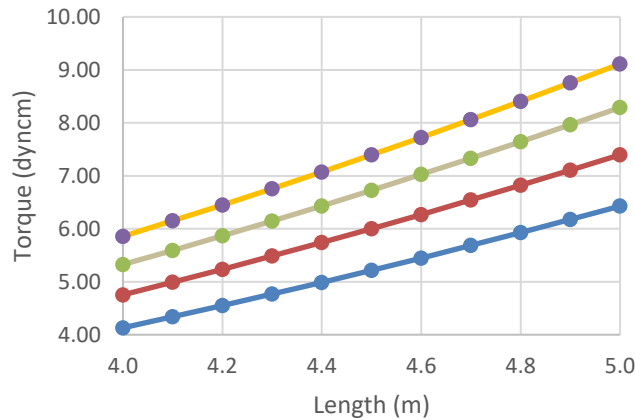


Boom and Mass

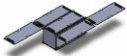
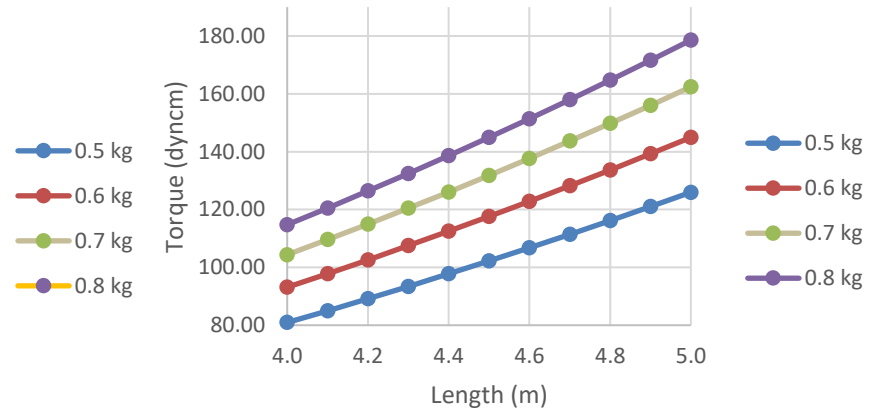
$$\tau = 3\Omega^2(I_{yy} - I_{xx})\sin\theta$$

$$\Omega^2 = \frac{Gm_E}{r^3}$$

Torque Generated at $\theta = 1^\circ$



Torque Generated at $\theta = 20^\circ$



Sample Calculations

Mass 0.5 kg					
Length (m)	I_{xx}	I_{yy}	$I_{xx} - I_{yy}$	Torque at 1°(dyncm)	Torque at 20°(dyncm)
4.0	6.446335	0.0265923	6.4197427	4.13	80.94
4.1	6.768008	0.0265923	6.7414157	4.34	85.00
4.2	7.0975385	0.0265923	7.0709462	4.55	89.15
4.3	7.4349267	0.0265923	7.4083344	4.77	93.41
4.4	7.7801725	0.0265923	7.7535802	4.99	97.76
4.5	8.1332759	0.0265923	8.1066836	5.22	102.21
4.6	8.4942369	0.0265923	8.4676446	5.45	106.76
4.7	8.8630555	0.0265923	8.8364632	5.69	111.41
4.8	9.2397317	0.0265923	9.2131394	5.93	116.16
4.9	9.6242656	0.0265923	9.5976733	6.17	121.01
5.0	10.0166571	0.0265923	9.9900648	6.43	125.96

Constants	
m_E (kg)	5.97E+24
r	6871000
G	6.67E-11
Omega^2	1.23E-06
Constants (1°)	6.43E-08
Constants (20°)	1.26E-06

Mass 0.5 kg		
Length(m)	Calculated Torque (1°)	Calculated Torque (20°)
1	-2.80	-54.85
2	-0.50	-9.83
3	1.80	35.18
4	4.09	80.20
5	6.39	125.21
6	8.69	170.23
7	10.98	215.24
8	13.28	260.26
9	15.58	305.27
10	17.87	350.29

Mass (kg)	Slopes (dyncm/m)		Intercepts	
	1°	20°	1°	20°
0.5	2.297	45.015	-5.096	-99.863
0.6	2.643	51.799	-5.863	-114.908
0.7	2.962	58.047	-6.571	-128.766
0.8	3.257	63.821	-7.224	-141.573

Components

- Boom and mass
- 6 Sun sensors
- 3-Axis Magnetometer
- Electromagnet
- 4 Hysteresis rods

Analyses and Budgets

- Sun Sensors
 - 3.3 V
 - 120 deg FOV
 - 5 deg of accuracy
- Magnetometer
 - 3.3 V or 5 V
 - Must be placed 10cm away from a strong magnetic field
 - Sensitivity is 10nT and dynamic range of +/- 800 uT
- Electromagnet
 - 5 V
 - Dipole moment of 0.2 Am²

Components – Electromagnet

- TensorMTQ-2
- Size: 20x20x150 mm³
- Weight: 130 g
- Power: 173 mW
- Cost: Unknown



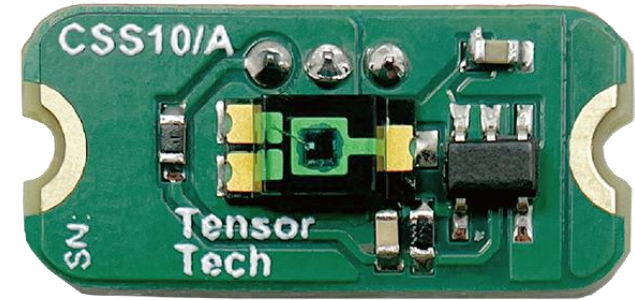
Components – Hysteresis Rods

- Size: 4x4x80 mm³
- Weight (1): 1-3 g
- Power: N/A
- Cost: Unknown



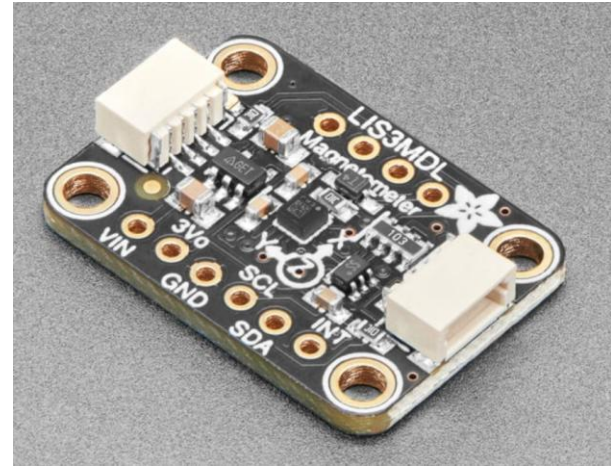
Components – Sun Sensors

- TensorCSS-10 (x6)
- Size: $15 \times 7 \times 6 \text{ mm}^3$
- Weight (1): 0.5 g
- Power (1): 0.33 mW
- Cost: Unknown



Components – Magnetometer

- Adafruit Triple-axis Magnetometer
- Size: 25.7x17.8x4.6 mm³
- Weight: 1.5 g
- Power: N/A
- Cost: \$9.95



- Magnetic Control with Momentum Bias
 - 3-axis magnetometer
 - 6 sun sensors
 - 3-axis torque coils
 - Variable spin rate momentum wheel
 - On-board computational resource
 - Dynamic simulation of spacecraft control

C&DH – Data Budget

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1 Detector, 1/2 FEC Rate

Frequency (Hz)	Sample Length (s)	1 Domino (bits / day)	1 Domino (KB / day)	1 Domino (KB / day * 2.6 mrg)	4800 bps Compliant?	Time to Downlink Data (secs)	Number of times a full downlink can be attempted in a day (12 min)	Number of Missed Passes Before Needing To Look At Compression	9600 bps Compliant?	Time to Downlink Data (secs)	Number of times a full downlink can be attempted in a day (12 min)	Number of Missed Passes Before Needing Compression
0.000555556	1800	18,048	2.26	5.87	Yes	9.78	73.65	74	Yes	4.888	147.30	147
0.001111111	900	36,096	4.51	11.73	Yes	19.55	36.82	37	Yes	9.776	73.65	74
0.001666667	600	54,144	6.77	17.60	Yes	29.33	24.55	25	Yes	14.664	49.10	49
0.003333333	300	108,288	13.54	35.19	Yes	58.66	12.27	12	Yes	29.328	24.55	25
0.008333333	120	270,720	33.84	87.98	Yes	146.64	4.91	5	Yes	73.32	9.82	10
0.017	60	541,440	67.68	175.97	Yes	293.28	2.45	2	Yes	146.64	4.91	5
0.033	30	1,082,880	135.36	351.94	Yes	586.56	1.23	1	Yes	293.28	2.45	2
0.05	20	1,624,320	203.04	527.90	No	879.84	0.82	1	Yes	439.92	1.64	2
0.1	10	3,248,640	406.08	1,055.81	No	1759.68	0.41	0	No	879.84	0.82	1
0.2	5	6,497,280	812.16	2,111.62	No	3519.36	0.20	0	No	1759.68	0.41	0
0.5	2	16,243,200	2,030.40	5,279.04	No	8798.40	0.08	0	No	4399.2	0.16	0
1	1	32,486,400	4,060.80	10,558.08	No	17596.80	0.04	0	No	8798.4	0.08	0
2	0.5	64,972,800	8,121.60	21,116.16	No	35193.60	0.02	0	No	17596.8	0.04	0
5	0.2	162,432,000	20,304.00	52,790.40	No	87984.00	0.01	0	No	4399.2	0.02	0
10	0.1	324,864,000	40,608.00	105,580.80	No	175968.00	0.00	0	No	8798.4	0.01	0

30% general margin + 2.0 encoding margin multiplier -> *2.6 margin



C&DH – Data Budget

197

2 Detectors, 1/2 FEC Rate

Frequency (Hz)	Sample Length (s)	2 Dominos (bits / day)	2 Dominos (KB / day)	2 Dominos (KB / day * 2.6 mrg)	4800 bps Compliant?	Time to Downlink Data (secs)	Number of times a full downlink can be attempted in a day (12 min)	Number of Missed Passes Before Needing Compression	9600 bps Compliant?	Time to Downlink Data (secs)	Number of times a full downlink can be attempted in a day (12 min)	Number of Missed Passes Before Needing Compression
0.000555556	1800	25,728	3.22	8.36	Yes	13.94	51.66	52	Yes	6.968	103.33	103
0.001111111	900	51,456	6.43	16.72	Yes	27.87	25.83	26	Yes	13.936	51.66	52
0.001666667	600	77,184	9.65	25.08	Yes	41.81	17.22	17	Yes	20.904	34.44	34
0.003333333	300	154,368	19.30	50.17	Yes	83.62	8.61	9	Yes	41.808	17.22	17
0.008333333	120	385,920	48.24	125.42	Yes	209.04	3.44	3	Yes	104.52	6.89	7
0.017	60	771,840	96.48	250.85	Yes	418.08	1.72	2	Yes	209.04	3.44	3
0.033	30	1,543,680	192.96	501.70	No	836.16	0.86	1	Yes	418.08	1.72	2
0.05	20	2,315,520	289.44	752.54	No	1254.24	0.57	1	Yes	627.12	1.15	1
0.1	10	4,631,040	578.88	1,505.09	No	2508.48	0.29	0	No	1254.24	0.57	1
0.2	5	9,262,080	1,157.76	3,010.18	No	5016.96	0.14	0	No	2508.48	0.29	0
0.5	2	23,155,200	2,894.40	7,525.44	No	12542.40	0.06	0	No	6271.2	0.11	0
1	1	46,310,400	5,788.80	15,050.88	No	25084.80	0.03	0	No	12542.4	0.06	0
2	0.5	92,620,800	11,577.60	30,101.76	No	50169.60	0.01	0	No	25084.8	0.03	0
5	0.2	231,552,000	28,944.00	75,254.40	No	125424.00	0.01	0	No	62712	0.01	0
10	0.1	463,104,000	57,888.00	150,508.80	No	250848.00	0.00	0	No	125424	0.01	0

30% general margin + 2.0 encoding margin multiplier -> *2.6 margin



C&DH – Data Budget

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6 Detectors, 1/2 FEC Rate

Frequency (Hz)	Sample Length (s)	6 Dominos (bits / day)	6 Dominos (KB / day)	6 Dominos (KB / day * 2.6 mrg)	4800 bps Compliant?	Time to Downlink Data (secs)	Number of times a full downlink can be attempted in a day (12 min)	Number of Missed Passes Before Needing Compression	9600 bps Compliant?	Time to Downlink Data (secs)	Number of times a full downlink can be attempted in a day (12 min)	Number of Missed Passes Before Needing Compression
0.000555556	1800	56,448	7.06	18.35	Yes	30.576	23.55	24	Yes	15.288	47.10	47
0.001111111	900	112,896	14.11	36.69	Yes	61.152	11.77	12	Yes	30.576	23.55	24
0.001666667	600	169,344	21.17	55.04	Yes	91.728	7.85	8	Yes	45.864	15.70	16
0.003333333	300	338,688	42.34	110.07	Yes	183.456	3.92	4	Yes	91.728	7.85	8
0.008333333	120	846,720	105.84	275.18	Yes	458.64	1.57	2	Yes	229.32	3.14	3
0.017	60	1,693,440	211.68	550.37	No	917.28	0.78	1	Yes	458.64	1.57	2
0.033	30	3,386,880	423.36	1,100.74	No	1834.56	0.39	0	No	917.28	0.78	1
0.05	20	5,080,320	635.04	1,651.10	No	2751.84	0.26	0	No	1375.92	0.52	1
0.1	10	10,160,640	1,270.08	3,302.21	No	5503.68	0.13	0	No	2751.84	0.26	0
0.2	5	20,321,280	2,540.16	6,604.42	No	11007.36	0.07	0	No	5503.68	0.13	0
0.5	2	50,803,200	6,350.40	16,511.04	No	27518.4	0.03	0	No	13759.2	0.05	0
1	1	101,606,400	12,700.80	33,022.08	No	55036.8	0.01	0	No	27518.4	0.03	0
2	0.5	203,212,800	25,401.60	66,044.16	No	110073.6	0.01	0	No	55036.8	0.01	0
5	0.2	508,032,000	63,504.00	165,110.40	No	275184	0.00	0	No	137592	0.01	0
10	0.1	1,016,064,000	127,008.00	330,220.80	No	550368	0.00	0	No	275184	0.00	0

30% general margin + 2.0 encoding margin multiplier -> *2.6 margin



Command and Data Handling



TakeMe2Space ZeroCube

Waiting to hear back on
device specs from
TakeMe2Space

	Daily data volume (bits)	Daily data volume (bytes)
30 sec heartbeat	2,949,120	368,640
60 sec heartbeat	1,474,560	184,320

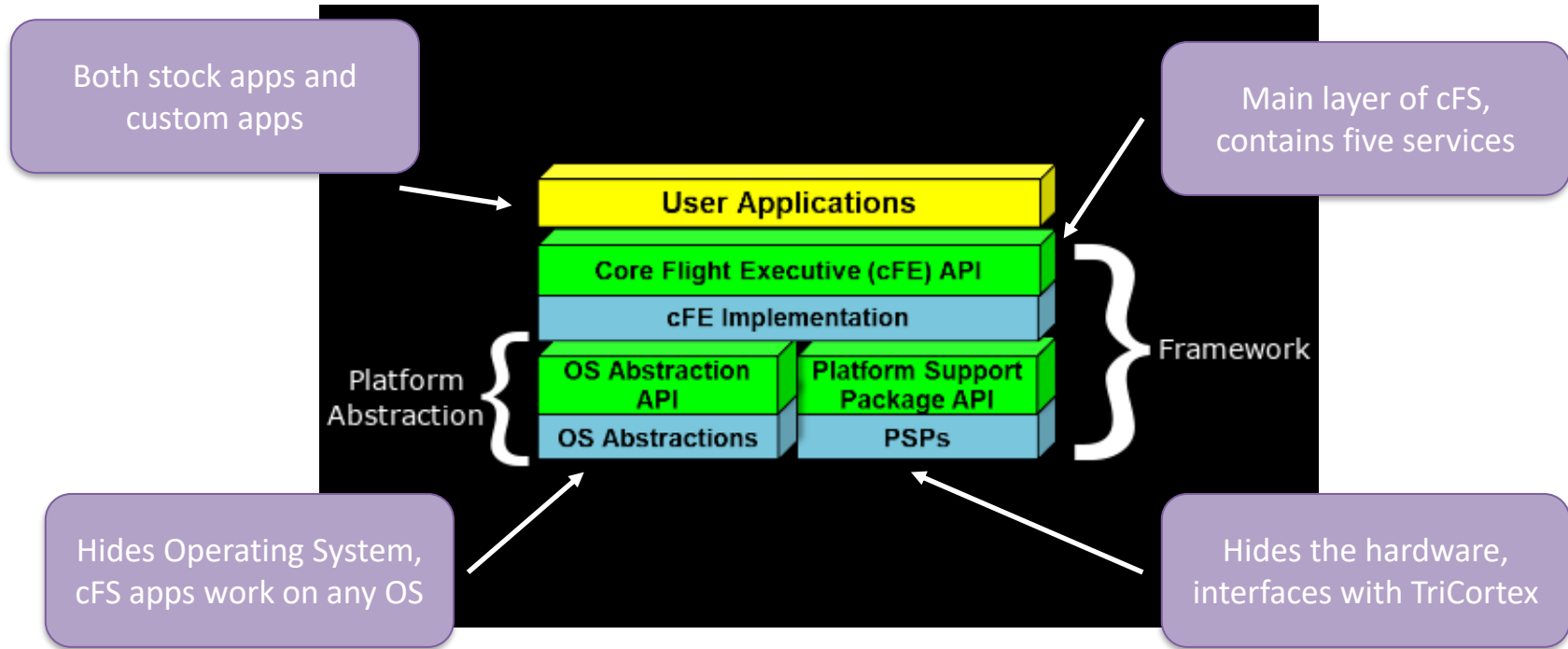
- Telemetry data volume upper bounds
- Assumes 128 byte packets to downlink all necessary information
- Each packet requires 0.21 sec of downlink time

Flight Computer

- Nextage NMC2
- Partnership, no cost
 - Collaboration agreement in progress
- Hybrid processor architecture
 - Single chip combining $4 \times$ RISC-V cores with reprogrammable FPGA fabric
- Radiation tolerant (30 kRad)
- 16 Gbit RAM and 500 GB SSD



core Flight System Block Diagram



Hardware Abstraction Layer

TriCortex Computer (FPGA), using TriCortex Real Time AI language



- Each CORE I processor permanently assigned to one hardware device
- Handle all sensor reading and actuator control
RISC-V cores never have to interact
- Results are written into AHB registers, RISC-V cores read
- Hardware faults isolated inside CORE I, never crash the RISC-V cores or cFS apps

Standards and Compliance

- NASA NPR 7150.2 – Software Engineering Requirements
- NASA-STD-8739.8 – Software Assurance
- NASA core Flight System

ADCS Critical Tech

COTS or Custom

- Fine Sun Sensor 1-6x — Custom – Cost, Precision
- Coarse Sun Sensor 6x — COTS – Precision
- Triaxial Magnetometer — COTS – Precision
- Navigation Satellite Receiver — COTS – Precision
- Magnetorquer 3x — COTS/Custom – Cost, Precision
- Onboard Computer — Custom – Cost
- TRIAD Angle Algorithm — Custom – Cost
- B-Dot Spin Algorithm — Custom – Cost
- GS Tracking Algorithm — Custom – Cost
- General Flight Algorithm — Custom – Cost
- Upset Recovery Algorithms — Custom – Cost
- Custom – Cost, Precision

EPD: PV-Cell Degradation Calculation

Using available data from the DHV Solar Panels, which use the same PV Cells as the EnduroSat 1U, the **Azur Space 3G30A**, to approximate the solar panel degradation.

$$\text{Degradation Factor} = \frac{P_{EoL}}{P_{BoL}}$$

Operating Temperature (Celsius)	Beginning-of-Life Power (W)	End-of-Life Power (W)	Degradation Factor
-50	2.48	2.41	97.18%
28	2.07	1.98	95.65%
125	1.5	1.4	93.33%

93.33% is used to provide the largest safety margin for extra power.

EPD: Reference for Preliminary Power Allocations

Subsystem	Minimum Power Consumption	% of Operating Power for Small Spacecraft	% of Operating Power for Medium to Large Spacecraft
Payload	20 – 50 W	40	40 – 80
Propulsion	0 W	0	0 – 5
Attitude Control	0 W	15	5 -10
Communications	15 W	5	5 -10
Command and Data Handling	5 W	5	5 -10
Thermal	0 W	5	0 – 5
Power	10 – 30 W	30	5 – 25
Structure	0 W	0	0

"New SMAD's Table 10-9" F. Zhu, *A Guide to CubeSat Mission and Bus Design*. Honolulu, HI:

Pressbooks, 2022.

Available: <https://pressbooks-dev.oer.hawaii.edu/epet302/>

NuKAT Preliminary Design Review (PDR) – Day Three

May 8th, 2026

This presentation has been approved for public release.
Limited distribution as approved by NuKAT team.



- **Dr. Amir Bahadori**
 - Professor and Program Director, Kansas State University Nuclear Engineering
 - Former NASA civil servant in the Space Radiation Analysis Group (SRAG)
- **Chris Brandt**
 - President, K-State's Research Foundation
- **Amy Betz**
 - Assistant Dean and Professor, Kansas State University Mechanical and Nuclear Engineering
- **Dave Copeland**
 - Space Communications System Engineer, John Hopkins University Applied Physics Laboratory
 - Co-Founder, Silversat

PDR Panelist Introduction

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- **Glen Fountain**
 - Project Manager, John Hopkins University Applied Physics Laboratory – Retired
 - Former Industry Advisor, Mike Wiegers Department of Electrical and Computer Engineering, Kansas State University
- **Chris Hersman**
 - Mission Systems Engineer, John Hopkins University Applied Physics Laboratory
- **Dr. Daniel Karkle**
 - Professor, Kansas State University Industrial and Manufacturing Systems Engineering
- **Joe Laurienti**
 - Founder and CEO, Ursa Major
 - Background experience in propulsion technology manufacturing and development



- Zach Metzinger

- Engineer, METECS, Lunar Gateway Mission
- Member, AMSAT Board of Directors
- Volunteer, AMSAT GOLF Program

- Dr. Matt O'Keefe

- Professor, Kansas State University Chemical Engineering

Jeffrey Sheldon

- Director, K-State's Sunderland Foundation Innovation Lab
- FCC Trustee, K-State's Amateur Radio Club (W0QQQ)

- Dr. Tim Steffensmeier

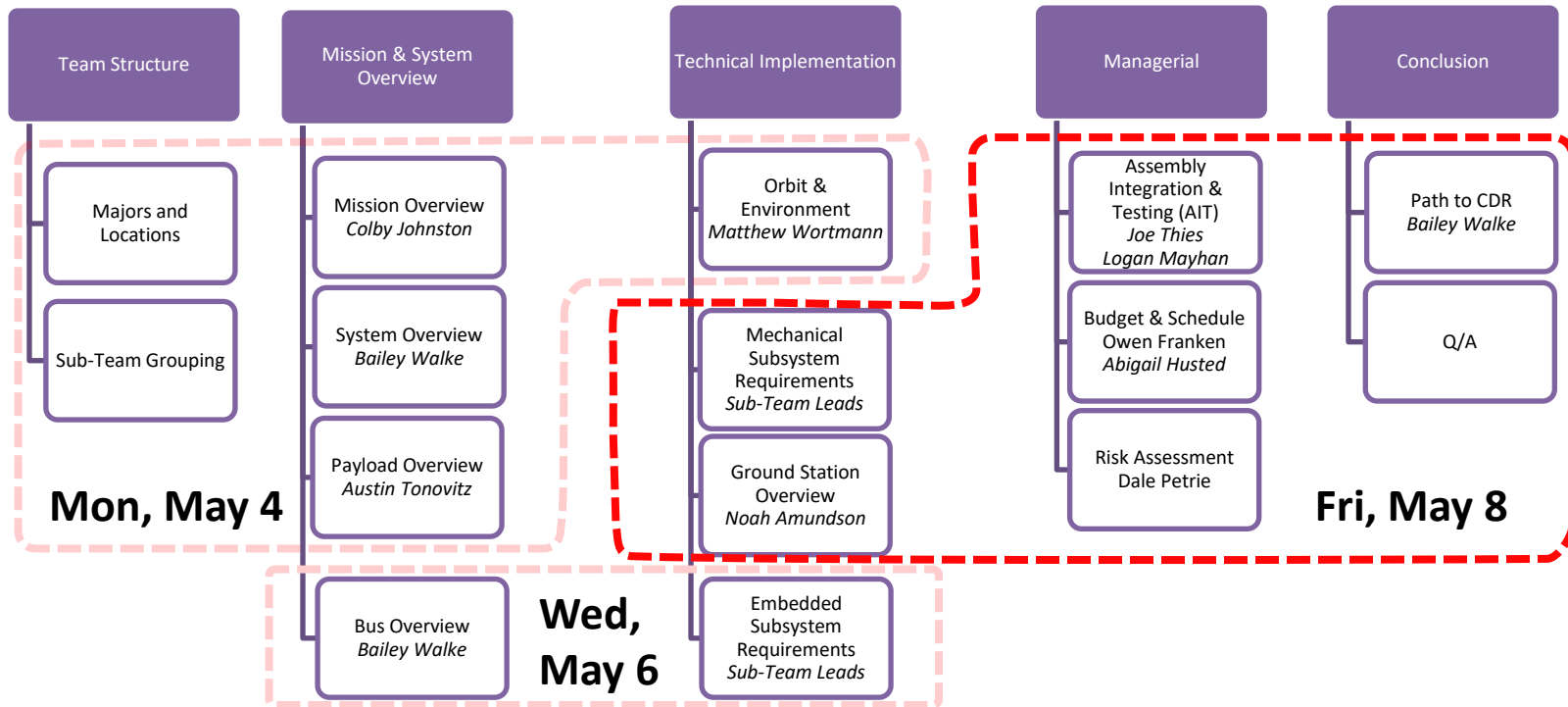
- Assistant Vice President and Director of Engagements and Outreach, Kansas State University

- Tim Trenkle

- Engineer (Retired), NASA Goddard Space Flight Center

Agenda

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Presenters

213

Bailey Walke	Chief Mechanical Systems Engineer
Trey Smoot	Structures Lead
Mitch Daues	Thermal Lead
Noah Amundson	Ground Station Lead
Joe Theis	Assembly, Integration, & Test Co-Lead
Logan Mayhan	Assembly, Integration, & Test Co-Lead
Owen Franken	Budget Lead
Abigail Husted	Schedule Lead
Dale Petrie	Risk Lead

STRUCTURES SUBSYSTEM

Trey Smoot – Structures Lead

Mechanical Engineering | Spring 2026

Bailey Walke – Mechanical Systems Lead

Mechanical Engineering Masters | Spring 2026



Rail redesign

- Moved tab locations / spacer geometry
- Increased room for solar panels

X and Y Panel redesign

- Accommodating new geometry

FEA and Vibrations Simulation

- Verifying design

Structure-Level 3 Requirements

216

Req. ID	Title	Requirement	Rationale	Risk	Compliance
STR-3.01	Dimensions and Units	The structure system shall be designed and analyzed in metric units (mm, kg, N), with the exception of fasteners and threaded features, which shall be designed with english units and specifications (in).	Ensures modeling consistency with CDS	3	Compliant
STR-3.02	Structural Coordinate System	The structure system shall be designed in accordance to the CDS coordinate system (C.S).	Deployer compatibility and consistent system integration	3	Compliant
STR-3.03	Structural Envelope	The structure system shall conform to the 2U external envelope defined by CDS.	Deployer compatibility	5	Compliant
STR-3.04	Structural Mass	The total mass of the structural subsystem shall not exceed 2kg.	50% of total allowed weight	3	Compliant
STR-3.05	Deployment Dispenser	The structure system shall interface with the launch dispenser rails and contact surfaces per CDS without interference.	Interface with LV	5	Compliant
STR-3.07	Payload Mounting	Detectors shall be mounted securely in the specified orientaton with restrained movement and no obstructions to it collecting its data at its proper rate	Allow safe transportation and proper functionality for detectors on board	10	Compliant
STR-3.08	Thermal Interface	The structure system shall provide defined conductive thermal paths between mounted components and exterior panels.	Supports thermal regulation and localized overheating	10	Compliant
STR-3.09	Structural Load Survival	The structure system shall survive quasi-static launch loads of 10g axial and 10g lateral	Required for launch qual	5	Partial Compliance
STR-3.10	Random Vibration Survival	The structure system shall survive random vibration enviornments defined by launch provider without structural failure.	Required for launch qual	5	Partial Compliance
STR-3.12	Primary Load Path	The structure system shall use deployment rails and chassis as the primary structural load path from launch vehicle. There shall be no primary load routed through the paneling.	Prevents thin panel loading	5	Compliant
STR-3.13	Field of View Compliance	The structure system shall comply with all specified fields of view (FOVs) from subsystems.	Ensures compliance wih launch integration, operational accessibility, and subsystem functionality	10	Compliant
STR-3.13c	Payload FOV	The structure system shall ensure moderated and unmoderated platforms do not cross within 70degrees of normal.	Science data accuracy	5	Compliant
STR-3.13e	ADCS Sensor FOV	The structure system shall maintain unobstructed FOV for all attitude determination sensors per interface definitions.	Ensures ADCS functionality	10	Compliant
STR-3.14	Keep-Out Zones	The structure system shall maintain designated keep-out zones defined below.	Ensures compliance wih launch integration, operational accessibility, and subsystem functionality	5	Compliant
STR-3.14a	Deployed Solar Panels	In the deployed state, the structure system shall ensure that there is no obstruction or interference of any components with the zenith-facing side of the deployed solar panels.	Operational requirement	10	Compliant



Structure-Level 4 Requirements

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Req. ID	Title	Requirement	Rationale	Risk	Compliance
STR-4.01	Chassis Size & Mass	The chassis shall conform to the 2U CDS external envelope and shall have a total mass not exceeding 1 kg.	Ensures deployer compatibility	3	Compliant
STR-4.02	Chassis Material	The chassis shall be manufactured from 6061-T6 aluminum.	Favorable strength-to-weight ratio, CubeSat standard	5	Compliant
STR-4.03	Stack Interface	The chassis shall include defined mounting features to constrain the E-Stacks in the X, Y, and Z directions while allowing removal in the +Z directions.	Ensures structural constraint without compromising removability.	4	Compliant
STR-4.04	Rail Material and Hardness	The material of the rails shall have a hardness equal or greater than the hardness of hard anodized aluminum. The material shall be 6061-T6 aluminum.	To prevent cold welding to the deployment chamber	5	Compliant
STR-4.05	Rail Dimensions	The contact surfaces of the rails shall extend the length of the bus (227mm) and be a 8.5mm by 8.5mm profile with 1mm chamfers.	Compliant with CDS	5	Compliant
STR-4.06	Rail Protrusion	The rails shall protrude at least 0.5mm from any features on +/- Z faces.	Prevents deployment interference	10	Compliant
STR-4.07	Rail Alignment & Spacing	The rails shall be spaced 100mm in the X and Y directions, run parallel to the X and Y axes at each corner, and be coplanar within 100 micrometers.	Interface with LV	10	Compliant
STR-4.08	Panel Quantity	The structure shall include TBD exterior panels to fully enclose the chassis while maintaining removable +Z faces.	Maintain environmental protection on all faces	10	Compliant
STR-4.09	Panel Material	All structural paneling shall be manufactured from 6061-T6 aluminum.	Maintains material consistency and predictable structural properties	5	Compliant
STR-4.10	Panel Thickness	All structural paneling shall be at least 1.5mm thick, being able to withstand 7 kRad	Must protect components from dangerous levels of radiation exposure	5	Compliant
STR-4.11	Exterior Feature Mounting Interface	Exterior panels shall provide defined mounting interfaces for external components (e.g., solar panels, antennas, sensors) without violating CDS protrusion limits.	Prevents deployment interference + external feature interference	5	Compliant
STR-4.12	Face Protrusions	The structure system shall not exceed CDS-allowed normal protrusion limits of 6.5mm on X and Y faces.	Prevents deployment interference	10	Compliant
STR-4.13	Fastener Specification	All structural fasteners shall conform to NAS-1351 (socket head cap screws) and NAS-1352 (nuts) or approved equivalent aerospace specifications.	Space vehicle best practice	5	Compliant
STR-4.14	Fastener Retention	All fasteners shall include locking features.	Prevents loosening under vibration	8	Compliant
STR-4.15	Fastener Material	All structural fasteners shall conform to NAS-1351 (socket head cap screws) and NAS-1352 (nuts) or approved equivalent aerospace specifications.	Prevents galvanic corrosion and ensures strength.	4	Compliant
STR-4.16	Fastener Sizing	Structural fasteners shall be sized to maintain a minimum factor of safety of TBD under worst-case launch loading.	Ensures joint integrity under launch acceleration/vibration loads	8	Compliant
STR-4.17	Tool Clearance	The structural design shall provide sufficient tool clearance for fastener installation	Ease of assembly	9	Compliant



Primary Structure - Rails

218

Requirements Compliance

- STR-4.05 - compliant - Length
- STR-4.07 - compliant - Spacing
- STR-4.04 compliant - Material

Manufacturing & Assembly

- CNC milling, solid block
- Rails connect X, Y, and Z panels as well as the Spacer

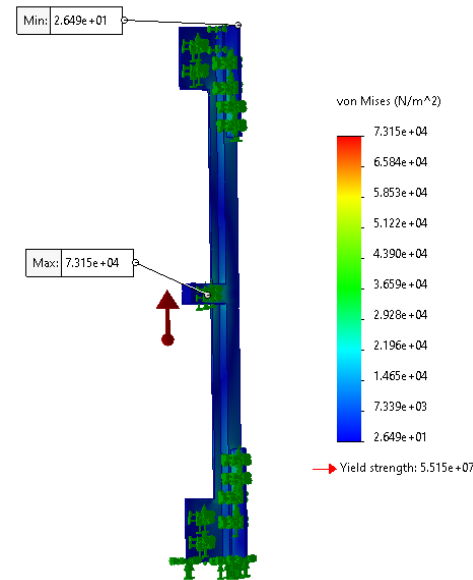
FEA Observations

- Currently passes FEA analysis, more testing recommended

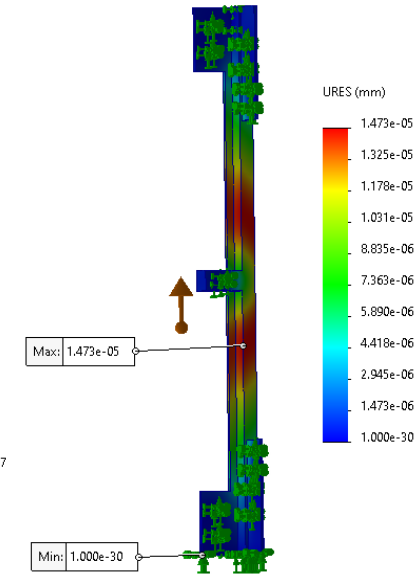
Acceleration Simulation Results:
10g's in +Z direction

Max Displacement (mm)	Max Stress (N/m^2)
0.00001473	7.315e+04

STRESS



DISPLACEMENT



Primary Structure: X-Panels

219

Requirements Compliance

- STR-4.10 compliant – 1.5mm Thick
- STR-3.12 compliant – No load through panels

Manufacturing & Assembly

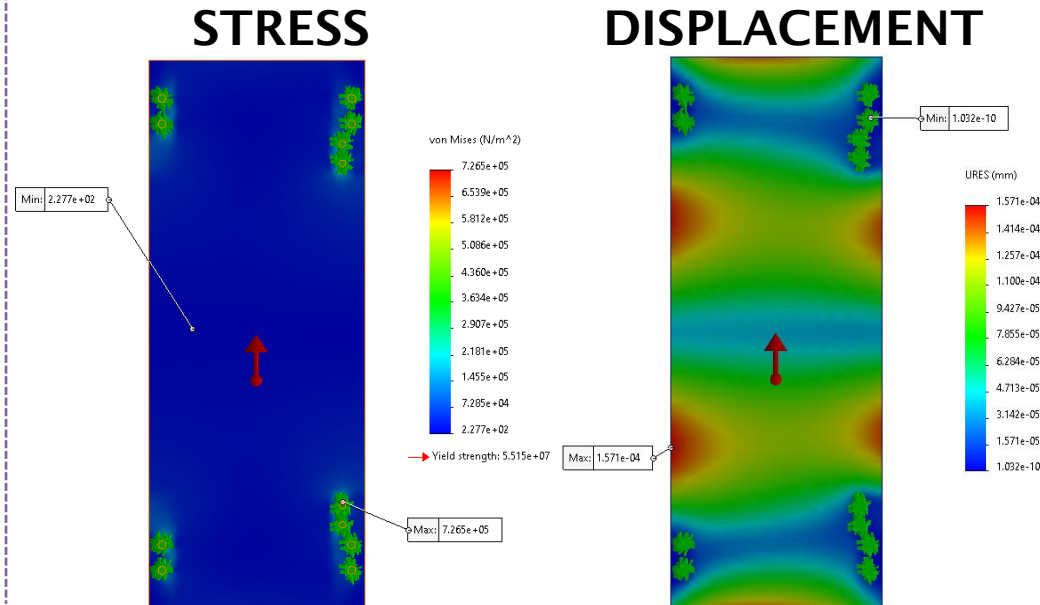
- CNC milling, solid block
- Bolts directly into the rails

FEA Observations

- Currently passes FEA analysis, more testing recommended

Acceleration Simulation Results:
10g's in +Z direction

Max Displacement (mm)	Max Stress (N/m^2)
0.0001571	7.265e+05



Primary Structure: Y-Panels

220

Requirements Compliance

- STR-4.10 compliant – 1.5mm Thick
- STR-3.12 compliant – No load through panels

Manufacturing & Assembly

- CNC milling, solid block
- Bolts directly into the rails

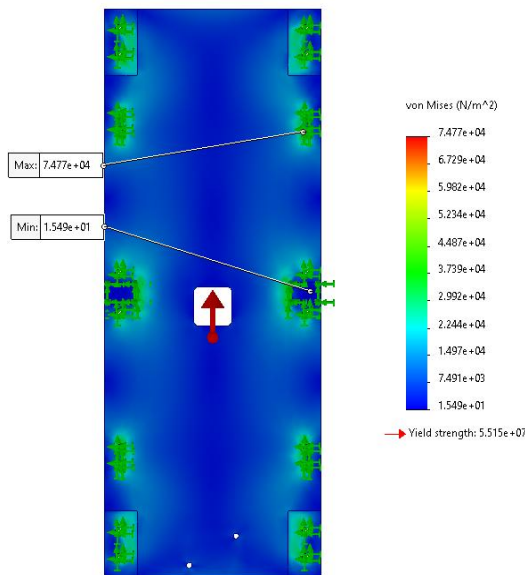
FEA Observations

- Currently passes FEA analysis, more testing recommended

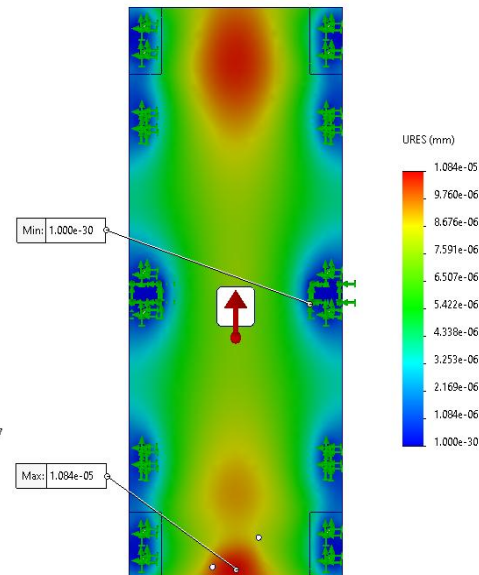
Acceleration Simulation Results:
10g's in +Z direction

Max Displacement (mm)	Max Stress (N/m^2)
0.00001084	7.477e+04

STRESS



DISPLACEMENT



Primary Structure: Z-Panels

221

Requirements Compliance

- STR-4.10 compliant – 1.5mm Thick
- STR-3.12 compliant – No load through panels

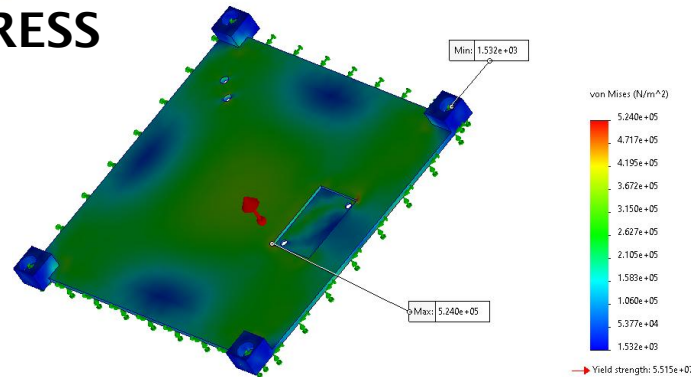
Manufacturing & Assembly

- CNC milling, solid block
- Bolts directly into the rails

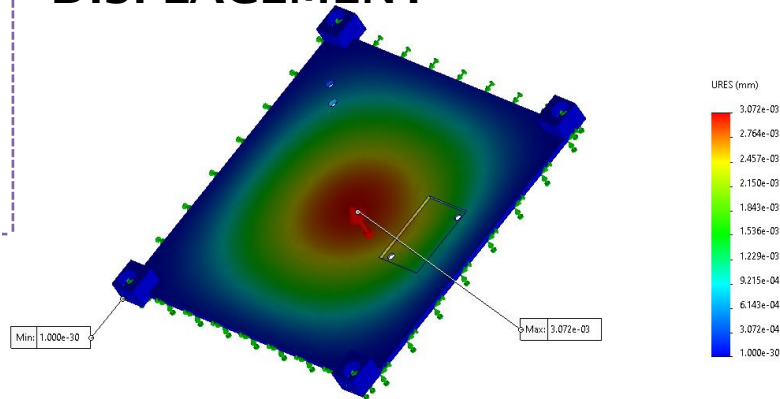
FEA Observations

- Currently passes FEA analysis, more testing recommended

STRESS



DISPLACEMENT

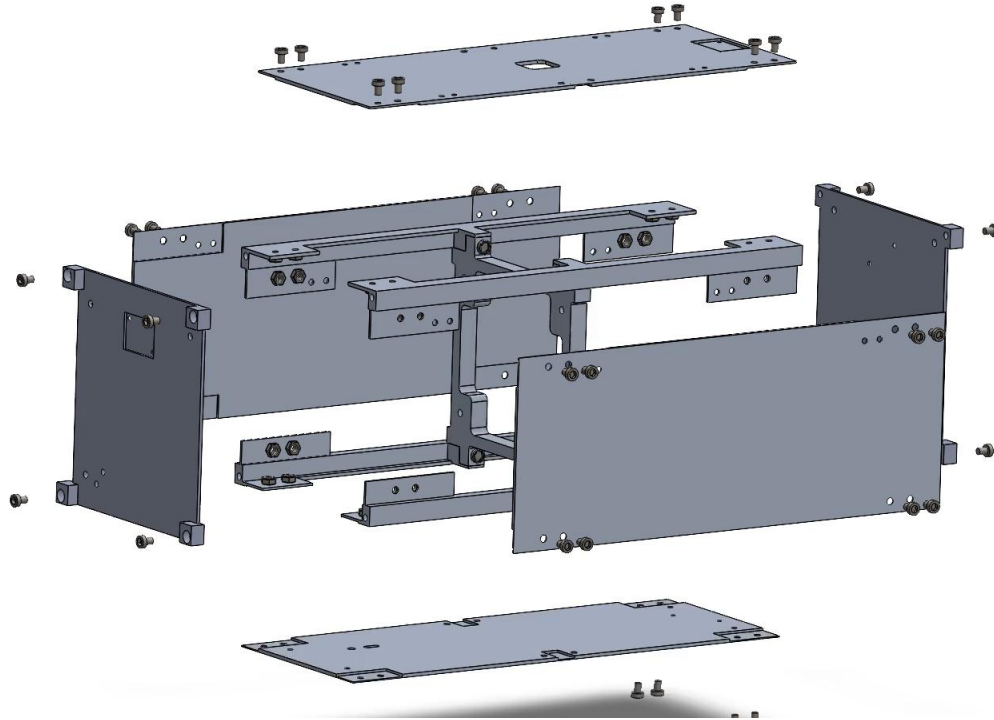


**Acceleration
Simulation Results:**
10g's in +Z direction

Max Displacement (mm)	Max Stress (N/m^2)
0.003072	5.240e+05

Mass Breakdown – Primary Structure 222

Component	Manufacturer	Number	Specifications	Mass(g)
Rails	K-State TDI	4	6061 Aluminum	146.09
X-Panels	K-State TDI	2	6061 Aluminum	132.86
Y-Panels	K-State TDI	2	6061 Aluminum	135.4
Z-Panels	K-State TDI	2	6061 Aluminum	79.63
Spacer	McMaster Carr	32	6061 Aluminum	54.31
M3 Nuts	McMaster Carr	32	18-8	16.32
M4 Nuts	McMaster Carr	8	18-8	4.88
M3 Bolts	McMaster Carr	32	18-8	11.52
Mass Total				581.01g



Primary Structure Vibration

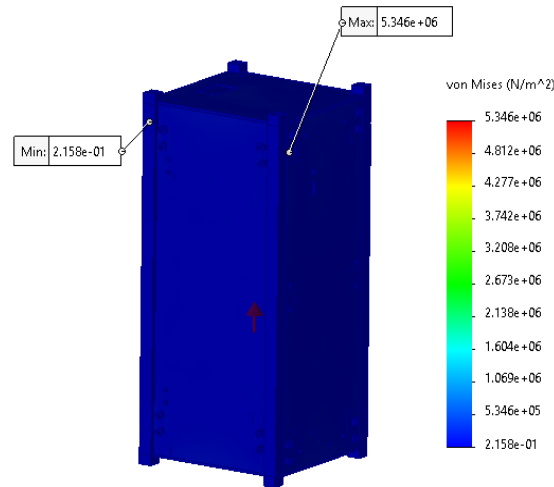
224

- Random Vibrations from 600-1500 Hz
- Fixed supports along rails
- No internal structures

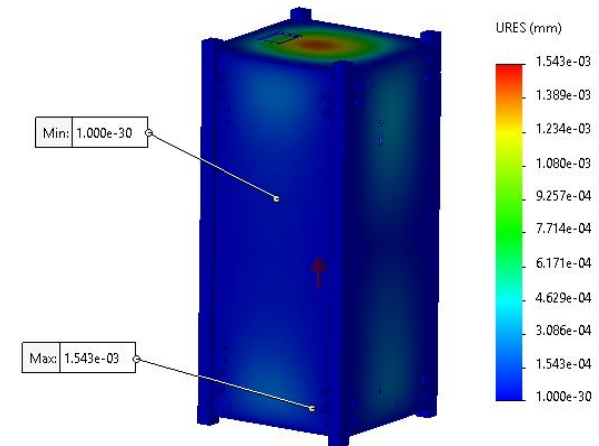
Simulation Results:

Max Displacement (mm)	Max Stress (N/m^2)
0.001543	5.346e+06

STRESS

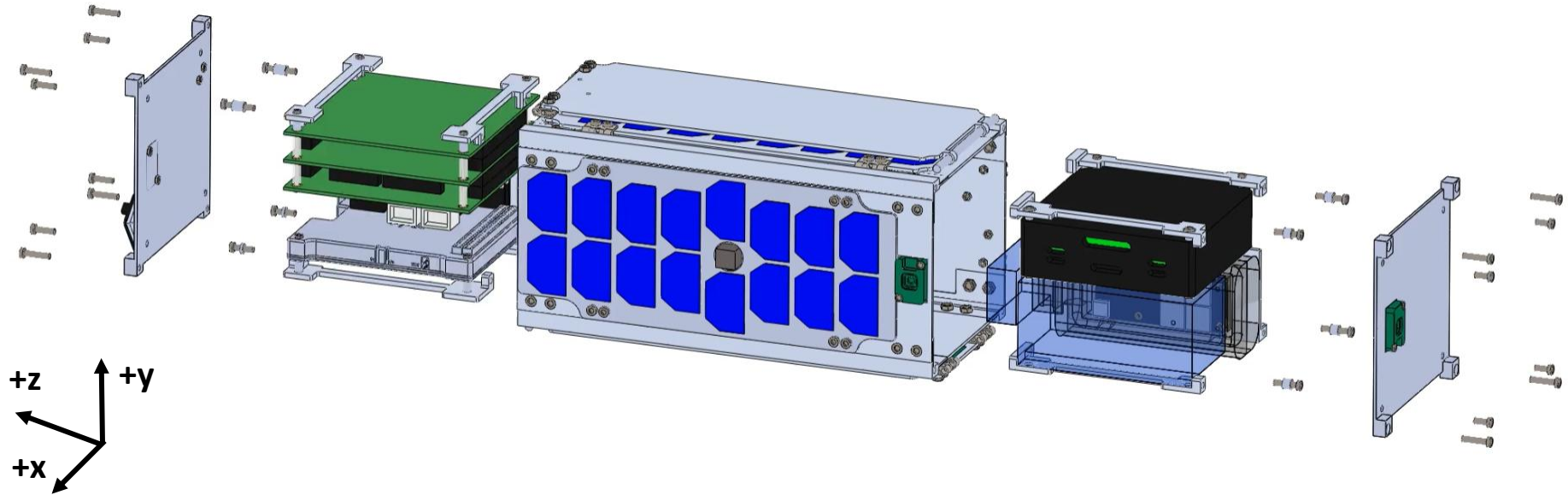


DISPLACEMENT



Removal & Assembly

225

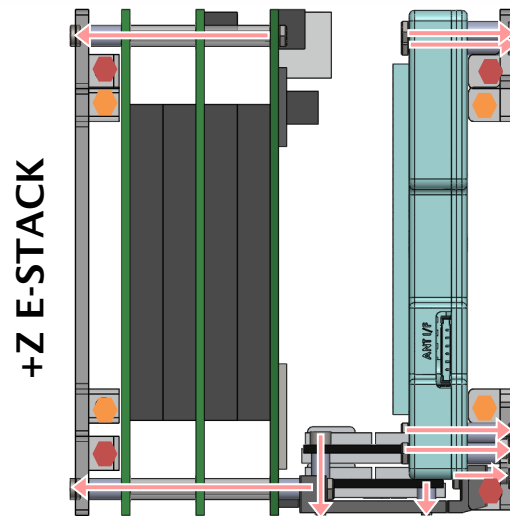


E-Stacks are removable with clearance through “z” panels without disassembly of chassis or x/y panels

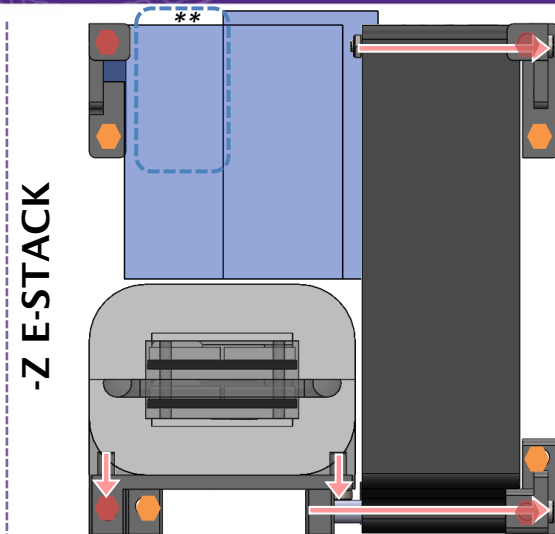
Secondary Structure

226

- 96mm x 86mm envelope
- All components constrained with 4 points of contact
- E-Stack constrained with 8 points of contact to chassis/panels



Component	Number	Specifications	Mass(g)
Bracket 1	1	6061	6.42
Bracket 2	1	6061	6.26
Bracket 3	1	6061	7.35
Bracket 4	1	6061	9.34
Payload Plate	1	6061	19.37
Spacers	12	M2, Delrin	0.83
Bolts	10	M2, 18-8	8.57
Nuts	10	M2, 18-8	1.16
Total Mass			59.30



Component	Number	Specifications	Mass(g)
Bracket 1	1	6061	6.77
Bracket 2	1	6061	6.07
Bracket 3	1	6061	6.78
Payload Plate	1	6061	33.94
Spacers	6	M2, Delrin	0.27
Bolts	8	M2, 18-8	6.86
Nuts	2	M2, 18-8	0.23
Total Mass			60.92



Requirement Compliance

- Primary Structure holds secondary structure effectively
- Primary Structure accommodates all external components

Subsystems Compatibility

- Full vibration study
- GD&T, tolerance stackup analysis

Structures has a Path Toward CDR

Structures Subsystem is Capable of Supporting NuKAT Mission

In light of this, Structures should proceed toward CDR with further effort to study structural and vibrations testing and determine part tolerancing.

THERMAL SUBSYSTEM

Mitch Daues – Thermal Lead

Mechanical Engineering | Spring 2026



Model Updates

- New Deployable Arrangement
- E-Stack Layout

Thermal Desktop Simulation

- Heating Rates, Temperatures, Hot/Cold Zones

Thermal – Level 3

230

NUKAT REQUIREMENTS VERIFICATION MATRIX

Req. ID	Title	Requirement	Risk Score	Status
3.8 Thermal Requirements				
THM-3.01	Thermal Control System Volume	The subsystem (heaters and heat strips/insulation) shall not take up more than [10.5%] of the NuKat's total volume	2	Compliant
THM-3.02	Thermal Control System Mass	The physical mass of the thermal subsystem (heaters, heat strips, insulation) shall not exceed [100] grams	3	Compliant
THM-3.03	Thermal Control System Power Usage	Energy usage per orbit shall not exceed 2 W. This includes measurement and active thermal regulation inside of components (Battery)	12	Partial Compliance
THM-3.04	Thermal Critical Systems and Components	Critical systems and components are defined as the following: Payload, Battery, OBC, ADCS, and Solar Array	1	Compliant
THM-3.05	Thermal Control - Payload	The payload shall be kept within a temperature range of -20 - +55 C	15	Compliant
THM-3.06	Thermal Control - EPDS	The EPDS shall be kept within a temperature range of -30 - +65 C	15	Partial Compliance
THM-3.07	Thermal Control - OBC	The OBC shall be kept within a temperature range of -40 - +85 C	15	Compliant



Thermal – Level 3

231

NUKAT REQUIREMENTS VERIFICATION MATRIX

Req. ID	Title	Requirement	Risk Score	Status
3.8 Thermal Requirements				
THM-3.08	Thermal Control - Battery	The battery components shall be kept within a temperature range of +5 - +60 C	15	Partial Compliance
THM-3.09	Thermal Control - Solar Array	The solar array shall be kept within a temperature range of -40 - +105 C	15	Not Compliant
THM-3.10	Thermal Control - Other	All other components shall be kept within a operatable temperature range defined by the "Operating Temps" table	12	Compliant
THM-3.11	Measurement Device Type	All measurement devices shall be thermistors capable of operating in a space environment, unless an individual component provides a built in measurement device	3	Compliant
THM-3.12	Measurement Device Quantity	a suffucient quantity of measurement devices shall monitor the status of all critical systems and components	8	Compliant
THM-3.13	Measurement Interval	Temperatue shall be recorded within 15 second intervals of components within the cubesat and relayed to controls.	4	Compliant
THM-3.14	Heater size	the heaters shall be of adauquate size to provide sufficient heat to the critical systems and components [TBD]	8	Not Compliant



Thermal – Level 4

232

NUKAT REQUIREMENTS VERIFICATION MATRIX

Req. ID	Title	Requirement	Risk Score	Status
4.8 Thermal Requirements				
THM-4.01	Thermistor Make and Model	Thermistors shall be model 0118MM manufactured by Collins Aerospace or equivalent	1	Compliant
THM-4.02	Thermistor Operating Temperature	The Thermistors shall operate with a temperature range of -50 °C - +155 °C	2	Compliant
THM-4.03	Thermistor Power Consumption	The thermistors shall operate with a power consumption no greater than 0.25 W	2	Compliant
THM-4.04	Thermistor Sampling Frequency	The thermistors shall be capable of being sampled at least once a second (1 Hz)	2	Compliant
THM-4.05	Thermistor Reading Resolution	The thermisors shall have a resolution of at least 0.1 °C	2	Compliant
THM-4.06	Thermistor Reading Accuracy	the thermistor reading shall be accuracate within +/-[5] %	8	Compliant
THM-4.07	Thermistor Mounting Method	the thermistors shall be mounted to ensure proper operation [Method TBD]	10	Not Compliant
THM-4.08	Thermistor/OBC communication	the thermistor shall be able to communicate with the OBC using GPIO Inputs as analog inputs	5	Compliant

Thermal – Level 4

233

NUKAT REQUIREMENTS VERIFICATION MATRIX

Req. ID	Title	Requirement	Risk Score	Status
4.8 Thermal Requirements				
THM-4.09	Heater Make and Model	Heaters shall be model HK6900 polyimide resistive heaters manufactured by Minco or equivalent	1	Compliant
THM-4.10	Heater Operating Temperature	The heaters shall operate with a temperature range of -32 °C - +100 °C	10	Compliant
THM-4.11	Heater Power Consumption	The heaters shall operate with a power consumption no greater than 2 W	8	Compliant
THM-4.12	Heater Response Time	the heaters shall be capable of providing sufficient heat within a set time [TBD]	8	Not Compliant
THM-4.13	Heater Mounting Method	the heaters shall be mounted using pressure sensitive adhesive to ensure proper thermal contact	10	Compliant
THM-4.14	Heater EPDS Connection	The heaters shall be supplied with [x] volts via [connection method]	4	Not Compliant
THM-4.15	Heater C&DH Connection	The heaters shall be capable of being toggled on/off by the OBC as needed	4	Compliant



Connections to other subsystems

EPDS:

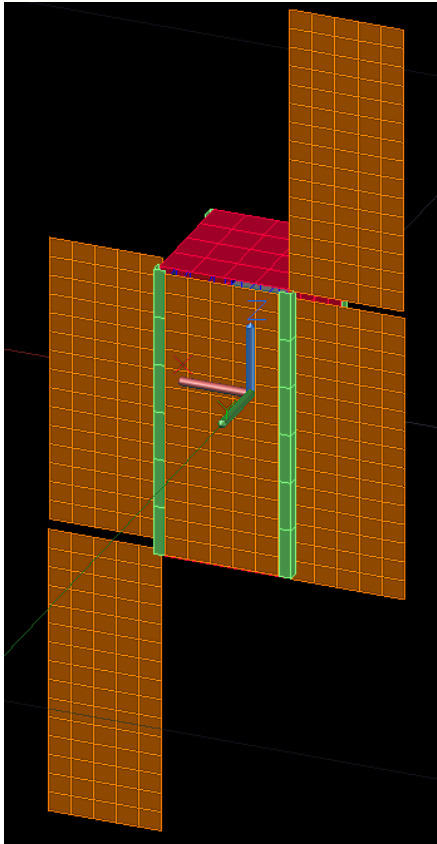
- Supply power to heaters

Structure:

- Access to Heat Sinks
- Anchoring

C&DH:

- Command Heaters on/off
- Read Thermistors



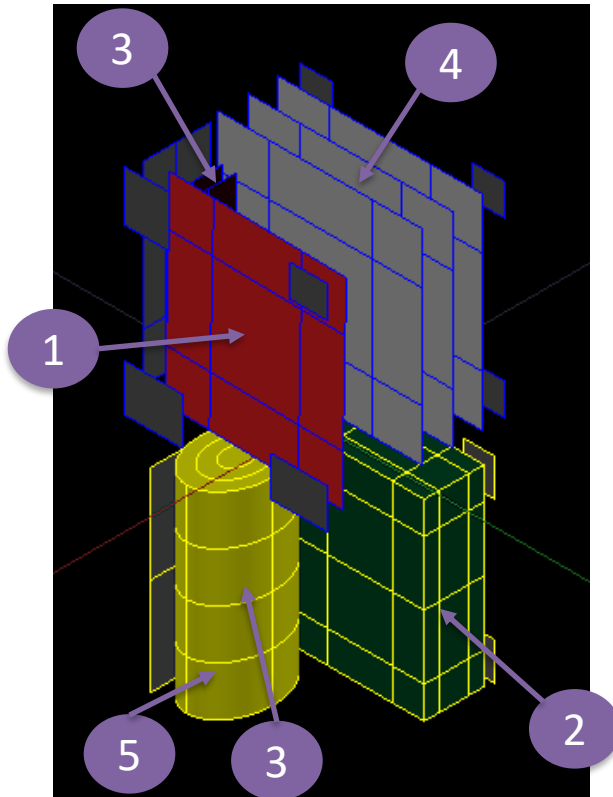
Use simplified geometry to predict thermal performance in orbit

Key Assumptions:

- Silver Teflon tape covering exterior panels ($\epsilon = 0.84$)
- Emissive white paint on back of solar panels ($\epsilon = 0.93$)
- All Components Operating
 - 25% waste heat
- E-Stacks insulated from structure

Thermal – TD Simulation

236

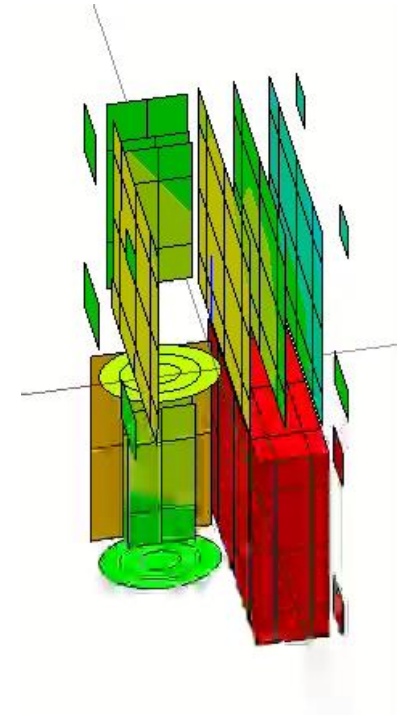
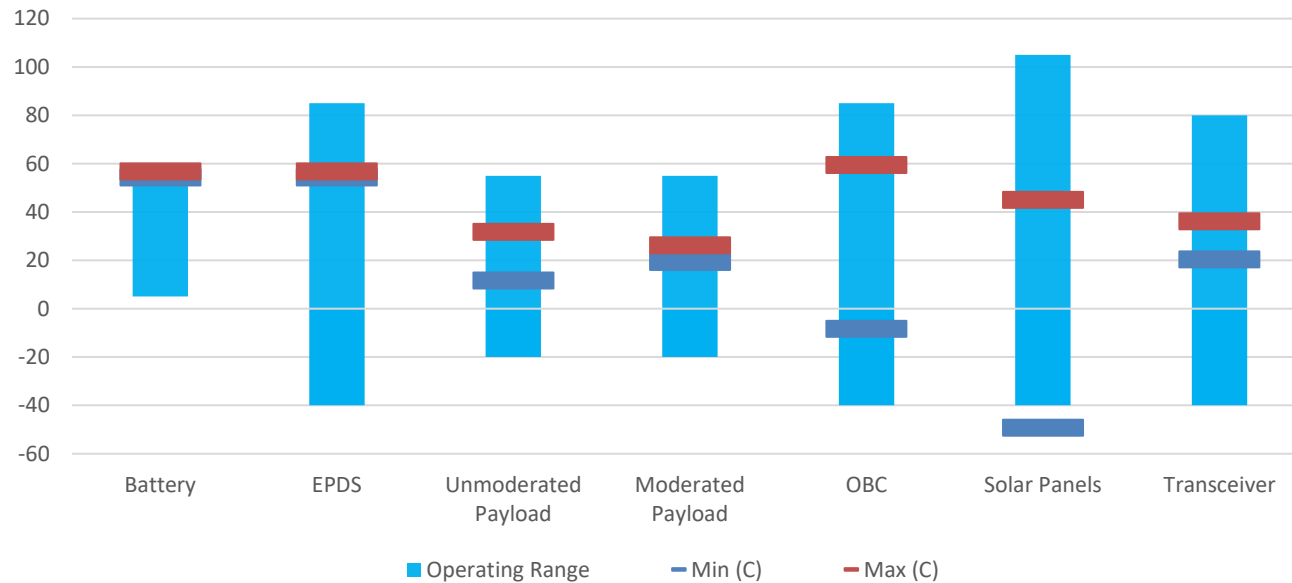


Submodel	# Nodes	Submodel	# Nodes
Solar Panels	375	(2)EPDS	48
Al-Panels	128	(3)Payload	12 (x2)
(1)Transceiver	16	(4)OBC	49
-Z E-Stack	8	(5)Moderator	22
+Z E-Stack	14		
Total		684	

Thermal – TD Simulation

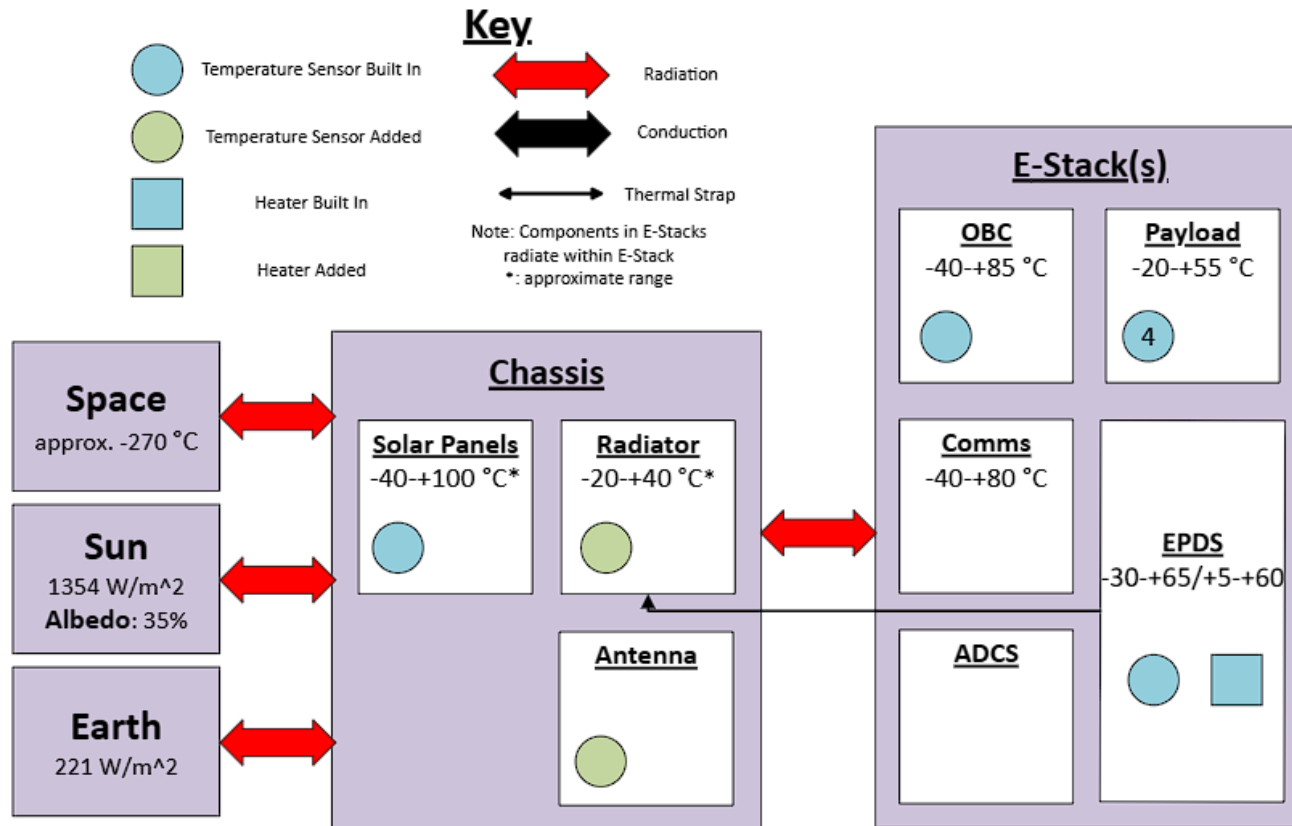
237

Predicted Temperatures vs Operating Temperatures for Critical Components



Thermal – Block Diagram

238



Requirement Compliance

- Solar Array under temperature
- Heater sizing/placement

Subsystems Compatibility

- Mounting of thermistors TBD
- Otherwise, the system is compatible with other subsystems

Thermal has a Path Toward CDR

Thermal Subsystem is Capable of Supporting NuKAT Mission

In light of this, thermal should proceed toward CDR with further effort to tighten thermal control and bring all components within required temperature ranges.

GROUND SYSTEM

Noah Amundson – GS Lead

Industrial Engineering | Spring 2026



International Amateur Radio Form

- IARU form is currently being completed and reviewed

Hardware Purchasing

- Plan in place to buy new hardware including antennas
- Funding through CubeSat account and K-State Foundation is confirmed

Ground Station Handoffs

- Beginning work on handing off project to next ground station lead

Ground Station – Level 3

242

Req. ID	Title	Requirement	Risk Score	Status
GSH-3.01	Location	The ground system shall have a primary and secondary location that allows for clear communication at TBD time intervals. Primary location shall be located within campus boundaries.	10	Partial Compliance
GSH-3.02	Contact Time	The ground system shall be aware of the satellite's location to achieve communication for 5.8 minutes. (Contact time per pass)	5	Compliant
GSH-3.03	Security	The ground station shall be in a secure location with restricted access.	9	Compliant
GSH-3.04	Power	The ground station shall be able to provide "x" amount of power to components.	9	Partial Compliance
GSH-3.05	Licensing	The ground station shall have licensing to operate UHF communication bands and meet FCC, IARU, AMSAT requirements.	10	Partial Compliance
GSH-3.06	Training	Personnel directly operating the ground station shall have expert level training on its systems.	2	Compliant
GSH-3.07	Operational Lifetime	The ground station shall have an operational lifetime that meets or exceeds mission requirements.	12	Compliant
GSH-3.08	FCC Specific Licensing	Students planning to work the ground station must pass the FCC exam offered through Jeff Sheldon at KSU Manhattan.	3	Compliant



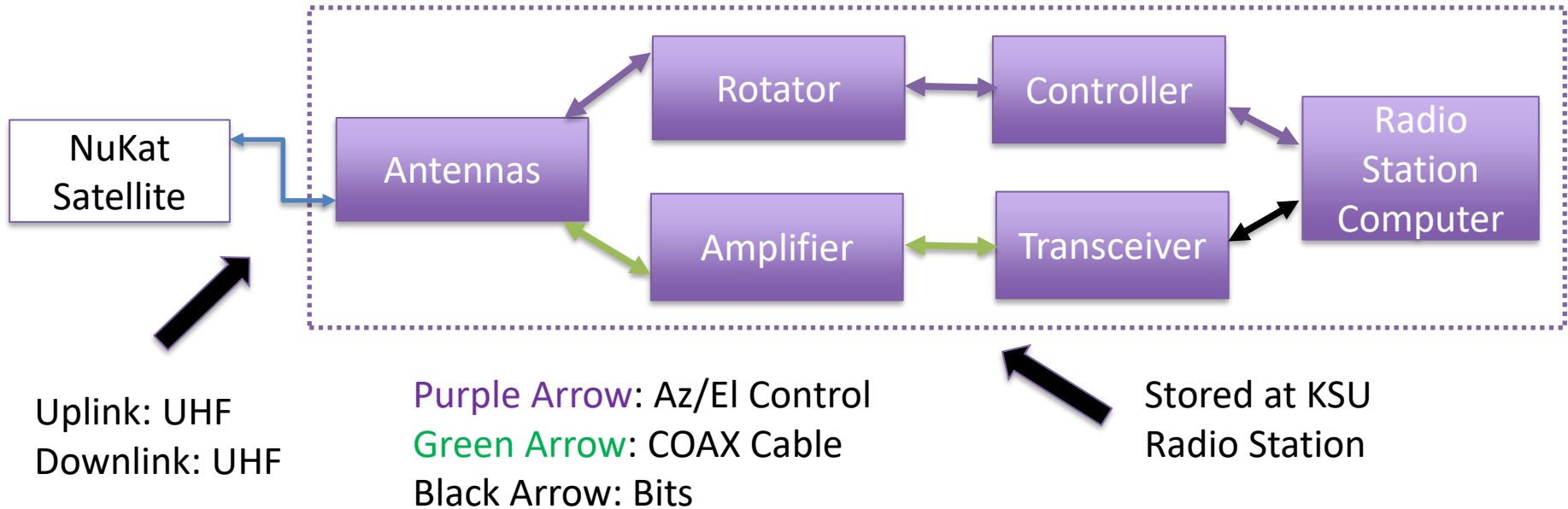
Ground Station – Level 4

243

Req. ID	Title	Requirement	Risk Score	Status
GSH-4.01	Satellite Antenna and Gain	Ground station antenna must be able to communicate and operate in UHF bands between 430-440 MHz and with an antenna gain between 10-15 dBi. (x2) M2 Antennas 436CP30 70cm Circularly Polarized Beams 436CP30	8	Compliant
GSH-4.02	SDR Transceiver	Must have UHF transmit and receive frequencies within 430-440 MHz. Radio station currently has VHF/UHF All Mode Transceiver IC-9700. Additional SDR transceiver will be procured.	4	Compliant
GSH-4.03	Rotator	Rotator will be needed for tracking NuKat in orbit. Radio station currently has Yaesu G-5500 rotator.	4	Compliant
GSH-4.04	Controller	Radio station currently has S.A.T controller. Will need controller to connect direct to P.C. for mission critical software. Viable option for new controller is Yaesu Rotator Computer Controllers GS-232B.	4	Compliant
GSH-4.05	Amplifier	Select the correct low-noise amplifier that meets the mission requirements. EME3-432 70cm Ultimate Split RX Relay Switches BPF VLNA	3	Compliant
GSH-4.06	Satellite Antenna Beamwidth	Ground station antenna will have an operational beamwidth of 30 degrees.	3	Compliant
GSH-4.07	Computer	Computer at radio station is sufficient and capable of meeting mission requirements.	2	Compliant
GSH-4.08	Cabling	Cabling will be needed to connect hardware together. COAX LMR-600 cable donated to radio station and will need COAX LMR-240 cable for turns, obstacles, and wrap around rotator. Lengths of cable needed are TBD.	2	Compliant
GSH-4.9	Ground Station Data Storage	Expected storage needed for 6 years is between 1.01×10^{10} bits and 2.03×10^{10} bits (4800-9600 BPS)	3	Compliant



Ground Station Block Diagram ²⁴⁴





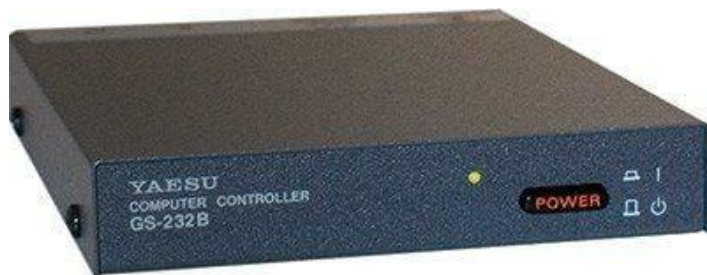
VHF/UHF All Mode Transceiver IC-9700

- Used to uplink and downlink data between satellite and ground station.
- Existing Cost: \$2,100



Yaesu G-5500 Rotator

- Rotator is used to aid the antenna as it tracks satellite in space.
- Existing Cost: \$830



Yaesu Rotator Computer Controllers GS-232B

- Aims a directional antenna from a computer instead of only using the manual rotator.
- New Cost: \$650

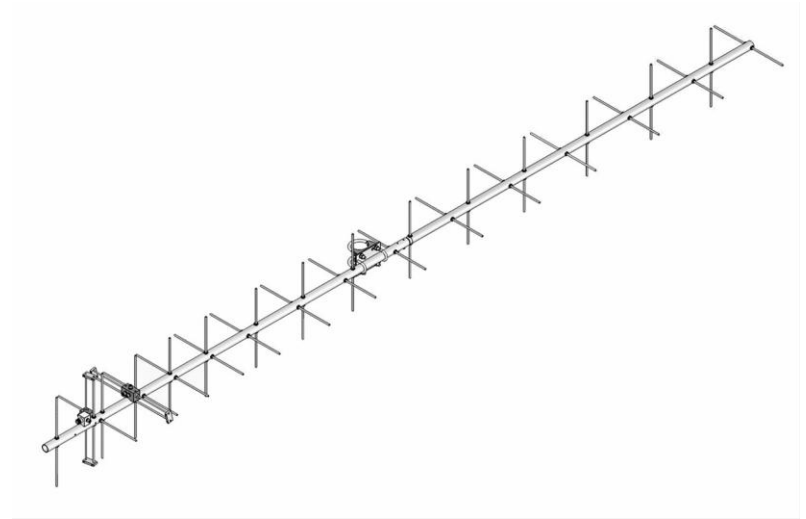


EME3-432 70cm Ultimate Split RX Relay

- Boosts downlink signals to survive travel through the atmosphere to be picked up by ground station.
- New Cost: \$247

M2 Antennas- 436CP30 70cm Circularly Polarized Beams 436CP30

- Satellite antennas are used to convert electrical signal from transceiver into radio waves.
- 30-degree beamwidth
- Beam Antenna Boom length: 9.75 ft
- New Costs: \$1,288 for 2 antennas



- LMR 600 Cabling
 - Stiff
 - Antenna to transceiver
 - Outdoors to indoors
- LMR 240 Cabling
 - Flexible
 - Rotator Loop “Pig Tail”
 - Transceiver inside the station



Operators and Ground Station ²⁴⁹



- Continue testing/experimenting at radio station.
- Continue to prepare for new ground station team lead next semester.
- New person to work on ground station database.
- Ground station software is being worked on, but less focused on this semester.

Requirement Compliance

- All requirements are partial/expected to be compliant or compliant

Subsystems Compatibility

- GSH 4.01 and GSH 4.06 about the new ground station antenna meet mission requirements of other subsystems.
- Other partially compliant requirements are pending review, need further investigation, and are expected to close.

Ground Station has a Path Toward CDR

Ground Station Hardware Subsystem is Capable of Supporting NuKAT Mission

In light of this, Ground Station Hardware should proceed toward CDR with further effort to completing compliance of partial compliant requirements and work to order all necessary new hardware and begin installation.

ASSEMBLY, INTEGRATION, & TESTING (AI&T)

Joe Theis – AI&T Co-Lead

Mechanical Engineering Technology | Spring 2026

Logan Mayhan – AI&T Co-Lead

Unmanned Aircraft Systems | Spring 2026



Further RVM Verification Method Development

- RVM entries at 396 in total
- Verification methods in place for all current entries

AI&T Plan Developed

- In-depth testing plan developed for NuKat
- Will guide all phases of AI&T

- RVM Verification Methods
 - Grouped by Analysis, Test, Inspect, Demonstrate (ATID)

Mission Parameter (Requirements)	Type	Description
Neutron Detector Payload Accuracy	T	Test the DSC Domino neutron detector and verify it has an accuracy within "x" percent
Core Capabilities (Nadir Tracking)	D	Demonstrate that Attitude Determination system can track the Nadir datum
Operational Life (Attitude Determination)	A	Analyze ADCS in environmental conditions of space for "x" years.
Neutron Detector Mounting Angle	I	Inspect the detector mounts and ensure correct orientation.

All 396 requirements have verification methods

RVM Verification Methods Distribution

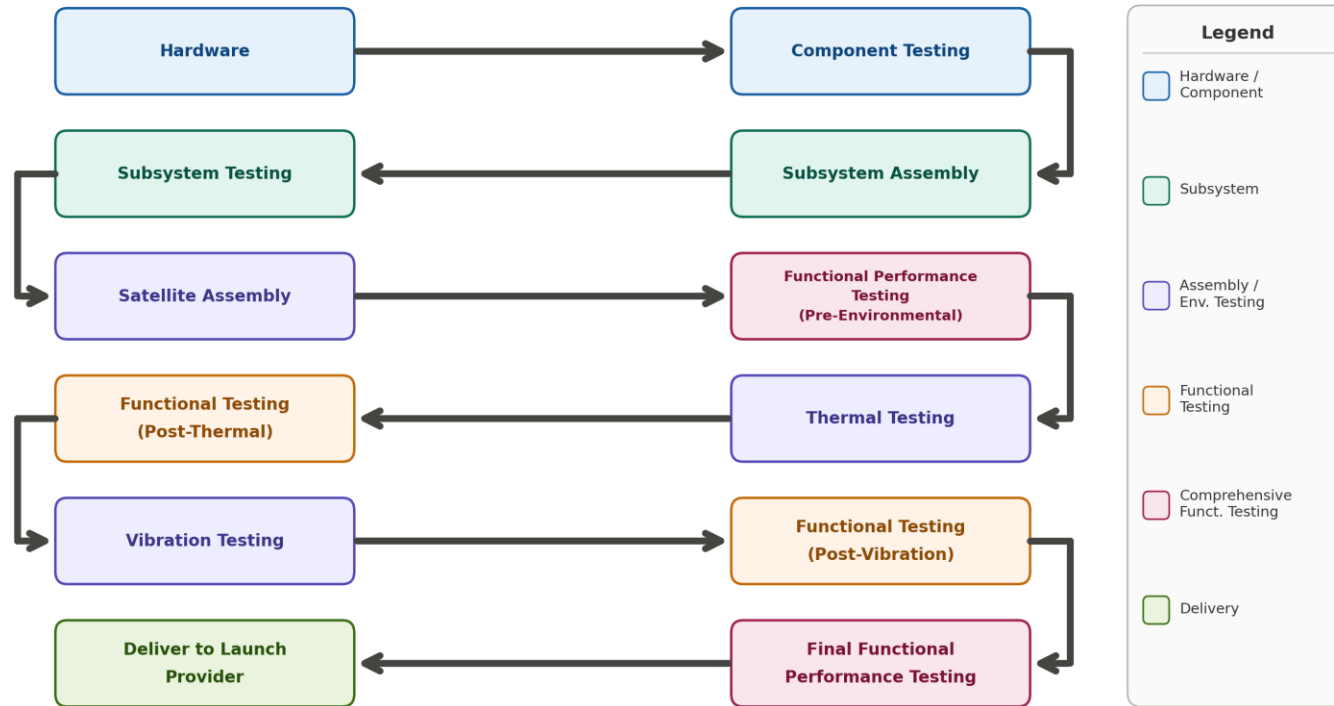
Analysis	Inspection	Demonstrate	Test	Combination
27	30	182	77	80

In-Depth Description of All AI&T Activities

- Approach and Philosophy of AI&T
- I&T Activity Flow Diagram
- I&T Activity Descriptions
- Procedure Descriptions
- Review Descriptions
- Designated Resources

Target Page Count ~ 35

<u>Assembly Levels</u>	<u>Testing Procedures</u>						
	Functional Performance	Functional	Vibration	Thermal	Subsystem Testing	Component Testing	
<u>Component</u>	X					X	X
<u>Subsystem</u>	X				X		X
<u>Satellite</u>	X	X	X	X			X



Located in Seaton Hall room 0004 with key code access control

- **Assembly Area:**
 - Dedicated area to satellite assembly
- **Testing Stations:**
 - Workstations for electrical and mechanical verification
- **Storage:**
 - Secure storage for hardware
- **Ground Support Equipment**
 - Electrical/Mechanical Support



Requirement Compliance

- No noncompliance

Subsystems Compatibility

- Compatible with all other systems

Assembly Integration & Testing has a Path Toward CDR

Assembly Integration & Testing Subsystem is Capable of Supporting NuKAT Mission

In light of this, Assembly Integration & Testing should proceed toward CDR.

BUDGET

Owen Franken – Budget Lead

Industrial Engineering | Spring 2026



Itemization of Budget

- Budget narrowed and updated from category estimates to specific budget for every subsystem and itemized costs.

Finalization of first draft of budget

Budget – Subteams

262

- Individual part orders will be tracked for budget control
- Values come from Subteam estimates and limited quotes
- Certain estimates taken from Arizona State University CubeSat 3U Budget Spreadsheet adjusted for inflation.
- Is compliant with requirements

Flight Hardware Cost

Subteam	Cost	Budget
PLD	\$5,000	\$5,000
EPDS	\$16,990	\$20,000
ADCS	\$10,000	\$10,000
C&DH	\$14,500	\$15,000
FSW	~\$6,000	\$10,000
STR	\$418	\$500
OBC	\$0	\$5000
THM	~\$1,000	\$1,000
Spares	\$3,500	\$3,500
Total	\$57,408	\$70,000

Budget – Hardware & System 263

System and Infrastructure	Cost	Budget
Contingency (~20%)	\$33,000	\$33,000
Integration & Testing	~\$12,000	\$12,000
Ground Station	\$3,573	\$5,000
Launch System	\$80,000	\$80,000
Total	\$128,573	\$130,000

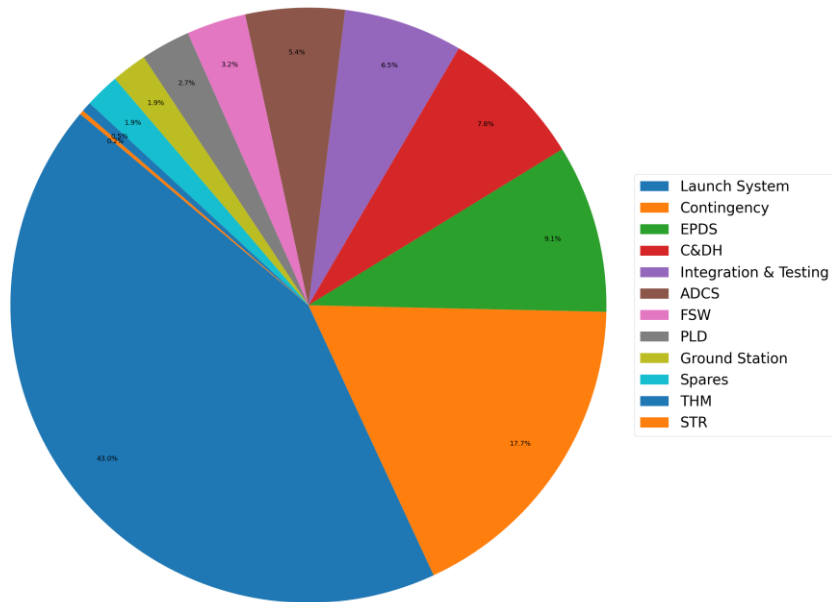
Categories	Cost	Budget
System and Infrastructure	\$128,573	\$130,000
Flight Hardware	\$57,408	\$70,000
Total	\$185,961	\$200,000

~\$185,961 total cost

Budget – Cost Breakdown

264

Total Project Cost Breakdown (Proportional)



- Launch and our Contingency are the largest two costs, taking up well over 50% of the budget.
- ~\$14,000 under budget with the current itemized list and estimates.

Requirement Compliance

- No noncompliance issues

Subsystems Compatibility

- No concerns
- Budget is compatible otherwise

Budget has a Path Toward CDR

Budget is Capable of Supporting NuKAT Mission

In light of this, Budget should proceed toward CDR with further effort to edit and adjust the budget as needed.

SCHEDULE

Abigail Husted – Schedule Lead

Industrial Engineering | Spring 2026



PERT Chart

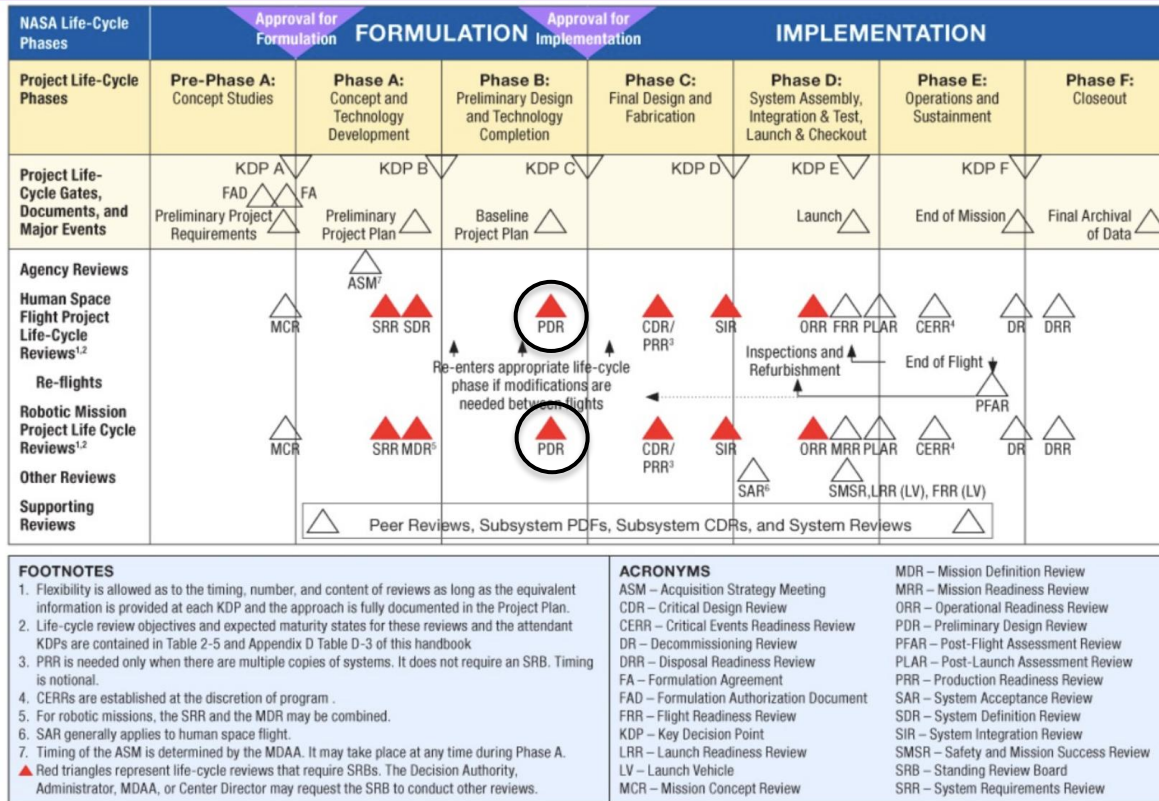
- Critical path defined
- Milestones for 3/26-5/26 tracked with all subsystems
- Milestones from PDR to CDR in development

Risk Analysis of Schedule

- Starting to use Python code (authored by Dr. Daniel Karkle) to run risk analysis on the PERT Chart Schedule

Schedule - Resources

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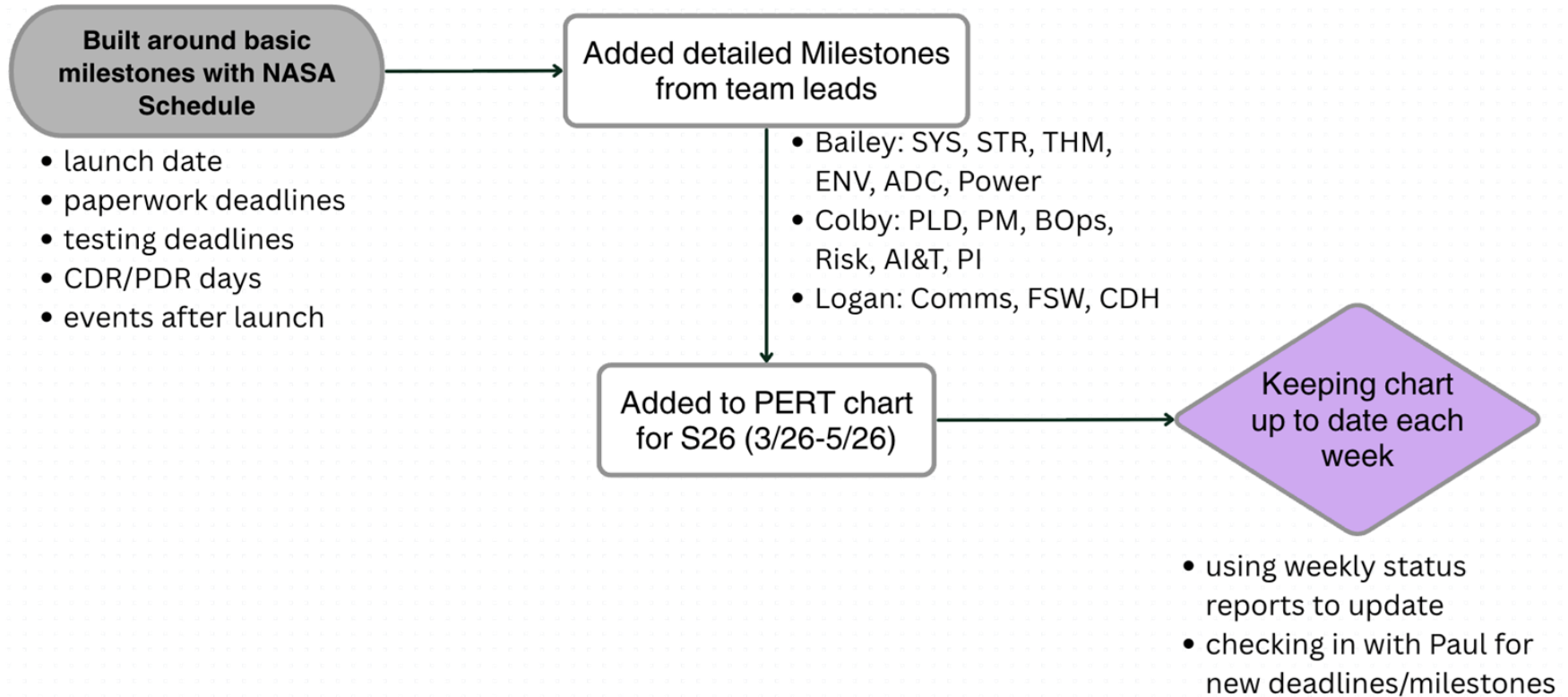


- Used NASA Space Flight Project Life Cycle & CubeSat Handbook to make a rough 'skeleton' of events on the PERT chart
- Black Circle = WE ARE HERE

NASA Space Flight Project Life Cycle from NPR 7120.5E

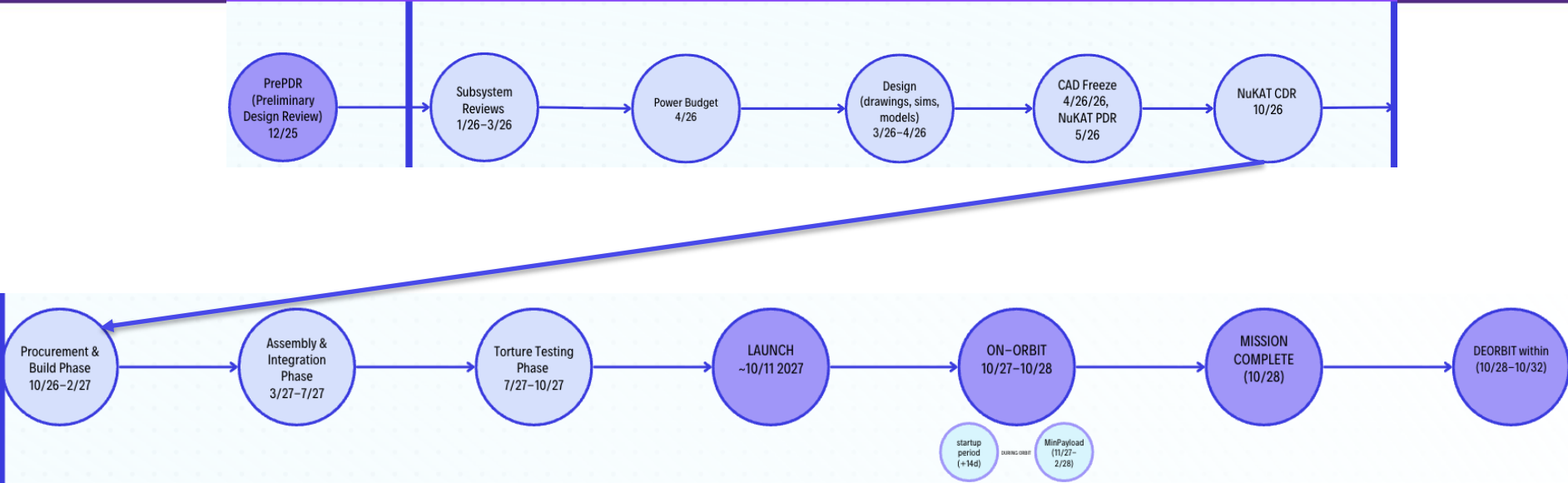
Schedule - Process

269



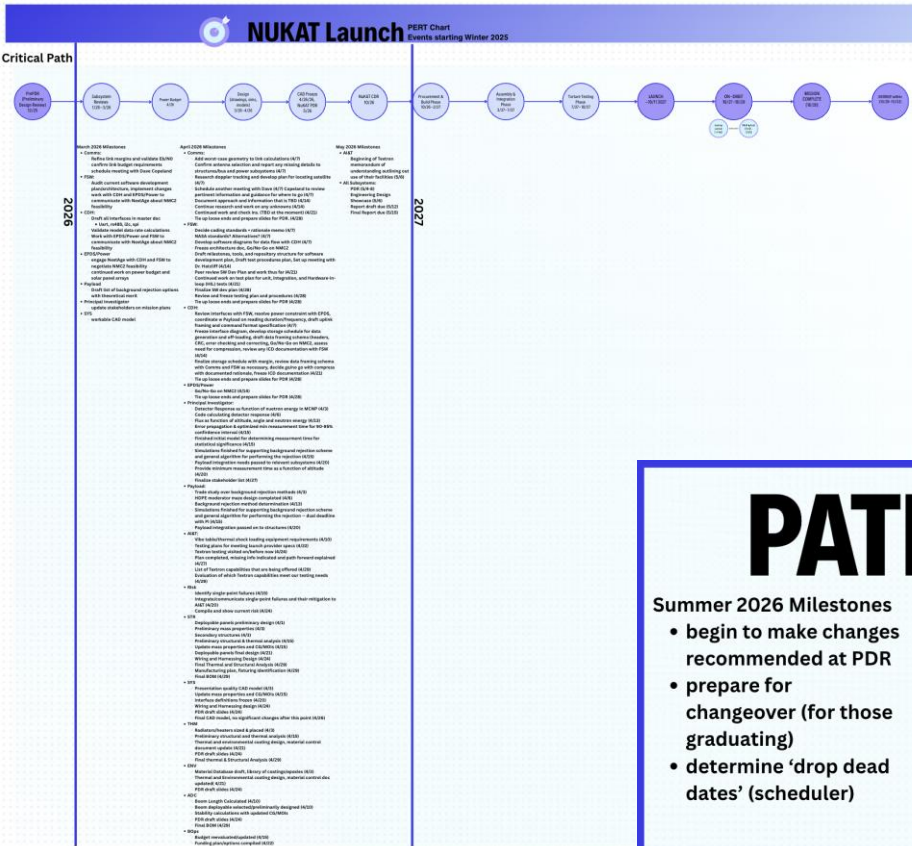
Schedule –PERT Critical Path

270



Schedule – PERT & CDR Path

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PATH FROM PDR TO CDR

Summer 2026 Milestones

- begin to make changes recommended at PDR
- prepare for changeover (for those graduating)
- determine 'drop dead dates' (scheduler)

August 2026 Milestones

- detailed drawings
- simulations
- build prototypes
- order items with 8-12 week lead time
- determine all vendors
- PDR recommendations complete - mid month

September 2026 Milestones

- CAD freeze (NO MORE CHANGES TO GEOMETRY)
- reaffirm power budget (from 4/26)
- testing runs
- fulfill promises made at PDR by end of September (Ideally)

October 2026 Milestones

- prepare for CDR
- complete all required documentation (stress reports, thermal analysis, ICDs)
- ALL PDR PROMISES MUST BE COMPLETE



KANSAS STATE UNIVERSITY

Carl R. Ice
College of Engineering

Requirement Compliance

- No noncompliance issues

Subsystems Compatibility

- No subsystem compatibility issues
- Schedule subsystem is compatible

Schedule has a Path Toward CDR

Schedule is Capable of Supporting NuKAT Mission

In light of this, Schedule should proceed toward CDR with further effort to develop the schedule & support all other subsystems.

RISK ASSESSMENT

Dale Petrie – Risk Lead

Industrial Engineering | Spring 2026



Updating of Risks

- The distribution has changed of red yellow and green risks

Wrote a Risk Management Plan

- Avoidance, Mitigation or Reduction Transfer or Share, Acceptance or Assumption, Research or Watch

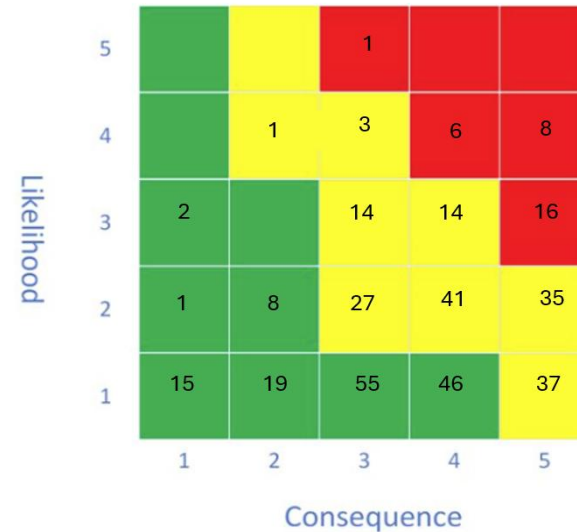
- Manage risk, not eliminate it
- Identify, assess, mitigate, and track risks
- Use RVM to log and score risks
- Focus on high-risk items, monitor lower ones
- Continuously update and review risks

Risk Assessment Strategy

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- Approximately 350 risks identified
- Each risk comes from a system requirement.
- Focus on red level risks
- Working with sub-teams to identify risks and create mitigation plans.

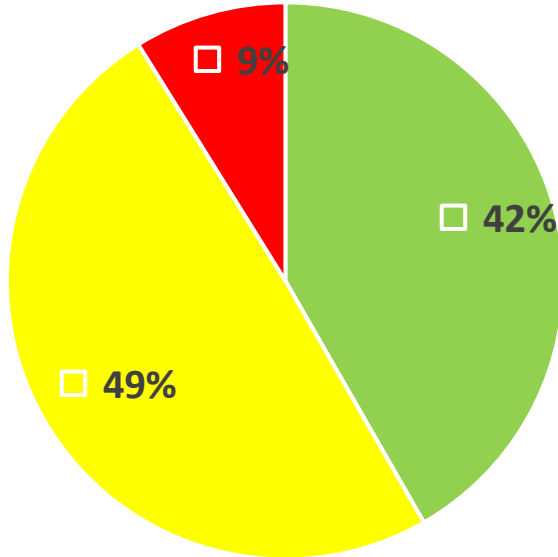
5x5 Matrix Number Breakdown of RVM



Risk Assessment Distribution

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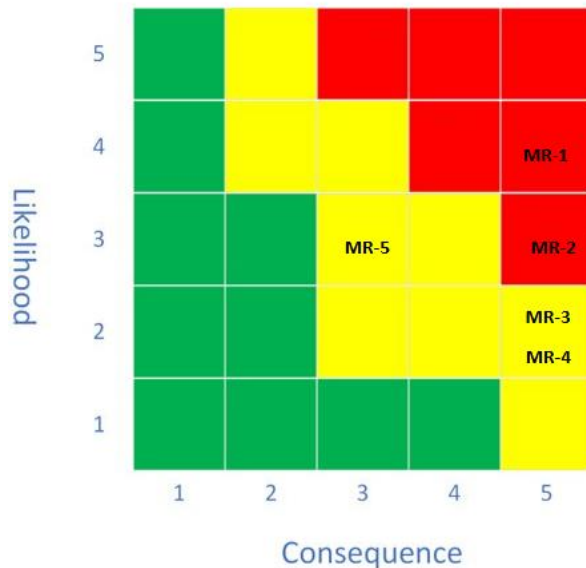
Risk Distribution



Risk Type	Frequency	Percent
Green	146	41.71%
Yellow	173	49.43%
Red	31	8.86%
Total	350	100.00%

Top 5 Mission-Level Risks (Level 1)

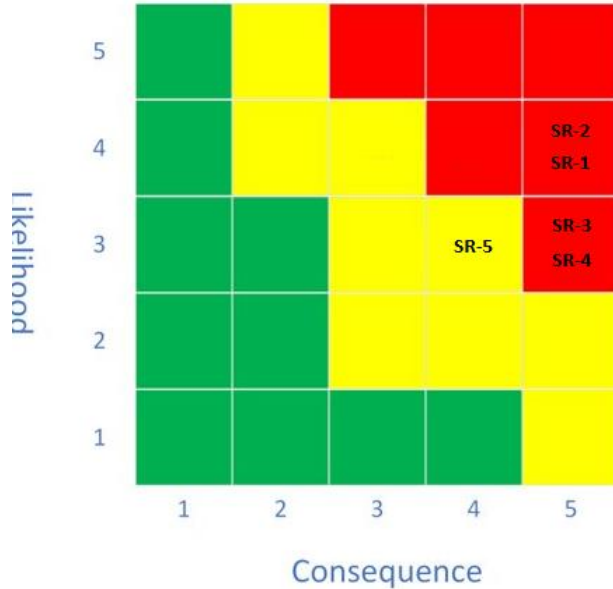
278



ID	Risk	Approach	Mitigation Strategy
MR-1	Satellite + payload may not remain fully integrated and operational for the minimum mission lifetime. (NU-1.03)	Mitigation	Validate integration early, run full mission simulations, test satellite + ground station together.
MR-2	Components may not survive radiation, vacuum, and temperature extremes in orbit. (NU-1.07)	Mitigation	Use space-rated parts, build environmental budgets, test with thermal vacuum and radiation simulations.
MR-3	Hardware may not survive launch vibration, shock, and transport. (NU-1.06)	Mitigation	Perform vibration and shock testing, reinforce structure, verify margins before launch.
MR-4	Satellite may not meet compliance requirements from launch providers and governing bodies. (NU-1.08)	Mitigation	Review requirements early, follow standards, perform compliance checks throughout design.
MR-5	Neutron detector may not achieve flight heritage or produce valid data. (NU-1.01)	Accept	Test and calibrate detector, validate data against expected orbital results.

Top 5 System-Level Risks (Level 2)

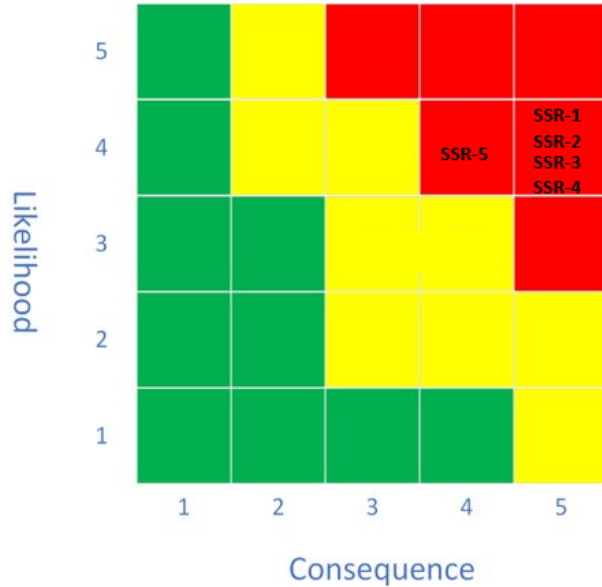
279



ID	Risk	Approach	Mitigation Strategy
SR-1	Orbital radiation environment may exceed design margins and damage on-orbit hardware. (SYS-2.01d)	Mitigate	Perform radiation analysis, size shielding and rad-hard components with margin, validate using simulation and testing.
SR-2	High-energy proton/electron environment may cause SEUs and system upsets. (SYS-2.01e)	Mitigate	Model EM environment, use tolerant components, add error correction and watchdog systems, validate with testing.
SR-3	Detector configuration may be obstructed and impact data accuracy. (PL-2.06)	Mitigate	Optimize detector placement, verify field of view, and test configurations before final integration.
SR-4	Bus system may not withstand required radiation over mission lifetime. (BUS-2.06)	Mitigate	Use space-rated components, simulate radiation exposure, and validate through testing and analysis.
SR-5	Satellite may not structurally survive launch and on-orbit environments. (SYS-2.16)	Mitigate	Perform FEA, vibration, shock, and thermal vacuum testing with proper safety margins.

Top 5 Subsystem-Level Risks (Level 3)

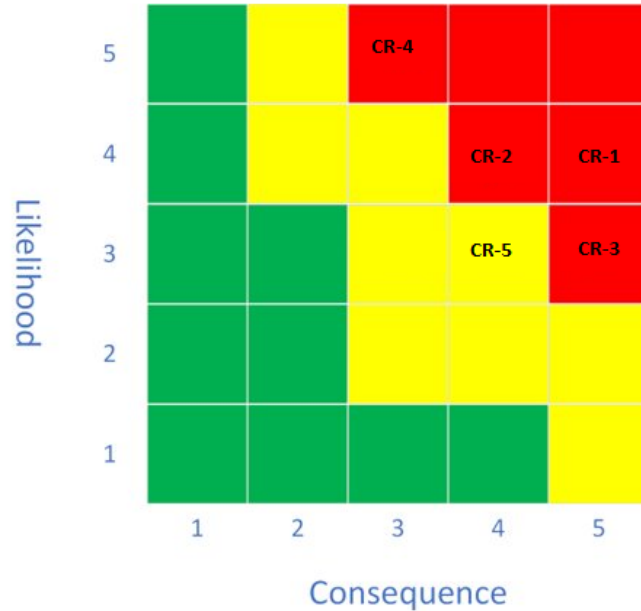
280



ID	Risk	Approach	Mitigation Strategy
SSR-1	Ground station compatibility TBD may prevent C&DH from transferring required telemetry. (SYS-3.06)	Mitigate	Finalize ground station infrastructure early and test all C&DH message formats with GS hardware.
SSR-2	FSW may not correctly control data interpretation and C&DH transmission. (SYS-3.15)	Mitigate	Validate FSW data handling logic and run end-to-end C&DH transmission tests.
SSR-3	FSW may not correctly control power distribution based on subsystem needs. (SYS-3.16)	Mitigate	Test power control logic under different operating modes and failure cases.
SSR-4	NuKAT structure may not meet launch system interface requirements. (SYS-3.19)	Mitigate	Verify launch provider requirements early and perform structure/dispenser compatibility checks.
SSR-5	Neutron detector assembly may exceed size limits and create integration issues. (PL-3.06)	Mitigate	Verify detector size in CAD and inspect fit before final integration.

Top 5 Component-Level Risks (Level 4)

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ID	Risk	Approach	Mitigation Strategy
CR-1	Ground station software hosting may not meet mission requirements. (GSW-4.01)	Mitigate	Verify server compatibility early, test hosting environment, and validate data handling and communication performance.
CR-2	Detector power distribution may not meet required 5V rail output. (PL-4.06)	Mitigate	Validate power budget, test voltage rail under load, and verify current supply meets detector needs.
CR-3	Facility may not provide reliable 120V regulated power for testing. (EGSE-4.02)	Mitigate	Test facility power stability, use regulated power supplies, and validate equipment performance under load.
CR-4	Lack of ESD protection may damage sensitive components during assembly. (EGSE-4.03)	Mitigate	Implement ESD protocols, use grounded mats and wrist straps, and verify compliance during handling.
CR-5	Thermistors may not accurately measure or report temperature. (THM-4.03)	Mitigate	Calibrate sensors, validate readings under test conditions, and compare against known references.

Requirement Compliance

- Address open requirement risks through the RVM and mitigation plans
- Focus on red-level risks first and track yellow/green risks as the design matures

Subsystems Compatibility

- Risk tracking is compatible across all NuKAT sub-teams
- Each subsystem can use the same RVM format to log impact, probability, risk score, and response
- Continue working with sub-teams to identify interface and compatibility concerns

Risk Management has a Path Toward CDR

Risk Management Subsystem is Capable of Supporting NuKAT Mission

In light of this, Risk Management should proceed toward CDR with further effort to close high-risk items, update mitigation plans, and continue tracking risks through the RVM.

CONCLUSION

Bailey Walke – Mechanical Systems Engineer

Mechanical Engineering Masters | Spring 2026



Path to Critical Design Review

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Focus Area	CDR Goal
Orbit & ConOps	Finalize mission operations timeline, ground operations concept, licensing approvals, and launch integration planning
Requirements	Baseline and verify all system/subsystem requirements with complete traceability and verification methods in the RVM
Architecture & Interfaces	Finalize flight architecture, interface definitions, ICDs, wiring/harness layouts, and subsystem integration compatibility
Detail Subsystem Designs	Complete detailed CAD, finalized structural/thermal analyses, manufacturing drawings, and validated mass, power, data, and link budgets
Risk Reduction	Retire major technical risks through analysis, prototyping, and subsystem testing; finalize mitigation strategies for remaining risks
Verification & Test Planning	Finalize system verification matrix, environmental and functional test procedures, AI&T flow, and reserve/confirm required test facilities per GEVS

NuKAT Critical Design Review to be presented next year



- What mission and design definition content is missing?
- What mission and design definition needs more development?
- What mission and design definition trade studies are needed?
- Feedback on path to compliance/noncompliances

BACKUP SLIDES – DAY THREE



Thermal – TD Simulation

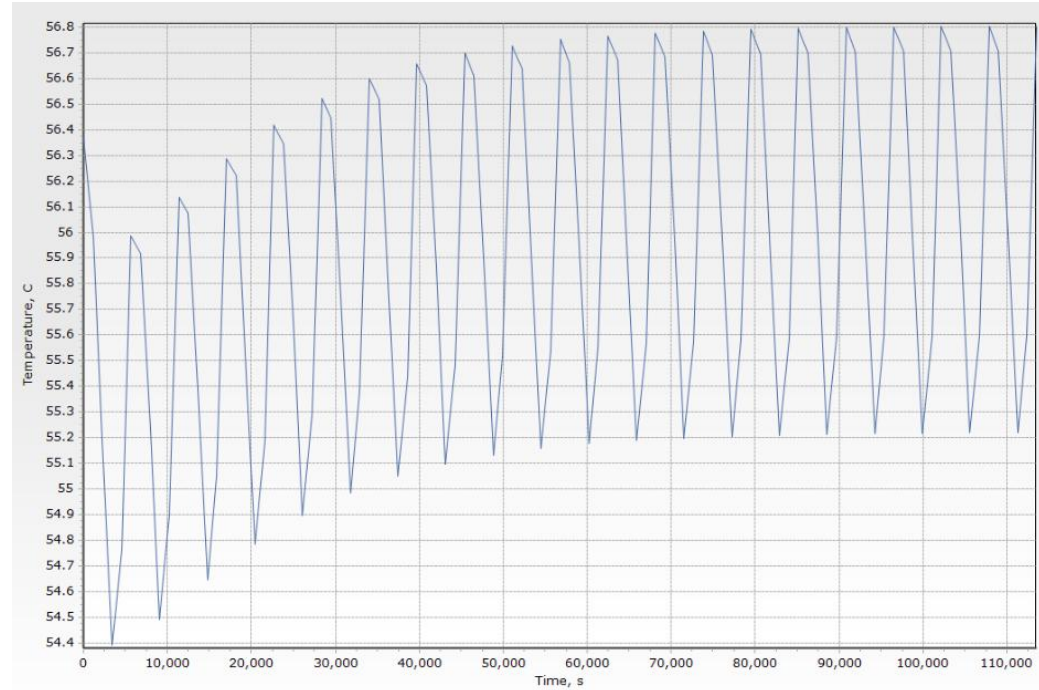
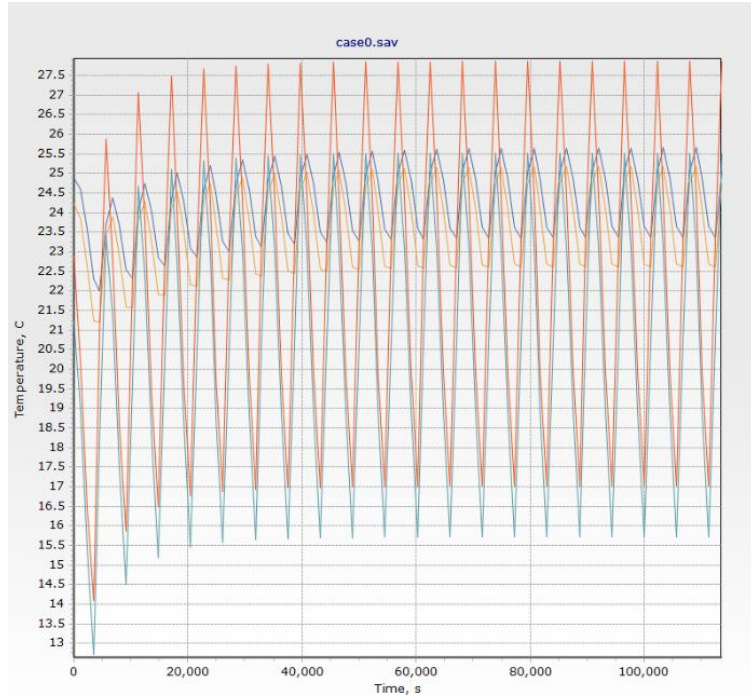
287

Problem	Solution
Orbit Uncertainty/variation	Simulate variety of possible scenarios
Placeholder/ Missing Components	Identify and integrate known components
Components Always On	Implement On/Off states
Component Waste Heat Estimate	Verify Accurate Values
Viability of Ignoring Conduction from Structure to E-Stacks	Research Feasibility and Make Adjustments



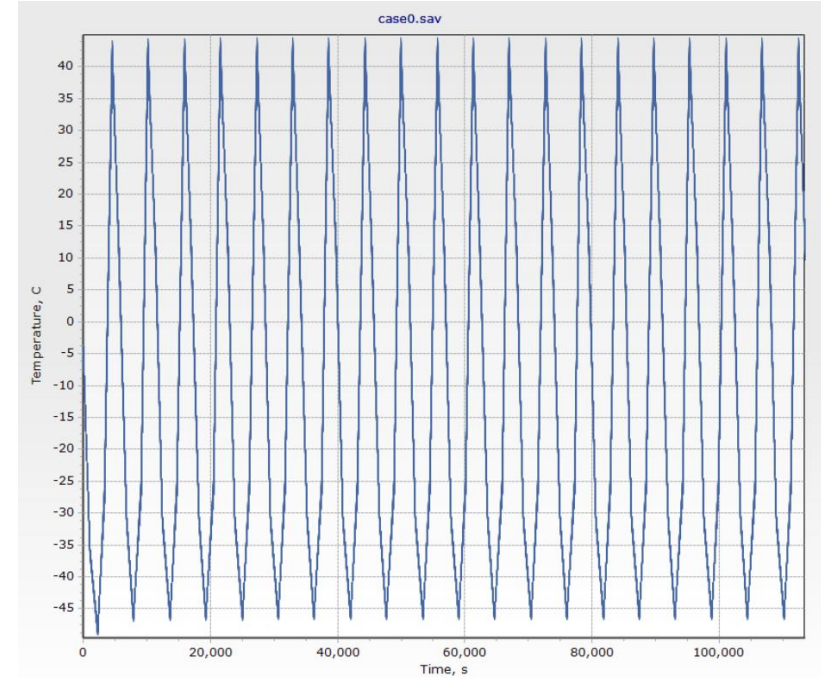
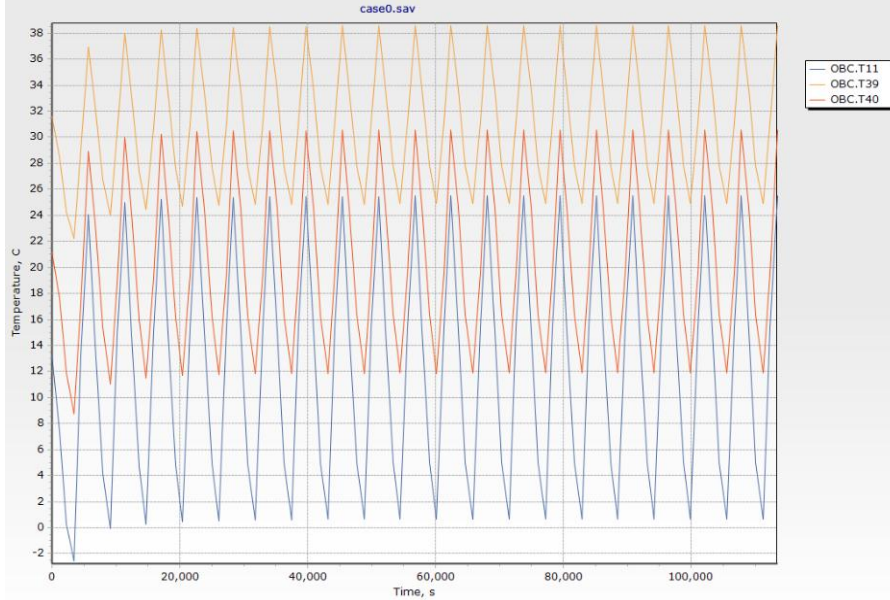
Thermal – Payload & EPDS

288



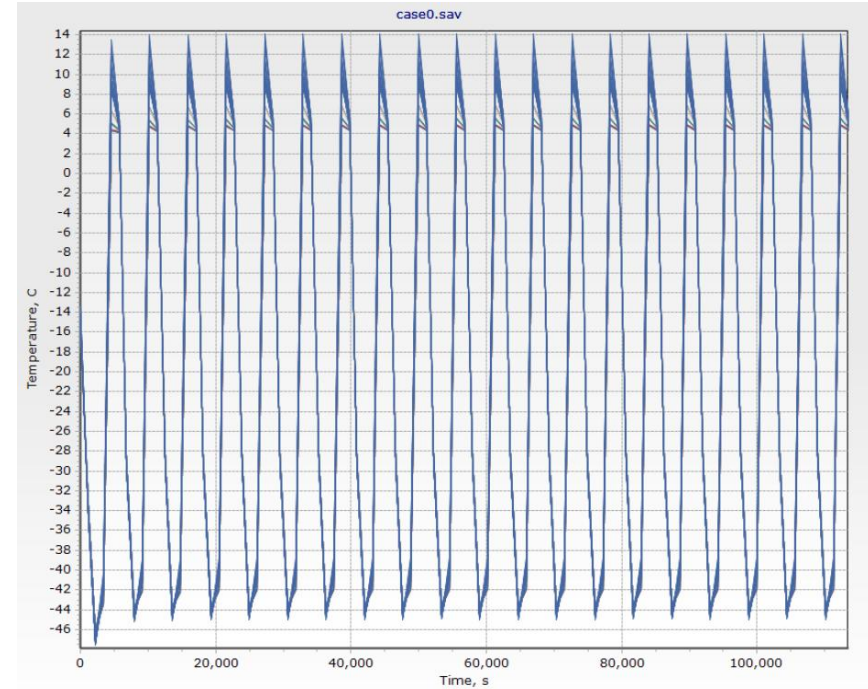
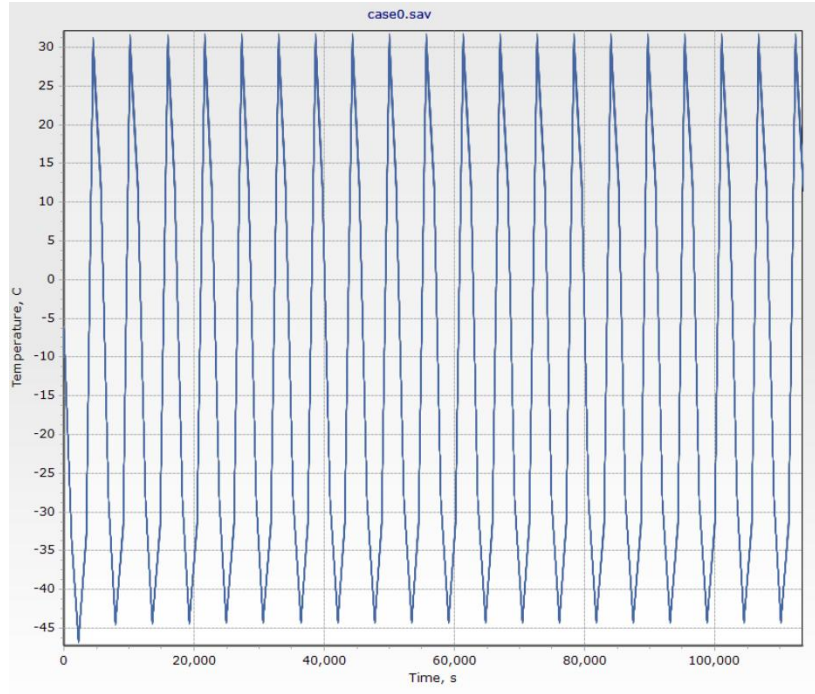
Thermal – OBC & Solar Panel

289



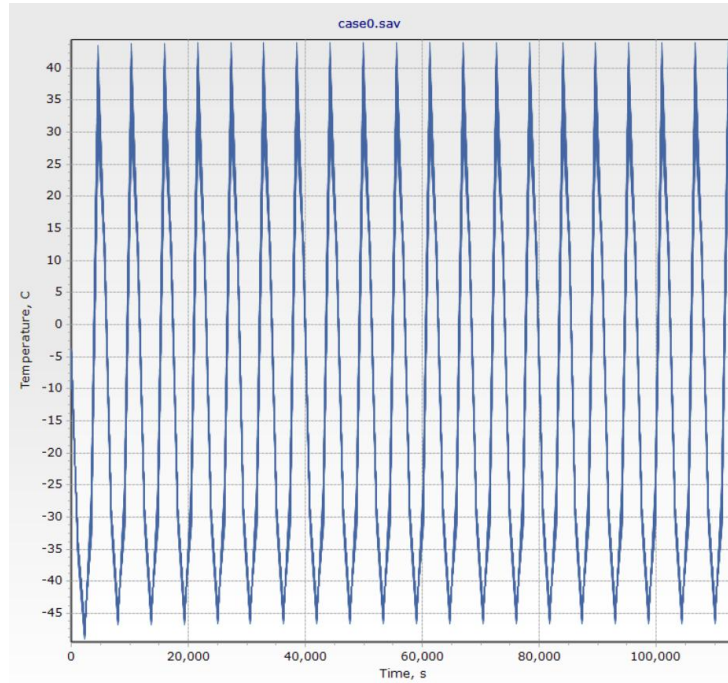
Thermal – Nadir & +/- Z

290



Thermal +/- Velocity

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Minco Thermofoil Heater	Collins Thermistor
Small & Lightweight	Fast Response Time
Variety of Sizes + Shapes	Easy to Implement
	Lightweight

Standards

- ASTM-e595 - Outgassing
- ESA 4006 – Thermistor reliability
- S-311-P-841 – Spacecraft Thermofoil Heaters

- VHF/UHF All Mode Transceiver IC-9700 currently used at radio station.
- Used to uplink and downlink data between satellite and ground station.
- Operates between 430-450 MHz, meeting mission requirements.
- Existing Cost: \$2,100



- Yaesu G-5500 rotator currently being used at radio station.
- Rotator is used to assist the antenna as it tracks satellite in space.
- Meets level 2 pointing accuracy requirement.
- Existing Cost: \$830



- Radio station currently uses S.A.T controller which does not meet mission requirements.
- Yaesu Rotator Computer Controllers GS-232B.
- Connects to PC to hosts mission critical software such as Cosmos.
- New Cost: \$650



Amplifier

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- EME3-432 70cm Ultimate Split RX Relay Switches BPF VLNA
- Boosts downlink signals to survive travel through the atmosphere to be picked up by ground station.
- New Cost: \$247



- IARU Request Form
- Part 97- Mission must provide value to amateur radio as a whole.
- Ideas:
 - FM Repeater
 - Linear Transponder Module
 - Packet Relay
 - Wildcat Space Program



The International Amateur Radio Union

Since 1925, the Federation of National Amateur Radio Societies
Representing the Interests of Two-Way Amateur Radio Communication

AMATEUR SATELLITE FREQUENCY COORDINATION REQUEST INSTRUCTIONS AND APPLICATION FORM¹

1. Amateur-satellite service. Amateur stations meet the requirements of the radio regulations², RR 1.56. and 1.57.

External Interfaces

- Operators
- Ground Station

Internal Interfaces

- Database

- Continue work on finishing IARU request form.
- Begin finalizing and ordering necessary equipment for radio station.
- Begin testing/experimenting at radio station.
- Continue to prepare for new ground station team lead next semester.
- New person to work on ground station database.

Level 3 Compliances:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
3.11 Ground Software Requirements					
GSW-3.01	Network	The ground software shall establish a network connection between what is hosting it and the ground station. TBD if the ground station has network sockets	Yes/No	Yes	Partial Compliance
GSW-3.02	File Transfer	The ground software shall reliably and securely be able to transfer information to the satellite.	Yes/No	Yes	Partial Compliance
GSW-3.03	Communication Protocol	The ground software shall establish an encryption protocol for uplinking commands during communications. Need of a compression protocol is TBD on data estimates	Yes/No	Yes	Compliant
GSW-3.04	Data Processing	The ground software shall be able to process and display data in a meaningful manner	Yes/No	Yes	Compliant
GSW-3.05	Storage/Data Pipeline	The ground software shall store and retrieve data from a database	Yes/No	Yes	Compliant
GSW-3.06	Distribution	The ground software shall distribute process data to users.	Yes/No	Yes	Compliant
GSW-3.07	Alert System	The ground software shall alert a user or admin when necessary	Yes/No	Yes	Compliant

Level 4 Compliances:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
4.11 Ground Software Requirements					
GSW-4.01	Host	The ground station software shall be hosted on a server in the Kansas State Network	Yes/No	Yes	Not Compliant
GSW-4.02	Software Package	The ground station software will be based on the open-source software Cosmos and additional functionality will be added on as needed.	Yes/No	Yes	Compliant
GSW-4.03	InfluxDB	The ground station software shall use InfluxDB as the database for storing telemetry data and neutron counts	Yes/No	Yes	Compliant
GSW-4.04	AES-256	The ground station software shall use Advanced Encryption Standard to encrypt uplink commands	Yes/No	Yes	Compliant
GSW-4.05	Data Distribution	The ground station software shall utilize the Ruby Mail package to send out data on request.	Yes/No	Yes	Compliant
GSW-4.06	Alert System	The ground station software shall build an alert system into COSMOS that will alert an admin or registered individual to anomalies or extreme events.	Yes/No	Yes	Compliant

Non-Compliant:

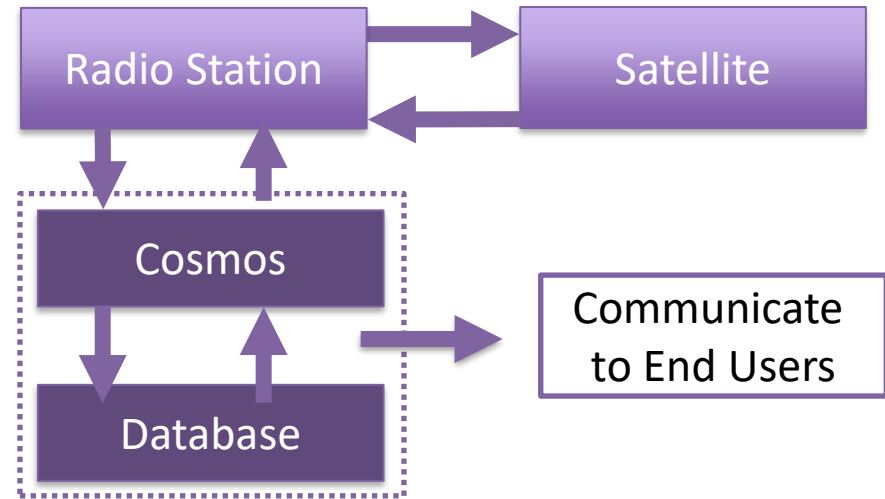
- Host
 - We are not currently cleared to host our software on the K-State network
- Mitigation
 - Formal request from our team
 - We will need to stress the importance of our mission, the value for students, and the value of the data we will return
 - Student-led team
 - Research
 - Interaction with Radio Club

External Interfaces

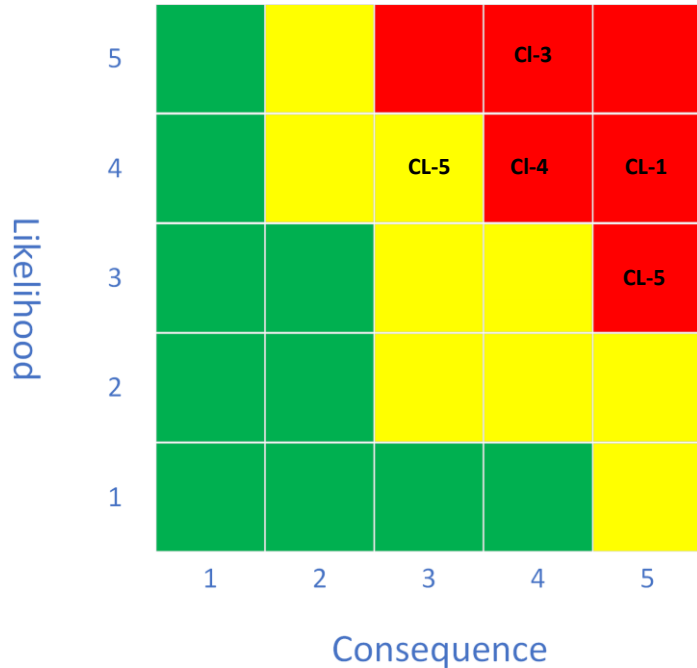
- Operators
 - Those who will be operating COSMOS through commands
- End Users
 - Those who will be requesting the data that we collect
- Connection to the ground station
 - This will be the network connection to the radio station
- Uplink/Downlink
 - This is the information that we will be sending and receiving from the radio station

Internal Interfaces

- Database
 - Connection between COSMOS and the InfluxDB

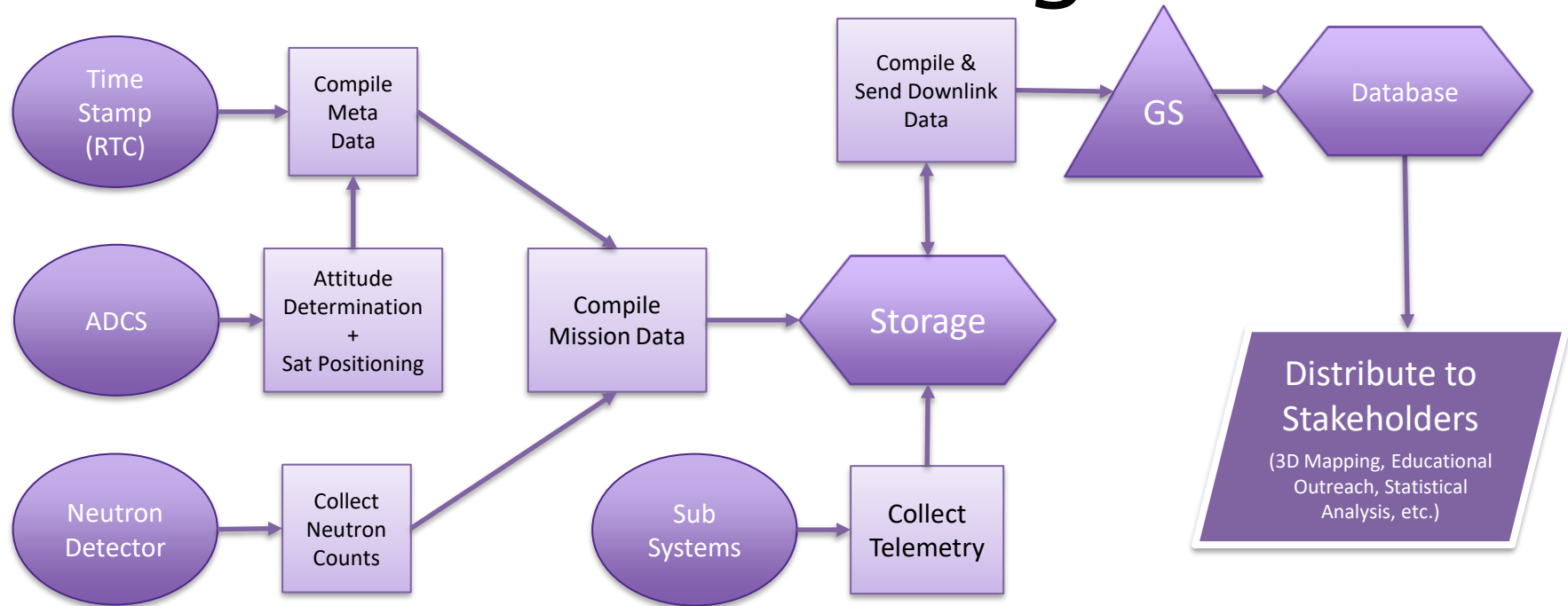


Component-Level Risk Slide (Level 4 Requirements)



ID	Risk	Mitigation Strategy	Approach
CL-1	Detectors may exceed thermal limits during high-load operation, causing noise or hardware failure.	Increase radiator efficiency, verify heater control logic, and perform thermal vacuum testing for worst-case conditions.	M
CL-2	Power distribution board may be unable to reliably provide the 5V rail required for all four detectors.	Validate rail loading with engineering models, add margin to regulator performance, test under max-draw scenarios.	M
CL-3	DSC-Domino C&DH interface may experience intermittent communication failures through USB-C under vibration or thermal cycling.	Perform vibration/thermal stress testing on connectors, consider strain-relief bracket, evaluate direct-solder alternative.	M
CL-4	Heater elements may not supply sufficient heat fast enough to prevent detectors from dipping below -20 °C.	Model heater sizing, improve mounting conductivity, run cold-start thermal TVAC tests to verify response speed.	M
CL-5	Some payload fastening points (Domino + moderator interface) may not maintain mechanical stability under launch loads.	Re-evaluate bolt selection, verify torque specs, and perform modal analysis + bench-level vibration testing.	M

Data Flow Diagram



Acronyms (A-L)

- ADC(S) – Attitude Determination and Control (Subsystem)
- ATID – Analysis, Test, Inspect, Demonstrate
- B-dot/S-dot – Method of angular velocity determination using change in Earth or Sun vector orientation over time
- BMS – Battery Management System
- bps – Bits per Second
- CAD – Computer Aided Design
- CAN – Controller Area Network
- CD(&)H – Communications and Data Handling
- ConOps – Concept of Operations
- CSS – Coarse Sun Sensor
- EGSE – Electrical Ground Station Equipment
- EM – Electromagnetic
- EMI – Electromagnetic Interference
- EPD – Energy and Power Distribution
- FSS – Fine Sun Sensor
- FSW – Flight Software
- GEVS – General Environmental Verification Standard
- GMSK – Gaussian Minimum Shift Keying
- GNSS – Global Navigation Satellite System
- GPS – Global Positioning System
- GS – Ground Station
- GSE – Ground Support Equipment
- HDPE – High Density Polyethylene
- HIL – Hardware In The Loop
- H&S – Health and Status
- I2C – Inter-Integrated Circuit
- KSU – Kansas State University
- KB – Kilobyte(s)
- LEO – Low Earth Orbit
- LV – Launch Vehicle

Acronyms (M-Z)

- MAB – Mission Advisory Board
- MCX – Micro Coaxial Connector
- MDR – Mission Definition Review
- MGSE – Mechanical Ground Station Equipment
- mrg - Margin
- MS – Microsoft
- MSN – Main Subsystem Network
- MSND – Microstructured Semiconductor Neutron Detector
- MTM – Magnetometer
- MTQ – Magnetorquer (either air or solid core)
- NASA – National Aeronautics and Space Administration
- NOAA – National Oceanic and Atmospheric Administration
- OBC – Onboard Computer
- PCDU – Power Control Distribution Unit
- PDR – Preliminary Design Review
- PID – Proportional-Integral-Derivative (controller)
- PLD – Payload
- PV - Photovoltaic
- RDT – Radiation Detection Technologies
- RF-SCOE – Radio Frequency Special Check-Out Equipment
- RS-422 – Recommended Standard 422
- RS-485 – Recommended Standard 485
- RTC – Real Time Clock
- RTD – Resistance Temperature Detector
- RVM – Requirement Verification Matrix
- Rx – Receive
- SAA – South Atlantic Anomaly
- SIL – Software In The Loop
- SMP - Small Push-On Connector
- SR – System Risk
- TBD – To Be Determined
- TRIAD – Tri-Axial Attitude Determination
- Tx – Transmit
- UHF – Ultra High Frequency